

Algebraic Geometry Notes

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Introduction

Rings are always commutative with unity.

Chapter 1

Affine Space

1.1 Affine Varieties and the Nullstellensatz

In algebraic geometry we are interested in studying spaces which are locally described by polynomial equations.

Definition 1.1

Let $F \in \mathbb{R}[x, y]$. Then the **vanishing set** or zero locus of F is

$$V(F) = \{(x, y) \in \mathbb{R}^2 : F(x, y) = 0\}$$

This extends the construction of many familiar spaces, such as the circle:

$$S^1 = V(f), \quad f(x, y) = x^2 + y^2 - 1$$

These equations can give information about the spaces themselves. For instance, we say that a point in the vanishing set is *singular* if $\nabla F = 0$ at that point. For S^1 , there are no singular points, while for the more complicated space

$$y^2 - x^2(x + 1) = 0$$

there is a singular point at $(0, 0)$, which is seen to be a crossing point of this curve. Thus, we intuitively see that singular points should correspond somehow to a failure to be smooth or a manifold.

This corresponds to the general situation in algebraic geometry, where we are able to apply tools from commutative algebra to analyze our spaces, as well as giving geometric context and intuition for problems in algebra.

It is worth noting that the solution sets to polynomial equations depend on the space we work over. For instance, $x^2 + y^2 - 1 = 0$ is satisfied at four discrete points over $\mathbb{Z}^2$, a circle over $\mathbb{R}^2$, and over $\mathbb{C}^2$ the solution set is basically a sphere minus one point. One could also

work over finite fields, and so on. For now we will be working over algebraically closed fields, in particular $\mathbb{C}$.

Definition 1.2

Let K be an algebraically closed field. Then we define the **affine n -space** over K to be

$$\mathbb{A}_K^n = \{(a_1, \dots, a_n) : a_i \in K\}$$

We may also denote affine n -space by $\mathbb{A}^n$ when K is obvious.

Definition 1.3

Let $S \subseteq K[x_1, \dots, x_n]$ be a collection of polynomials. Then their **vanishing locus** is

$$V(S) = \{x \in \mathbb{A}^n : f(x) = 0 \text{ for all } f \in S\} \subseteq \mathbb{A}^n$$

Any set of this form is called an **affine variety**.

Proposition 1.1

If $S_1 \subseteq S_2 \subseteq K[x_1, \dots, x_n]$ then $V(S_1) \supseteq V(S_2)$.

Proposition 1.2

If f, g are polynomials then

$$V(f, g) = V(f) \cap V(g)$$

and

$$V(fg) = V(f) \cup V(g)$$

More generally, for collections $S_1, S_2 \subseteq K[x_1, \dots, x_n]$,

$$V(S_1 \cup S_2) = V(S_1) \cap V(S_2)$$

and

$$V(S_1 S_2) = V(S_1) \cup V(S_2)$$

where $S_1 S_2$ is the pointwise product

$$S_1 S_2 = \{fg : f \in S_1, g \in S_2\}$$

Even more generally, we can take arbitrary unions of collections $\bigcup_{\lambda \in \Lambda} S_\lambda$ (equivalently, arbitrary intersections of the varieties $\bigcap_{\lambda \in \Lambda} V(S_\lambda)$), but only *finite* products (equivalently, unions).

We note that the pointwise product result requires the fact that K is a domain (since we need the fact that $fg = 0 \implies f = 0$ or $g = 0$).

Proposition 1.3

If $X \subseteq \mathbb{A}^n, Y \subseteq \mathbb{A}^m$ are affine varieties, then $X \times Y \subseteq \mathbb{A}^{n+m}$ is also an affine variety.

Proof. Let $S_X \in K[x_1, \dots, x_n]$ and $S_Y \in K[y_1, \dots, y_m]$ be such that $X = V(S_X), Y = V(S_Y)$. Then project these naturally into $K[x_1, \dots, x_{n+m}]$ by converting each formal variable y_i to x_{n+i} , and it follows that $X \times Y = V(S'_X, S'_Y)$. $\square$

Example 1.1

In $\mathbb{A}^1$, the only affine varieties are $\mathbb{A}^1$, and finite sets. This can be seen by noting that we immediately have

$$V(S) \subseteq V(f)$$

whenever $f \in S$. Then, either S only contains the zero polynomial, or else f can be chosen to be nonzero, hence to have finitely many roots.

This logic also shows that any variety in $\mathbb{A}^1$ can be described by a single polynomial equation $(z - z_1) \cdots (z - z_n) = 0$, or the zero polynomial. (For general S this polynomial need not actually be in S , but if S is an ideal then it is.) We would like to have a similar finiteness result for varieties in $\mathbb{A}^n$. The following will demonstrate this.

Definition 1.4

Recall that if $S \subseteq K[x_1, \dots, x_n]$, then the generated **ideal** is

$$\langle S \rangle = \{r_1 f_1 + \dots + r_m f_m : r_i \in K[x_1, \dots, x_n], f_i \in S\}$$

Proposition 1.4

$$V(S) = V(\langle S \rangle).$$

Proof. To show that $V(\langle S \rangle) \subseteq V(S)$, we just note that $S \subseteq \langle S \rangle$.

To see that $V(S) \subseteq V(\langle S \rangle)$, at any element $x \in V(S)$, each $f \in S$ vanishes, so any polynomial in $\langle S \rangle$ vanishes as well. $\square$

Theorem 1.5: Hilbert's Basis Theorem

If a ring R is Noetherian then $R[x]$ is as well.

Proof. See Section A.1. $\square$

In particular note that fields are always Noetherian, so we see that $\langle S \rangle$ itself is finitely generated.

Corollary 1.6

Any affine variety can be written as $X = V(f_1, \dots, f_m) \subseteq \mathbb{A}^m$.

Thus we see that the theory of varieties can be written entirely in terms of ideals, which allows us to enjoy finiteness properties.

Definition 1.5

If J is an ideal in R then the **radical** of J is

$$\sqrt{J} = \{x \in R : \exists m \text{ s.t. } x^m \in J\}$$

Lemma 1.7

For I, J two ideals, $\sqrt{IJ} = \sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J}$.

Proof. First, we have $IJ \subseteq I \cap J$, since $SI \subseteq I$ for any subset $S \subseteq R$ and any ideal I . Therefore $\sqrt{IJ} \subseteq \sqrt{I \cap J}$.

If $x \in \sqrt{I \cap J}$ with $x^m = z$, $z \in I \cap J$, then plainly $x \in \sqrt{I}, \sqrt{J}$ as well so $\sqrt{I \cap J} \subseteq \sqrt{I} \cap \sqrt{J}$.

Lastly, if $x \in \sqrt{I} \cap \sqrt{J}$, then $x^m = y \in I$ and $x^n = z \in J$ for some y, z . It follows that $x^{m+n} = yz \in IJ$. So $\sqrt{IJ} \subseteq \sqrt{I} \cap \sqrt{J}$, completing the circular inclusion. $\square$

Lemma 1.8

If J, J_1, J_2 are ideals in $K[x_1, \dots, x_n]$, then

- (a) $V(\sqrt{J}) = V(J)$.
- (b) $V(J_1) \cup V(J_2) = V(J_1 J_2) = V(J_1 \cap J_2)$.
- (c) $V(J_1) \cap V(J_2) = V(J_1 + J_2)$.

Proof. (a) $V(\sqrt{J}) \subseteq V(J)$ because $J \subseteq \sqrt{J}$. In the other direction, if $x \in V(J)$, then for any $g \in \sqrt{J}$, $g^m(x) = 0$ for some m . Since K is a field, $g(x) = 0$ so $x \in V(\sqrt{J})$.

(b) The first equality is a special case of the version for subsets. The second follows from the lemma $\sqrt{IJ} = \sqrt{I \cap J}$.

(c) This is the previous version, except that we note that $J_1 + J_2 = \langle J_1 \cup J_2 \rangle$. $\square$

At this point, we have constructed a mapping that takes ideals to affine varieties. To better understand this correspondence, we now construct the inverse map.

Definition 1.6

Let $X \subseteq \mathbb{A}^n$ be a subset. Then **vanishing ideal** of X is

$$I(X) = \{f \in K[x_1, \dots, x_n] : f|_X = 0\}$$

For similar reasons as before, this is indeed an ideal. In particular it is a radical ideal. Also, the map $X \mapsto I(X)$ is inclusion-reversing, so if $X_1 \subseteq X_2$ then $I(X_1) \supseteq I(X_2)$.

While $X \mapsto I(X)$ and $J \mapsto V(J)$ are maps from sets to ideals and vice versa, we need to clarify the manner in which they compose, since they are not strictly inverses of each other.

Theorem 1.9: Nullstellensatz

- (a) If $J \subseteq K[x_1, \dots, x_n]$ is an ideal then $I(V(J)) = \sqrt{J}$.
- (b) If $X \subseteq \mathbb{A}^n$ is an affine variety then $V(I(X)) = X$.

Proof. (a) It is clear that any f with $f^m \in J$ must vanish on $V(J)$ by our previous work, so $\sqrt{J} \subseteq V(J)$. The other direction is the actually difficult part of this proof and is not demonstrated. It is equivalent to the statement that if f vanishes on $V(J)$ then $f^m \in J$ for some m .

- (b) Suppose $X = V(J)$. Then $J \subseteq \sqrt{J} = I(X)$ (by part a) so $V(I(X)) \subseteq X$. In the other direction, $I(X)$ is defined to be functions which vanish on X , so necessarily X lies in their vanishing locus. $\square$

This shows that if we restrict the previous mapping between ideals and varieties to only the radical ideals, then we get a bijection. This bijection is inclusion-reversing.

Corollary 1.10

There is a bijective correspondence between maximal ideals and points $a = (a_1, \dots, a_n) \in \mathbb{A}^n$.

Proof. Maximal ideals are maximal among radical ideals (since the radical of a proper ideal is proper). By the inclusion reversing bijection, they correspond to *minimal* nonempty varieties, which are singular points. $\square$

In particular the bijection can be written as $a \mapsto \langle x_1 - a_1, \dots, x_n - a_n \rangle$.

Lemma 1.11

Let $X_1, X_2 \subseteq \mathbb{A}^n$ be affine varieties. Then

- (a) $I(X_1 \cup X_2) = I(X_1) \cap I(X_2)$,

$$(b) \ I(X_1 \cap X_2) = \sqrt{I(X_1) + I(X_2)}.$$

Proof. (a) By the inclusion reversing bijection,

$$\begin{cases} X_1 \subseteq X_1 \cup X_2 \\ X_2 \subseteq X_1 \cup X_2 \end{cases} \implies I(X_1 \cup X_2) \subseteq I(X_1) \cap I(X_2)$$

and any function which vanishes on X_1, X_2 vanishes on their union, so $I(X_1) \cap I(X_2) \subseteq I(X_1 \cup X_2)$.

(b) We have

$$I(X_1 \cap X_2) = I(V(I(X_1)) \cap V(I(X_2))) = I(V(I(X_1) + I(X_2))) = \sqrt{I(X_1) + I(X_2)} \quad \square$$

Example 1.2

Writing $z = x + iy$ in $\mathbb{A}_{\mathbb{C}}^1$, let X_1, X_2 be the subsets $\{x = 0\}, \{y = 0\}$, respectively. Note that as before these are *not* algebraic varieties, so the above Lemma does not hold. In fact it fails, since the vanishing ideal of both is the zero ideal, but their intersection is the origin, so:

$$I(X_1 \cap X_2) = \langle x \rangle \neq 0 = \sqrt{I(X_1) + I(X_2)}$$

Corollary 1.12: Weak Nullstellensatz

If $J \subseteq K[x_1, \dots, x_n]$ is an ideal, then the following are equivalent:

- (a) $V(J) = \emptyset$,
- (b) $J = K[x_1, \dots, x_n]$,
- (c) $1 \in J$.

Proof. (b) $\iff$ (c) is basic ring theory. Now note that if $1 \in J$ then clearly $V(J) = \emptyset$, and if $V(J) = \emptyset$ then 1 vanishes on $V(J)$, so $1 \in \sqrt{V(J)} \implies 1 \in J$. $\square$

1.2 Subvarieties

We now study varieties which live inside of other varieties, instead of simply $\mathbb{A}^n$.

Definition 1.7

Let $X \subseteq \mathbb{A}^n$ be an affine variety. Then the **coordinate ring** of X is

$$A(X) = \{f|_X : f \in K[x_1, \dots, x_n]\} \subseteq K^{(K^n)}$$

Note that the elements of $A(X)$ are considered to be functions represented by polynomials, not polynomials, and only need to be equal as functions. For instance, $x^2 + x$ is the zero function on $\mathbb{F}_2$, but is not equal as a polynomial to the zero polynomial. ($\mathbb{F}_2$ is not algebraically closed, but this is just for illustration.)

Lemma 1.13

$A(X)$ is a K -algebra.

Recall that a K -algebra can equivalently be defined either as a K -vector space with a bilinear product, or as a ring A with a ring homomorphism from K into $Z(A)$.

Proof. $A(X)$ is clearly a ring, and the homomorphism takes $\lambda \in K$ to the constant polynomial $f \equiv \lambda$. $\square$

Example 1.3

If $X \subseteq \mathbb{A}^1$ is a finite set with r points, then by Lagrange interpolation we can define a polynomial with any prescribed values on these points. So $A(X) = K^r$.

Proposition 1.14

Let $X \subseteq \mathbb{A}^n$ be an affine variety. Then $f \mapsto f|_X$ is a surjective ring homomorphism with kernel $I(X)$, so that

$$A(X) = K[x_1, \dots, x_n] / I(X)$$

Proof. Surjectivity is by definition. Then, we have injectivity since $f|_X = 0$ if and only if f vanishes on X , which is to say that $f \in I(X)$. $\square$

In particular if $X = V(J)$ then

$$A(V(J)) = K[x_1, \dots, x_n] / \sqrt{J}$$

Corollary 1.15

$A(\mathbb{A}^n) = K[x_1, \dots, x_n]$.

Proof. We have $\mathbb{A}^n = V(\langle 0 \rangle)$, so $I(\mathbb{A}^n) = I(V(\langle 0 \rangle)) = \sqrt{\langle 0 \rangle} = \langle 0 \rangle$. Hence

$$A(\mathbb{A}^n) = K[x_1, \dots, x_n]$$

$\square$

This shows that the theory of polynomials on $\mathbb{A}^n$ is really the theory of polynomial *functions* on $\mathbb{A}^n$, so there is no generality of studying polynomial functions on varieties. In this way, we can see that our previous geometric correspondence from ideals in $A(\mathbb{A}^n)$ should also give some meaning to ideals in general $A(X)$. These turn out to be the subvarieties.

Definition 1.8

Fix an affine variety $Y \subseteq \mathbb{A}^n$. Then for any $S \subseteq A(Y)$, define its **vanishing locus** to be

$$V(S) = V_Y(S) = \{x \in Y : f(x) = 0 \forall f \in S\}$$

A subset which is the vanishing locus of some $S \subseteq A(Y)$ is called an **affine subvariety**.

Correspondingly, for any subset $X \subseteq Y$, we define the **vanishing ideal** of X in Y to be

$$I(X) = I_Y(X) = \{f \in A(Y) : f|_X \equiv 0\}$$

This new correspondence behaves almost exactly the same as the one for varieties in $\mathbb{A}^n$.

Proposition 1.16

Let $Y \subseteq \mathbb{A}^n$ be an affine variety.

- (a) $I_Y(X)$ is a (radical) ideal,
- (b) any subvariety $X \subseteq Y$ is a variety in $\mathbb{A}^n$,
- (c) any variety $X \subseteq \mathbb{A}^n$ such that $X \subseteq Y$ is also an affine subvariety of Y ,
- (d) whenever $X \subseteq Y$ is a subvariety,

$$A(X) \cong A(Y)/I(X)$$

- (e) I_Y, V_Y give an inclusion reversing, bijective correspondence between affine subvarieties of Y and radical ideals of $A(Y)$,
- (f) $V(J_1) \cup V(J_2) = V(J_1 J_2) = V(J_1 \cap J_2)$.
- (g) $V(J_1) \cap V(J_2) = V(J_1 + J_2)$.
- (h) $I(X_1 \cup X_2) = I(X_1) \cap I(X_2)$,
- (i) $I(X_1 \cap X_2) = \sqrt{I(X_1) + I(X_2)}$.

Proof. (a) Clear by lifting to polynomials.

- (b) Let $X = V_Y(J)$ for an ideal $J \subseteq A(Y)$. Let π be the quotient map from $K[x_1, \dots, x_n]$ by $I(Y)$, and let $I = \pi^{-1}(J)$. Then since $X \subseteq Y$, for $x \in X$, $\pi g(x) = 0$ if and only if

$g(x) = 0$. So

$$\begin{aligned} X = V_Y(J) &= \{x : g'(x) = 0 \forall g' \in J\} = \{x : \pi g(x) = 0 \forall g \in \pi^{-1}(J)\} \\ &= \{x : g(x) = 0 \forall g \in \pi^{-1}(J)\} = V(\pi^{-1}(J)) \end{aligned}$$

and X is a variety.

(c) Let $X = V(J)$. Then by the same logic as above,

$$\begin{aligned} X = V(J) &= \{x : g(x) = 0 \forall g \in J\} = \{x : \pi g(x) = 0 \forall g \in J\} \\ &= \{x : g'(x) = 0 \forall g' \in \pi(J)\} = V_Y(\pi(J)) \end{aligned}$$

so X is an affine subvariety of Y .

(d)

□

If we are given two varieties $X \subseteq \mathbb{A}^n, Y \subseteq \mathbb{A}^m$ which are cut out by

$$I(X) \subseteq K[x_1, \dots, x_n], \quad I(Y) \subseteq K[y_1, \dots, y_m]$$

then we know the product is a variety $\mathbb{A}^{n+m}$, which is cut out by taking

$$I_{X \times Y} = I(X)K[x_1, \dots, x_n, y_1, \dots, y_m] + I(Y)K[x_1, \dots, x_n, y_1, \dots, y_m]$$

This is a reflection of a more general fact concerning tensor products, since

$$K[x_1, \dots, x_n, y_1, \dots, y_m] = K[x_1, \dots, x_n] \otimes_K K[y_1, \dots, y_m]$$

Proposition 1.17

Let $I_A \subseteq A, I_B \subseteq B$ be ideals, and let C be a ring with morphisms into A, B . Then

$$I_{A,B} = I_A(A \otimes_C B) + I_B(A \otimes_C B)$$

is an ideal of $A \otimes_C B$, and the map

$$\begin{aligned} (A/I_A) \otimes_C (B/I_B) &\rightarrow A \otimes_C B / I_{A,B} \\ (a + I_A) \otimes_C (b + I_B) &\mapsto a \otimes_C b + I_{A,B} \end{aligned}$$

is an isomorphism.

Proposition 1.18

Let $X \subseteq \mathbb{A}^n, Y \subseteq \mathbb{A}^m$ with $I_{X \times Y}$ defined as above. Then $I(X \times Y) = I_{X \times Y}$ and

$$A(X \times Y) = A(X) \otimes_K A(Y)$$

Proof. It is clear that $V(I_{X \times Y}) = X \times Y$, so by the Nullstellensatz,

$$I(X \times Y) = \sqrt{I_{X \times Y}}$$

So we need to show $I_{X \times Y}$ is radical, which is equivalent to saying that

$$K[x_1, \dots, x_n, y_1, \dots, y_m] / I_{X \times Y}$$

is reduced. By the previous proposition we have that

$$\begin{aligned} & K[x_1, \dots, x_n, y_1, \dots, y_m] / I_{X \times Y} \\ & \cong \left(K[x_1, \dots, x_n] / I(X) \right) \otimes_K \left(K[y_1, \dots, y_m] / I(Y) \right) = A(X) \otimes_K A(Y) \end{aligned}$$

This proves the second claim contingent on the first, but also we know that $A(X), A(Y)$ are reduced since $I(X), I(Y)$ are radical. Then the fact that $A(X) \otimes_K A(Y)$ is reduced follows from the general fact that $U \otimes_K V$ is reduced whenever U, V are reduced K -algebras for K a perfect field. $\square$

1.3 The Zariski Topology

We previously saw that arbitrary intersections and finite unions of affine varieties were again varieties. This is the same property as the closed subsets of a topological space, which leads us to define the Zariski topology on $\mathbb{A}^n$. One immediate goal with this definition will be to decompose varieties into their irreducible components.

Definition 1.9

The **Zariski topology** on $\mathbb{A}^n$ is the topology whose closed sets are the affine varieties in $\mathbb{A}^n$. The Zariski topology on a variety $X \subseteq \mathbb{A}^n$ is the subspace topology of $\mathbb{A}^n$, whose closed sets are the subvarieties of X .

Example 1.4

The Zariski topology on $\mathbb{A}^1$ is the cofinite topology.

The Zariski topology is fairly pathological as a topological space.

Example 1.5

Let $a_1, a_2, \dots$ be a non-repeating sequence of points in $\mathbb{A}^1$. Fix $a \in \mathbb{A}^1$ and an open neighborhood U containing a . Then the complement of U is finite, so eventually every point in the sequence lies in U . Therefore (a_i) converges to every point of $\mathbb{A}^1$.

Example 1.6

Let $f : \mathbb{A}^1 \rightarrow \mathbb{A}^1$ be injective. Then any closed set is finite, and thus has a finite preimage, which is closed. So f is continuous.

Example 1.7

The Zariski topology on $\mathbb{A}^2$ is not the product topology on $\mathbb{A}^1 \times \mathbb{A}^1$. To see this, let

$$\Delta = \{(a, a) : a \in K\} \subseteq \mathbb{A}^2$$

Δ is clearly closed in the Zariski topology on $\mathbb{A}^2$. If Δ is closed in the product topology, then the complement can be written as the union of rectangular sets:

$$\mathbb{A}^2 \setminus \Delta = \bigcup_i (U_i \times V_i), \quad U_i, V_i \in \mathcal{O}(\mathbb{A}^1)$$

But for any U, V open, U, V are cofinite and $\mathbb{A}^1$ is infinite, so there is $x \in U \cap V$. This means that $(x, x) \in U \times V$, contradicting the decomposition of $\mathbb{A}^2 \setminus \Delta$.

Proposition 1.19

A space X is Hausdorff if and only if the diagonal is closed in the product topology on $X \times X$.

Proof. We want to show that $\Delta^c = (X \times X) \setminus \Delta$ is open. Let $(x, y) \in \Delta^c$. Since X is Hausdorff there are $U \ni x, V \ni y$ disjoint, so that $U \times V$ contains (x, y) and does not intersect the diagonal. Unioning over all pairs in Δ^c , Δ^c is open so Δ is closed. The same logic works in the other direction. $\square$

This gives a faster proof that the Zariski topology is not the product topology. Also, we will later see that this extends to higher products $\mathbb{A}^{n+m}$, since $U \cap V$ will always be nonempty for appropriate U, V , even outside the cofinite topology.

Definition 1.10

Let X be a topological space. X is **reducible** if it can be written as the union $X = X_1 \cup X_2$ for two proper closed subsets. It is **disconnected** if it can be written as the union of two nonempty disjoint closed subsets.

If X is disconnected then it is reducible, so if X is irreducible then it is connected.

Proposition 1.20

Let X be irreducible. Then

- (a) Any two nonempty open sets have nonempty intersection.
- (b) Any nonempty open set is dense in X .

Proof. (a) If U_1, U_2 are nonempty and open with empty intersection, then $X \setminus U_1, X \setminus U_2$ are proper closed subsets whose union is all of X .

- (b) A set is dense if and only if it intersects every nonempty open set, so this follows from (a).

□

We want ways to describe the irreducible and connected ideals cleanly. Topologically this is difficult, so we will convert their definitions to algebraic ones.

Proposition 1.21

A nonempty affine variety $X \subseteq \mathbb{A}^n$ is irreducible if and only if $A(X)$ is a domain (which is equivalent to $I(X)$ being a prime ideal).

Proof. ($\implies$) Let X be irreducible, take $f, g \in A(X)$ with $fg = 0$, and let $X_1 = V_X(\langle f \rangle), X_2 = V_X(\langle g \rangle)$. Then $X = X_1 \cup X_2$ which are both closed, so $X_1 = X$ or $X_2 = X$. In the first case $f = 0$ and in the second $g = 0$.

($\impliedby$) Let $X = X_1 \cup X_2$ with X_1, X_2 closed. Then X_1, X_2 are both subvarieties of X , so let $I_1 = I_X(X_1), I_2 = I_X(X_2)$. Then $I_1 I_2$ is the zero ideal in $A(X)$, which implies that at least one is zero. But if I_1 is zero then $X_1 = X$ and similarly for I_2 . □

This tells us that the Nullstellensatz restricts to a correspondence between nonempty irreducible affine subvarieties of $Y \subseteq \mathbb{A}^n$ to prime ideals of $A(Y)$.

Example 1.8

A finite set is irreducible if and only if it is a single point. In this case $A(X) = K$ which is a domain.

Proposition 1.22

$\mathbb{A}^n$ is irreducible (hence connected).

Proof. $A(\mathbb{A}^n) = K[x_1, \dots, x_n]$ is an integral domain. □

1.4 Irreducible Decompositions

To show that we can decompose any affine variety into irreducible components, we will prove this for more general topological spaces. This will allow us to easily extend to other forms of varieties later.

Definition 1.11

A topological space X is **Noetherian** if there is no infinite strictly decreasing chain of closed subsets

$$X_0 \supsetneq X_1 \supsetneq X_2 \supsetneq \dots$$

Recall that Noetherian rings satisfy the *ascending* chain condition on ideals; here we have a *descending* chain condition since we work over closed (equivalently, a Noetherian space satisfies the ascending condition on opens).

Proposition 1.23

Quotients of Noetherian rings are Noetherian.

Proof. Ideals in the quotient correspond to ideals in the original ring (containing the quotiented ideal), so an ascending chain in the quotient lifts to an ascending chain in the original. $\square$

Lemma 1.24

An affine variety $X \subseteq \mathbb{A}^n$ is Noetherian.

Proof. Suppose there is a descending chain

$$X_0 \supsetneq X_1 \supsetneq X_2 \supsetneq \dots$$

Then by the inclusion reversing bijection, we get

$$I(X_0) \subsetneq I(X_1) \subsetneq \dots \subseteq A(X)$$

But $A(X)$ is a quotient of $K[x_1, \dots, x_n]$, which is Noetherian, hence $A(X)$ is Noetherian. This is a contradiction. $\square$

Proposition 1.25

Any subset of a Noetherian space is Noetherian under the subspace topology.

Proof. Let $X \subseteq Y$ with Y Noetherian. Suppose there is some descending chain

$$X_0 \supsetneq X_1 \supsetneq X_2 \supsetneq \dots$$

of relatively closed sets in X . Then by definition there are $Y_i \subseteq Y$ closed such that $Y_i \cap X = X_i$. It follows that

$$Y_0 \supseteq (Y_0 \cap Y_1) \supseteq \dots$$

is a descending chain (not necessarily strict) in Y . To see that this is closed, we have

$$\left(\bigcap_{i=0}^n Y_i \right) \cap X = \bigcap_{i=0}^n (Y_i \cap X) = \bigcap_{i=0}^n X_i = X_n$$

so we actually have

$$Y_0 \supsetneq (Y_0 \cap Y_1) \supsetneq \dots$$

This is a strict descending chain of closed sets in Y , contradicting the assumption that it is Noetherian. $\square$

Proposition 1.26

Any Noetherian topological space X can be written as the finite union of nonempty irreducible closed subsets

$$X = \bigcup_{i=1}^r X_i$$

If we also assume that $X_i \not\subseteq X_j$ for $i \neq j$, then this is unique up to permutation.

Proof. If X is empty then we take $r = 0$ and are done. Otherwise, assume that X is not a finite union of irreducibles. Then X itself is certainly not irreducible, so there are proper nonempty closed subsets X_1, X_2 with $X_1 \cup X_2 = X$. These cannot both be irreducible, so one (say X_1) can also be written as the union of two proper nonempty closed subsets. This gives a strictly descending chain of closed subsets, contradicting Noetherianity. So X must have been a finite union of irreducibles.

To see uniqueness, take two decompositions

$$X = \bigcup_{i=1}^r X_i = \bigcup_{j=1}^r Y_j$$

Then for any X_i we can intersect against the other decomposition and apply irreducibility:

$$X_i = \bigcup_{j=1}^r (Y_j \cap X_i)$$

By irreducibility there is j such that $Y_j \cap X_i = X_i$, or $X_i \subseteq Y_j$. Then applying the decomposition in the other direction we get i' s.t. $Y_j \subseteq X_{i'}$. But $X_i \subseteq X_{i'}$ is only satisfied by $i = i'$. Thus we have $X_i = Y_j$, giving a bijection between the decompositions, so they are unique up to permutation. $\square$

Now that we have proven that X can be decomposed into irreducible components in the topological sense, we return to affine varieties and the question of how to actually compute this decomposition.

Definition 1.12

A **primary ideal** I of a ring R is an ideal such that whenever $xy \in I$, either x or a power of y is in I .

Equivalently, $xy \in I$ if and only if either x or y is in I , or if both x, y are in $\sqrt{I}$.

Proposition 1.27

Radicals of primary ideals are prime.

Proof. Let $xy \in \sqrt{I}$ for I primary. Then $(xy)^m \in I \implies x^m y^m \in I$. We either have $x^m \in I$ or $y^m \in I$, or else both x^m, y^m are in $\sqrt{I}$. In every case at least one of x, y is in $\sqrt{I}$. $\square$

In Section A.2, we show that in a Noetherian ring, every ideal (in particular, $I(X)$) has a *primary decomposition*

$$I(X) = \bigcap_{i=1}^r Q_i$$

of primary ideals Q_i . Then this is

$$I(X) = \sqrt{I(X)} = \sqrt{\bigcap_{i=1}^r Q_i} = \bigcap_{i=1}^r \sqrt{Q_i}$$

since if y lies in the radical of the intersection, then y^{m_i} lies in each Q_i for some m_i , hence $y \in \sqrt{Q_i}$ for each i .

Define $P_i = \sqrt{Q_i}$. These ideals are prime since the radicals of prime ideals are prime. So

$$I(X) = \bigcap_{i=1}^r P_i \implies X = V(I(X)) = \bigcup_{i=1}^r V(P_i)$$

Each $V(P_i)$ is irreducible since P_i is prime. Discard any P_i which are contained in another P_j (i.e. those maximal w.r.t. inclusion within the decomposition). Then this is a decomposition into irreducible closed subsets, which do not contain each other, and is therefore the unique decomposition.

Proposition 1.28

If X is an affine variety, then the irreducible components X_i of X are the maximal (w.r.t. inclusion) irreducible subvarieties of X .

Proof. Suppose not. Then writing

$$X = \bigcup_{i=1}^r X_i$$

there must be some X_j which is not maximal. Suppose without loss of generality that this is X_1 . Then let $X_1 \subsetneq X'_1$ where X'_1 is also an irreducible subvariety. Then

$$X = X'_1 \cup \bigcup_{i=2}^r X_i$$

is another decomposition into irreducibles, and after possibly discarding any subvarieties contained in X'_1 , we get a distinct decomposition, contradiction. $\square$

Corollary 1.29

There is a bijection between irreducible components of X and minimal prime ideals of $A(X)$.

Note that starting from the irreducible decomposition of a Noetherian space, we can obtain a decomposition into its connected components. We do this by making a graph where each vertex is an irreducible component, and adding edges between components which intersect. Then the path connected components of the graph are the connected components of X .

As before, we would like to translate this to an algebraic condition.

Theorem 1.30: Chinese Remainder Theorem

If I_1, I_2 are ideals in R that are coprime ($I_1 + I_2 = R$), then the natural ring homomorphism

$$R/I_1 \cap I_2 \rightarrow (R/I_1) \times (R/I_2)$$

is an isomorphism.

Proposition 1.31

Suppose X is a disconnected affine variety with disjoint closed subsets $X_1 \cup X_2 = X$. Then the map $A(X) \rightarrow A(X_1) \times A(X_2)$

$$f \mapsto f|_{X_1} \times f|_{X_2}$$

is an isomorphism.

Proof. We know that

$$A(X_i) = A(X)/I_X(X_i)$$

The map given is the natural ring homomorphism

$$A(X) \rightarrow (A(X)/I_X(X_1)) \times (A(X)/I_X(X_2))$$

Since $X = X_1 \cup X_2$, we have $0 = I(X) = I(X_1) \cap I(X_2)$. Since $\emptyset = X_1 \cap X_2$ we have $A(X) = I(\emptyset) = \sqrt{I(X_1) + I(X_2)}$ so $I(X_1) + I(X_2) = A(X)$. Then we apply the Chinese

remainder theorem to see that

$$A(X)/_{I(X_1) \cap I(X_2)} = A(X) \rightarrow \left(A(X)/_{I(X_1)} \right) \times \left(A(X)/_{I(X_2)} \right)$$

is an isomorphism. $\square$

So we see that functions on disconnected varieties are just arbitrary combinations of functions on their connected components, so we can essentially just study the rings of functions on connected components.

1.5 Dimensions of Algebraic Varieties

So far we have not formally defined dimension for affine varieties, although examples like curves and surfaces may give us some intuition. Here, as with our previous work, we define this notion topologically and then calculate it using algebra.

Definition 1.13

Let X be a nonempty topological space. Then the **dimension** of X is the supremum over all n such that there is a chain of length n of irreducible closed subsets:

$$\emptyset \subsetneq Y_0 \subsetneq Y_1 \subsetneq \dots \subsetneq Y_n \subseteq X$$

If $Y \subseteq X$ is a closed irreducible subset then its **codimension** is the supremum over n such that there is a chain of length n of irreducible closed subsets containing Y :

$$Y \subseteq Y_0 \subsetneq Y_1 \subsetneq \dots \subsetneq Y_n \subseteq X$$

Example 1.9

The only irreducible closed subsets of $\mathbb{A}^1$ are points and $\mathbb{A}^1$, so it has dimension 1.

Note that a general Noetherian topological space is not necessarily of finite dimension:

Example 1.10

Consider $\mathbb{N}$ where the closed sets are of the form $\{1, \dots, n\}$. This is Noetherian but it has an infinite chain of irreducible closed subsets.

However, for affine varieties, the ring properties will force us to have finite dimension:

Definition 1.14

Let R be a ring. Its **Krull dimension** is the supremum over all n such that there is a chain of length n of prime ideals

$$R \supseteq P_0 \supsetneq P_1 \supsetneq \dots \supsetneq P_n$$

The **height** of a prime ideal P is the supremum over all n such that there is a chain of length n of prime ideals containing P :

$$R \supseteq P_0 \subsetneq P_1 \subsetneq \dots \subsetneq P_{n-1} \supseteq P_n$$

Lemma 1.32

If X is a nonempty irreducible subvariety of an affine variety X , then

- (a) $\dim X$ is the Krull dimension of $A(X)$,
- (b) $\text{codim}_X Y$ is the height of $I_X(Y)$ in $A(X)$.
- (c) Both are finite numbers.

Proof. Any chain in $A(X)$ bijects with a chain in X . The lengths are bounded because $A(X)$ is finitely generated as a K -algebra. $\square$

Lemma 1.33

If R, S are finitely generated K -algebras which are also domains, then

$$\dim(R \otimes_K S) = \dim R + \dim S$$

We just give a sketch; the point is that we can produce a **Noether normalization** which is an injection from

$$K[x_1, \dots, x_n] \rightarrow R$$

which makes R a finitely generated module over the image (where $n = \dim R$). We do this for both R, S , giving a map

$$K[x_1, \dots, x_n, y_1, \dots, y_m] \rightarrow R \otimes_K S$$

which is a Noether normalization.

Proposition 1.34

Let X, Y be nonempty irreducible affine varieties. Then:

- (a) $\dim(X \times Y) = \dim X + \dim Y$,
- (b) If $Y \subseteq X$ then $\dim X = \dim Y + \text{codim}_X Y$,
- (c) If $f \in A(X)$ is nonzero, then every irreducible component of $V_X(f)$ has codimension 1.

Proof. (a) We know that $A(X \times Y) = A(X) \otimes_K A(Y)$. Then the result follows by the Lemma.

- (b) $A(X)$ is nice enough that all of the maximal chains of prime ideals have the same length. So then since Y is irreducible, $I_X(Y)$ is prime and the chains in X with $I_X(Y)$ as a step can be decomposed into a length $\text{codim}_X Y$ segment and a length $\dim Y$ segment.
- (c) Krull's principal ideal theorem says that every minimal prime ideal containing a (nonunit, non-zero divisor) f has height 1. $\square$

In particular, if X is irreducible then $\text{codim}_X \{a\} = \dim X$ for any point $a \in X$. Now, we develop a similar relation for reducible varieties.

Proposition 1.35

Let X be a topological space and $A \subseteq X$. Then $\dim A \leq \dim X$.

Proof. Any chain of closed irreducibles in A can be written as

$$\emptyset \subsetneq (A \cap X_0) \subsetneq \dots \subsetneq (A \cap X_n) \subseteq A$$

where the X_i are closed. The X_i can also be taken to be irreducible, and then we have that

$$\emptyset \subsetneq \overline{A \cap X_0} \subsetneq \dots \subsetneq \overline{A \cap X_n} \subseteq A$$

The inclusions are strict because we can take points in $X_n \setminus X_{n-1}$. $\square$

Proposition 1.36

Let X be a Noetherian space and Y an irreducible subset of X . Then one of the irreducible components of X contains Y .

Proof. Suppose the decomposition is

$$X = X_1 \cup \dots \cup X_n$$

Then

$$Y = \overline{Y \cap X_1} \cup \dots \cup \overline{Y \cap X_n}$$

Since Y is irreducible, there is some i such that $\overline{Y \cap X_i} = Y$. But $\overline{Y \cap X_i} = Y \cap \overline{X_i} = Y \cap X_i$ so we have $Y \subseteq X_i$. $\square$

Proposition 1.37

- (a) If X has irreducible decomposition $X_1 \cup \dots \cup X_n$, and X is a Noetherian topological space, then

$$\dim X = \max \{\dim X_i\}$$

(b) For any topological space,

$$\dim X = \sup_{a \in X} \{\operatorname{codim}_X \{a\}\}$$

Proof. (a) We automatically have $\dim X \geq \max \{\dim X_i\}$. Now, take a chain of length $r = \dim X$:

$$\emptyset \subsetneq Y_0 \subsetneq \dots \subsetneq Y_r \subseteq X$$

Then Y_r is contained in some X_i , so we get $r \geq \dim X_i$.

(b) This is clear from the formula. □

Definition 1.15

A Noetherian topological space is of **pure dimension** if every irred. component has the same dimension. An affine variety of dimension 1 is a **curve**, of dimension 2 a **surface**, and of codimension 1 a **hypersurface**.

Chapter 2

Sheaves and Morphisms

2.1 Regular Functions on Affine Varieties

As when developing a theory of categories like topological spaces, differentiable manifolds, and so on, we would like to describe what the morphisms in the category of affine varieties should be. We will begin with functions $f : X \rightarrow \mathbb{A}^1$, since these are the mostly closely related to polynomial functions, hence easiest to describe in language related to what we have already seen. Taking continuous and differentiable maps as our examples, we note that the property of being a continuous or differentiable map is something that can be checked locally, and is preserved under restrictions to smaller domains. Moreover, if two such maps agree on their open intersection, then they “glue” into a map on the union of their domains. We will define maps from $X \rightarrow \mathbb{A}^1$ in such a way that these properties also hold.

Perhaps the most immediately obvious maps to consider are the polynomial functions. However, if we work only with maps defined on open sets, then we have access to a richer class of maps. Recall that by definition, open sets are the complement of the vanishing locus of some functions. As a result, if U is open in X and $f \in I(X \setminus U)$, then for $g \in A(X)$, the expression

$$\varphi = \frac{g}{f}$$

has nonzero denominator in U , so at least this expression makes sense to write down. In order to mimic the local and gluing properties of continuous and differentiable maps, though, we will only require that the functions are *locally* given in the form of g/f , as opposed to globally. This isn’t necessarily justification for why these are the right maps to look at, but it turns out that they are.

Definition 2.1

Let X be an affine variety and $U \subseteq X$ open. Then a **regular function** on U is a function $\varphi : U \rightarrow \mathbb{A}^1$ such that for every $a \in U$ there is $f, g \in A(X)$ and an open

subset $a \in U_a \subseteq U$ such that

$$\varphi(x) = \frac{g(x)}{f(x)}$$

on U , with $f(x) \neq 0$ on U . We denote the set of regular functions on U by $\mathcal{O}_X(U)$. This carries the structure of a K -algebra.

Example 2.1

Let $X = V(x_1x_4 - x_2x_3)$ and let $U = X \setminus V(x_2, x_4)$. Then define

$$\varphi(x_1, x_2, x_3, x_4) = \begin{cases} \frac{x_1}{x_2}, & x_2 \neq 0 \\ \frac{x_3}{x_4}, & x_4 \neq 0 \end{cases}$$

When x_2, x_4 both nonzero then $x_1x_4 - x_2x_3$ implies that this is well defined. But neither quotient makes sense on all of $V(x_1x_4 - x_2x_3)$ (and more generally, there is no global representation).

Lemma 2.1

Let $U \subseteq X$ be open and let $\varphi \in \mathcal{O}_X(U)$. Then

$$V(\varphi) = \{x \in U : \varphi(x) = 0\}$$

is closed in U .

Proof. We recall that if $\{U_i : i \in I\}$ is an open cover of U , then $V \subseteq U$ is closed in U if and only if $V \cap U_i$ is closed in U_i for each i . Now, for each $a \in U$ we have $U_a \subseteq U$ open and f, g with $\varphi = g/f$ on U_a . Since f is nonvanishing, we have

$$V(\varphi) \cap U_a = V(g) \cap U_a$$

which is closed in U_a . Since the collection of all such U_a covers U , $V(\varphi)$ is closed in U . $\square$

As desired, we have maintained the restriction property for regular functions. Precisely, if $U \subseteq V \subseteq X$ and U, V are open, then the restriction map $\mathcal{O}_X(V) \rightarrow \mathcal{O}_X(U)$ takes regular functions to regular functions. However, this is not surjective in general. For instance, the element $\varphi(x) = 1/x$ is a regular function in $\mathcal{O}_{\mathbb{A}^1}(\mathbb{A}^1 \setminus \{0\})$ but there is no regular function which restricts to φ from $\mathcal{O}_{\mathbb{A}^1}(\mathbb{A}^1)$. We will later see a theorem that implies this occurs for a broad class of removed singularities. Nevertheless, the restriction is actually injective under the right assumptions.

Proposition 2.2: Identity Theorem

Let X be an irreducible affine variety and let $\emptyset \subsetneq U \subseteq V \subseteq X$ be open. Then if $\varphi_1, \varphi_2 \in \mathcal{O}_X(V)$ and $\varphi_1|_U = \varphi_2|_U$, it follows that $\varphi_1 = \varphi_2$.

Proof. The set $V(\varphi_1 - \varphi_2)$ on which φ_1, φ_2 is closed. It contains U , which is nonempty and open in V . V is irreducible since it is an open subset of X irreducible. So the closure of U in V is V , hence $V \subseteq V(\varphi_1 - \varphi_2)$. Thus $\varphi_1 = \varphi_2$ on V . $\square$

This resembles analytic continuation from complex analysis in spirit. However, it is a less surprising result because we already know that open sets are “large” in the Zariski topology, so it is natural that they extend uniquely.

Now, we compute $\mathcal{O}_X(U)$ for a class of simple open sets.

Definition 2.2

Let X be an affine variety and $f \in A(X)$. Then the **distinguished open subset** for f in X is

$$D(f) = \{x \in X : f(x) \neq 0\}$$

A general open subset is called a distinguished open subset if it is of the form $D(f)$.

Proposition 2.3

The intersection of distinguished open subsets is a distinguished open subset.

Proof. Let $f, g \in A(X)$. Then

$$D(f) \cap D(g) = (X \setminus V_X(f)) \cap (X \setminus V_X(g)) = X \setminus (V_X(f) \cup V_X(g)) = X \setminus V(fg) = D(fg) \quad \square$$

Proposition 2.4

Any open subset in X is a finite union of distinguished subsets.

Proof. Let $U \subseteq X$ be open. Then $U = X \setminus V(I)$ for some ideal $I \subseteq A(X)$, and by Hilbert’s basis theorem we know this can be written as

$$U = X \setminus V_X(\langle f_1, \dots, f_r \rangle)$$

for some finite generators. Then

$$U = X \setminus \bigcap_{i=1}^r V_X(f_i) = \bigcup_{i=1}^r (X \setminus V_X(f_i)) = \bigcup_{i=1}^r D(f_i) \quad \square$$

Thus the distinguished open sets form a basis for the Zariski topology. This is helpful because we showed earlier that being a regular function is a local property. Hence, we can write an open set as a union of distinguished opens, and check if the restriction to each is regular. Thus, we just need to understand $\mathcal{O}_X(D(f))$.

Lemma 2.5: Partition of Unity

Let X be an affine variety and suppose

$$D(f) = \bigcup_{i \in I} D(f_i)$$

Then there is $m \in \mathbb{N}$ such that

$$f^m = \sum_{i \in I_0} r_i f_i$$

where I_0 is a finite subset of I and each $r_i \in A(X)$.

This is basically the Nullstellensatz, so we remark that algebraic closure of the ground field is critical for the existence of partitions of unity. Since each f_i is supported on $D(f_i)$, this breaks f^m into terms which are supported on distinguished open sets. The utility of this is clear from analogy with smooth partitions of unity.

Proof. By the Nullstellensatz,

$$f \in I_X(V_X(f))$$

We want to show that this is the radical of an ideal which is the sum of some principal ideals. Note that

$$V_X(f) = X \setminus D(f) = \bigcap_{i \in I} V_X(f_i) = V_X(\langle f_i : i \in I \rangle)$$

So

$$f \in \sqrt{\langle f_i : i \in I \rangle} \implies f^m = \sum_{i \in I_0} r_i f_i$$

for some finite subset I_0 . □

Functions on distinguished open sets are locally quotients of polynomials except on $V_X(\langle f \rangle)$, which we know has dimension at most 1 by Krull's principal ideal theorem. In some sense, we should expect that this will allow the singularities to act similarly to removable singularities.

Proposition 2.6

For any distinguished open subset $D(f) \subseteq X$, the regular functions on $D(f)$ take the form

$$\mathcal{O}_X(D(f)) = \left\{ \frac{g}{f^m} : g \in A(X), m \in \mathbb{N} \right\}$$

Thus, we see that regular functions on distinguished open subsets are given globally as a polynomial quotient, not just locally. In particular, taking $f = 1$ we get $D(f) = X$ so the regular functions on X are all of $A(X)$.

Proof. ($\supseteq$) This is clear, since any quotient g/f^m makes sense on $D(f)$ and is clearly locally a quotient.

($\subseteq$) Let $\varphi \in \mathcal{O}_X(D(f))$.

Claim 1: for any $a \in D(f)$, φ can be written locally as a quotient

$$\varphi = \frac{g_a}{f_a}$$

on U_a where $D(f_a) = U_a$ and $g_a = 0$ on $V_X(f_a)$.

To see this, first note that we can generally write

$$\varphi = \frac{g_a}{f_a}$$

on U_a for some g_a, f_a with $a \in U_a$. But since the distinguished opens form a basis, there is a distinguished open subset $D(h_a)$ contained in U_a which contains a . On $D(h_a)$,

$$\varphi = \frac{g_a h_a}{f_a h_a}$$

Since $D(h_a) \subseteq D(f_a)$, $V_X(h_a) \supseteq V_X(f_a)$, so h_a vanishes wherever f_a vanishes. So on $V_X(f_a h_a)$, $g_a h_a = 0$. Thus taking $g'_a = g_a h_a$, $f'_a = f_a h_a$, we have that $g'_a = 0$ on $V_X(f'_a)$, proving Claim 1.

Claim 2: For $a, b \in D(f)$ and f_a, g_a, f_b, g_b as constructed above, $g_a f_b = g_b f_a$ as elements of $A(X)$.

To see this, $U_a \cap U_b$ is a nonempty open set, on which φ has two representations

$$\varphi = \frac{g_a}{f_a} = \frac{g_b}{f_b}$$

Clearing denominators, we have $g_a f_b = g_b f_a$ on $U_a \cap U_b$. On the other hand, if $x \notin U_a \cap U_b$, then either $x \in V_X(f_a)$ or $x \in V_X(f_b)$ (since by Claim 1, we chose $U_i = X \setminus V_X(f_i)$). Say $x \in V_X(f_a)$. Then by Claim 1, $g_a(x) = f_a(x) = 0$, so both sides are zero. So everywhere we have $g_a f_b = g_b f_a$, proving Claim 2.

Finally, with g_a, f_a, U_a defined as in Claim 1,

$$\bigcup_{a \in D(f)} U_a = \bigcup_{a \in D(f)} D(f_a) = \bigcup_{a \in D(f)} D(f_a)$$

So by the partition of unity, there is m and a finite subset $A \subseteq D(f)$ such that

$$f^m = \sum_{a \in A} r_a f_a$$

Now, define

$$g = \sum_{a \in A} r_a g_a$$

It follows that for any $x \in D(f)$,

$$gf_x = \sum_{a \in A} r_a g_a f_x \stackrel{(2)}{=} \sum_{a \in A} r_a g_x f_a = g_x \sum_{a \in A} r_a f_a = g_x f^m$$

Since f, f_x are nonzero at x , we have

$$\frac{g(x)}{f^m(x)} = \frac{g_x}{f_x} = \varphi(x)$$

□

We can view this construction in a completely algebraic sense.

Definition 2.3

Let $S \subseteq R$ be a multiplicative subset. Then the **localization** of R at S is the ring of fractions

$$S^{-1}R = R[S^{-1}] = \left\{ \frac{r}{s} : r \in R, s \in S \right\} / \sim$$

where the equivalence relation $\sim$ is such that

$$\frac{r}{s} \sim \frac{r'}{s'}$$

if and only if there is $t \in S$ such that

$$t(rs' - r's) = 0$$

In particular, when S is the subset $S = \{1, f, f^2, \dots\}$ for some $f \in R$, we denote $R_f = R[S^{-1}]$.

Corollary 2.7

Let $f \in A(X)$. Then the ring of regular functions on the distinguished subset $D(f)$ is

$$\mathcal{O}_X(D(f)) \cong A(X)_f$$

which are isomorphic as K -algebras.

Proof. We can consider the homomorphism of K -algebras $\varphi : A(X)_f \mapsto \mathcal{O}_X(D(f))$ given by

$$\varphi \left(\left[\frac{g}{f^m} \right] \right) = \frac{g}{f^m}$$

We check that this is well defined, injective, and surjective. First, if

$$\left[\frac{g}{f^m} \right] = \left[\frac{g'}{f'^m} \right]$$

then there is k such that

$$f^k (gf'^m - g'f^m) = 0$$

Since $f \neq 0$ on $D(f)$, we have $gf'^m - g'f^m = 0$ on $D(f)$, so

$$\frac{g}{f^m} = \frac{g'}{f'^m}$$

as functions on $D(f)$. So this map is well defined. Since $D(f)$ is a distinguished subset, every element of $\mathcal{O}_X(D(f))$ is of the form g/f^m , so this map is surjective. To prove injectivity, if

$$\varphi\left(\left[\frac{g}{f^m}\right]\right) \equiv 0$$

on $D(f)$ then $g = 0$ on $D(f)$, so $fg = 0$ on $D(f)$. Then

$$\left[\frac{g}{f^m}\right] = \left[\frac{0}{1}\right]$$

since $f(g \cdot 1 - f^m \cdot 0) = fg = 0$ on $D(f)$. □

So we have computed the ring of regular functions on distinguished open subsets. As an example of a subset which is not distinguished, we consider $\mathbb{A}^2 \setminus \{(0, 0)\}$.

Example 2.2

We will show that

$$\mathcal{O}_{\mathbb{A}^2}(\mathbb{A}^2 \setminus \{(0, 0)\}) \cong \mathbb{K}[x_1, x_2] \cong \mathcal{O}_{\mathbb{A}^2}(\mathbb{A}^2)$$

which is to say that every regular function on $U = \mathbb{A}^2 \setminus \{(0, 0)\}$ can be extended over the origin. To see this, note that $U = D(x_1) \cup D(x_2) \supseteq D(x_1x_2)$. Letting $\varphi \in \mathcal{O}_{\mathbb{A}^2}(U)$, φ restricts to regular functions on $D(x_i)$, which then restrict to regular functions on $D(x_1x_2)$. This means there are m, n such that

$$\varphi = \frac{g_1}{x_1^m} = \frac{g_2}{x_2^n}$$

for $g_1, g_2 \in K[x_1, x_2]$. The equality holds on $D(x_1x_2)$, so by our characterization of $\mathcal{O}_X(D(x_1x_2))$ we know the two are equal as equivalence classes. Thus there is r such that

$$(x_1x_2)^r(g_1x_2^n - g_2x_1^m) = 0$$

as elements of $K[x_1, x_2]$. Since $K[x_1, x_2]$ is a domain, we have

$$g_1x_2^n = g_2x_1^m$$

Since $K[x_1, x_2]$ is a UFD, we get that x_1^m divides x_2^n or g_1 . By assuming $\gcd(g_1, x_1) = \gcd(g_2, x_2) = 1$ if necessary, which is harmless, we see that $m = 0, n = 0$, so $\varphi = g_1 = g_2$ extends to a polynomial function on $\mathbb{A}^2$.

Proposition 2.8

Let Y be a nonempty irreducible subset of an affine variety X such that $A(X)$ is a UFD. Let $U = X \setminus Y$. Then $\mathcal{O}_X(U) = A(X)$ if and only if $\text{codim}_X(Y) \geq 2$.

Proof. ($\Leftarrow$) Clearly any element of $A(X)$ can be restricted to a regular function on U , so we just show that this is the form of all regular functions. First we show this when $\text{codim}_X Y = 2$. In this case, the height of $I_X(Y)$ in $A(X)$ is 2, so there is a maximal chain of prime ideals

$$(0) \subseteq \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}_2 \subseteq I_X(Y)$$

Since Y is irreducible, $I_X(Y)$ is prime, and since $A(X)$ is a domain, (0) is prime. So by maximality this chain is of the form

$$(0) \subsetneq \mathfrak{p} \subsetneq I_X(Y)$$

Now, take any $q \in I_X(Y) \setminus \mathfrak{p}$ and set $\mathfrak{q} = \langle q \rangle$. $\mathfrak{q}$ is not all of $I_X(Y)$, since then $I_X(Y)$ would be principal and would have height at most 1 by Krull's principal ideal theorem. But $I_X(Y)$ has height $\text{codim}_X Y = 2$.

$\sqrt{\mathfrak{p} + \mathfrak{q}}$ is the intersection of prime ideals containing $\mathfrak{p} + \mathfrak{q}$. Any such ideal strictly contains $\mathfrak{p}$, so by maximality of the chain, they also must contain $I_X(Y)$. $I_X(Y)$ is a prime ideal containing $\mathfrak{p} + \mathfrak{q}$, so $\sqrt{\mathfrak{p} + \mathfrak{q}} = I_X(Y)$. Also, $\mathfrak{p} \cap \mathfrak{q} = (0)$, because it is a prime ideal strictly contained in $\mathfrak{p}$ (as $\mathfrak{q}$ is not $I_X(Y)$).

Now, by the inclusion reversing bijection, we have

$$Y \subsetneq V_X(\mathfrak{p}), V_X(\mathfrak{q})$$

where both $V_X(\mathfrak{p}), V_X(\mathfrak{q})$ are irreducible subvarieties of X such that

$$V_X(\mathfrak{p}) \cap V_X(\mathfrak{q}) = V_X(\sqrt{\mathfrak{p} + \mathfrak{q}}) = V_X(I_X(Y)) = Y$$

$\mathfrak{p}, \mathfrak{q}$ are both finitely generated, so let

$$\begin{aligned} \mathfrak{p} &= \langle p_1, \dots, p_n \rangle \subseteq A(X) \\ \mathfrak{q} &= \langle q_1, \dots, q_m \rangle \subseteq A(X) \end{aligned}$$

In particular, the requirement that $\mathfrak{p} \cap \mathfrak{q} = (0)$ guarantees that $p_i \nmid q_j$ and $q_j \nmid p_i$ for any i, j . We have

$$U = X \setminus Y = X \setminus (V_X(\mathfrak{p}) \cap V_X(\mathfrak{q})) = (X \setminus V_X(\mathfrak{p})) \cup (X \setminus V_X(\mathfrak{q})) \supsetneq D(p_1 \cdots p_n) \cup D(q_1 \cdots q_m)$$

Write $D_p = D(p_1 \cdots p_n)$, $D_q = D(q_1 \cdots q_m)$, $D = D_p \cap D_q$. D is nonempty since it is the intersection of nonempty open sets.

Pick $\varphi \in \mathcal{O}_X(U)$. Then $\varphi|_{D_p}, \varphi|_{D_q}$ are both regular functions on distinguished open subsets, so there are $k, l \in \mathbb{N}, g_p, g_q \in A(X)$ such that

$$\varphi|_{D_p} = \frac{g_p}{(p_1 \cdots p_n)^k}, \quad \varphi|_{D_q} = \frac{g_q}{(q_1 \cdots q_m)^l}$$

(We assume that $\gcd(g_p, p_i) = \gcd(g_q, q_j) = 1$, which is harmless.) Both representations are equal to φ on D , so on D we have

$$g_p(q_1 \cdots q_m)^l = g_q(p_1 \cdots p_n)^k \in A(X)$$

The locus for this equality is closed, so in particular it contains $\overline{D}$.

(since D is nonempty and open, and X is irreducible). The p_i are coprime with the q_j , and we assumed that g_p was coprime with the p_i and g_q coprime with the q_j . So since $A(X)$ is a UFD, we have $l = k = 0$, and thus $\varphi = g_p = g_q$ on D . Since this is an agreement of regular functions on an open set, by the identity theorem we have that $\varphi = g_p = g_q$ on all of U . So $\mathcal{O}_X(U) = A(X)$ when $\text{codim}_X Y = 2$.

To see the general case, let $\text{codim}_X Y \geq 2$. Then since $\dim X$ is finite, there is $Y' \supseteq Y$ with $\text{codim}_X Y' = 2$. Let $\varphi \in \mathcal{O}_X(U)$. Then φ restricts to a regular function on $X \setminus Y' \subseteq U$. By the theorem for codimension 2, we have

$$\varphi|_{X \setminus Y'} = g|_{X \setminus Y'}$$

for some $g \in A(X)$. Since φ, g agree on an open subset of U , $\varphi = g$ on U as well. So $\mathcal{O}_X(U) = A(X)$ in the general case.

($\implies$) Now suppose that $\mathcal{O}_X(U) = A(X)$. Clearly we cannot have $\text{codim}_X Y = 0$, since then $Y = X$ and $\mathcal{O}_X(U)$ is the zero ring. So all that we need to show is that $\text{codim}_X Y \neq 1$.

Suppose for contradiction that $\text{codim}_X Y = 1$. $\text{codim}_X Y$ is the height of $I_X(Y)$ in $A(X)$, so $I_X(Y)$ has height 1. It is prime since Y is irreducible. Then since $A(X)$ is a Noetherian UFD, every prime ideal of height 1 is principal (for instance, Gathmann Proposition 2.37). So we have $I_X(Y) = \langle f \rangle$ for some nonunit $f \in A(X)$. Then $1/f \in \mathcal{O}_X(U)$ is a valid representation of a regular function. But if $1/f = g \in \mathcal{O}_X(U)$ for some $g \in A(X)$, then we would have

$$1 = fg$$

on U . Then $V(1 - fg) \supseteq \overline{U} = X$, so $fg = 1 \in A(X)$. But f is not a unit, contradiction. Thus $\text{codim}_X Y \geq 2$. $\square$

2.2 Sheaves

We defined regular functions in such a way that:

1. Restrictions of regular functions are regular,
2. Regularity can be checked locally, meaning that if a function is regular on each open set in an open cover, then it is a regular function everywhere,
3. Constructing regular functions by “gluing” together functions on open subsets is unique (by the identity theorem).

The notion of a sheaf is an abstract view of how data of this type can be collected into a single object. We will formalize this so that we can discuss mappings from an affine variety

into another one (other than $\mathbb{A}^1$, which we have just developed), seeing affine varieties as not simply a set in $\mathbb{A}^n$ but as such a set together with the data given by the sheaf.

Definition 2.4

Let (X, τ) be a topological space. Then a **presheaf** of rings on X is a collection of rings $\{\mathcal{F}(U)\}_{U \in \tau}$, together with ring homomorphisms $\rho_{V \rightarrow U} : \mathcal{F}(V) \rightarrow \mathcal{F}(U)$ for each inclusion of opens $U \subseteq V$. We also require that $\rho_{V \rightarrow V}$ is the identity on $\mathcal{F}(V)$, and that whenever $U \subseteq V \subseteq W$, the following diagram commutes:

$$\begin{array}{ccccc} \mathcal{F}(U) & \xrightarrow{\rho_{U \rightarrow V}} & \mathcal{F}(V) & \xrightarrow{\rho_{V \rightarrow W}} & \mathcal{F}(W) \\ & & \searrow \rho_{U \rightarrow W} & & \nearrow \end{array}$$

Elements of $\mathcal{F}(V)$ are called **sections** of $\mathcal{F}$ on V , and elements of $\mathcal{F}(X)$ are called **global sections**. The maps $\rho_{V \rightarrow U}$ are also called **restriction maps**, and $\rho_{V \rightarrow U}(\varphi)$ is denoted $\varphi|_U$.

$\mathcal{F}(U)$ should intuitively be thought of as a ring of functions with the restriction maps as actual restrictions, but we do not require this.

Definition 2.5

A presheaf $\mathcal{F}$ is called a **sheaf** if it satisfies the gluing properties:

1. Whenever $\{U_i\}_{i \in I}$ is an open cover of $U \subseteq X$ open, if $\varphi_i \in \mathcal{F}(U_i)$ for each i and $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j} \in \mathcal{F}(U_i \cap U_j)$ for each i, j , then there is $\varphi \in \mathcal{F}(U)$ such that $\varphi|_{U_i} = \varphi_i$,
2. Such a φ is unique.

In essence, compatible data can be glued together on small pieces to produce larger data.

We also note that this makes sense in many categories (basically any reasonable category which is a set with structure). For instance, we can have a sheaf of K -algebras, R -modules, abelian groups, and so on.

Example 2.3

The continuous ring-valued functions on a topological space form a sheaf of rings. If they are K -valued, then it is also a sheaf of K -algebras. The differentiable functions on $\mathbb{R}^d$ also form a sheaf of K -algebras.

Example 2.4

If X is any space and R a ring, then the constant functions $f : U \rightarrow R$, $U \subseteq X$ form the **constant presheaf**. However, this is in general not a sheaf, because if we can take two disjoint open sets, then we can prescribe different values on each. This is compatible data since the intersection is empty, but it does not glue into a constant function. Intuitively, being constant is not a local property. The *locally* constant functions do form a sheaf, though.

Proposition 2.9

If X is an affine variety, then the sheaf of regular functions $\mathcal{F}(U) = \mathcal{O}_X(U)$ with the restriction maps given by function restriction form a sheaf.

Proof. That this is a presheaf is clear. Since being regular is a local property, compatible regular functions can be glued into regular functions, and by the identity theorem they are unique. $\square$

Definition 2.6

Let $W \subseteq X$ be open and $\mathcal{F}$ a presheaf (resp. sheaf) on X . Then the restriction presheaf (sheaf) on W is

$$\mathcal{F}|_W(U) = \mathcal{F}(U)$$

for $U \subseteq W$ open, with the same restriction maps as in $\mathcal{F}$.

If sheaves of functions collect data of functions, then we can look at the behavior of functions around a given point in the topological space. The possible behaviors of these functions are described by stalks.

Definition 2.7

Let $\mathcal{F}$ be a presheaf over X and $a \in X$. Then the **stalk** of $\mathcal{F}$ at a is

$$\mathcal{F}_a = \{(U, \varphi) : \varphi \in \mathcal{F}(U)\} / \sim$$

with the equivalence relation

$$(U, \varphi) \sim (V, \psi)$$

if and only if there is $W \subseteq U \cap V$ open containing a such that $\varphi|_W = \psi|_W$. An element of $\mathcal{F}_a$ is called a **germ** at a , or sometimes a **local section**.

A stalk has the structure of a ring if $\mathcal{F}$ is a presheaf of rings, of K -algebras if $\mathcal{F}$ is a presheaf of K -algebras, and so on. For instance, we have

$$\begin{aligned} [(U_1, \varphi_1)] + [(U_2, \varphi_2)] &= [(U_1 \cap U_2, \varphi_1 + \varphi_2)] \\ [(U_1, \varphi_1)] \cdot [(U_2, \varphi_2)] &= [(U_1 \cap U_2, \varphi_1 \varphi_2)] \end{aligned}$$

When we work with the sheaf of regular functions, we denote this $\mathcal{O}_{X,a}$. The information contained in a germ or stalk is the local behavior of a function. The strength of this information is dependent on the specific sheaf or presheaf.

Example 2.5

Consider the sheaf of continuous functions on X and consider a germ $\varphi_a \in \mathcal{F}_a$ for $a \in X$. Then every representative of the germ agrees when evaluated at a , so $\varphi_a(a)$ is well defined (though $\varphi_a(a)$ does not uniquely determine a germ). However, a germ cannot be evaluated anywhere else, since there are continuous functions in the same germ with different values there.

Example 2.6

Consider the sheaf of holomorphic functions on $\mathbb{C}$ and consider a germ $\varphi_a \in \mathcal{F}_a$. Then any two representatives agree on an open set containing a . Up to ignoring representatives defined on disconnected sets, regularity tells us that each germ corresponds to exactly one holomorphic function. Thus any section on, say, $\mathbb{D}$, is entirely determined by its germ at 0, and the germ can be evaluated anywhere on its maximal domain.

Example 2.7

For the sheaf of regular functions on an affine variety, there is a well-defined evaluation map $\varphi_a \mapsto \varphi(a)$.

We will now see how to compute the stalks of regular functions over affine varieties. This algebraic operation is called localization, because it reflects the fact that we are looking at the properties of these functions locally.

Definition 2.8

Let R be a ring and $\mathfrak{p} \subseteq R$ an ideal. Let $S_{\mathfrak{p}} = R \setminus \mathfrak{p}$ (which is multiplicative). Then the **localization** of R around $\mathfrak{p}$ to be the localization $R_{\mathfrak{p}} = R[S_{\mathfrak{p}}^{-1}]$.

Lemma 2.10

Let $\pi : R \rightarrow R_{\mathfrak{p}}$ be the natural inclusion map $r \mapsto \left[\frac{r}{1} \right]$. Then the maps

$$\{\mathfrak{q} \subseteq R_{\mathfrak{p}} \text{ prime}\} \xrightleftharpoons[\mathfrak{q}' \cdot R_{\mathfrak{p}} \leftarrow \mathfrak{q}']{\mathfrak{q} \mapsto \pi^{-1}(\mathfrak{q})} \{\mathfrak{q}' \subseteq R \text{ prime, contained in } \mathfrak{p}\}$$

are inclusion preserving bijections which are inverses.

Definition 2.9

A **local ring** is a ring with a unique maximal ideal.

Corollary 2.11

If $\mathfrak{p}$ is prime then $R_{\mathfrak{p}}$ is a local ring with maximal ideal $\mathfrak{p} \cdot R_{\mathfrak{p}}$.

Proof. The unique prime ideal contained in $\mathfrak{p}$ which is maximal among such ideals is $\mathfrak{p}$ itself. Passing through the bijection, the unique maximal ideal in $R_{\mathfrak{p}}$ is $\mathfrak{p} \cdot R_{\mathfrak{p}}$. $\square$

Lemma 2.12

Let $a \in X$ be a point in an affine variety. The map from the localization of the coordinate ring to the stalk at a

$$A(X)_{I_X(a)} \rightarrow \mathcal{O}_{X,a}$$

given by

$$\left[\frac{g}{f} \right]_{A(X)_{I_X(a)}} \mapsto \left[\left(D(f), \frac{g}{f} \right) \right]_{\mathcal{O}_{X,a}}$$

is an isomorphism of K -algebras.

Proof. We check that this map is well defined, surjective, and injective. Being a K -algebra morphism is clear.

To see that it is well defined, if

$$\left[\frac{g}{f} \right]_{A(X)_{I_X(a)}} = \left[\frac{g'}{f'} \right]_{A(X)_{I_X(a)}}$$

then there is $h \notin I_X(a)$ such that

$$h(fg' - gf') = 0 \in A(X)$$

So on

$$D(h) \cap D(f) \cap D(f')$$

we have

$$\frac{g}{f} = \frac{g'}{f'}$$

pointwise. This is a subset of both $D(f), D(f')$, so

$$\left[\left(D(f), \frac{g}{f} \right) \right]_{\mathcal{O}_{X,a}} = \left[\left(D(f'), \frac{g'}{f'} \right) \right]_{\mathcal{O}_{X,a}}$$

For surjectivity, pick

$$[(U, \varphi)]_{\mathcal{O}_{X,a}}$$

Then φ is regular so there is U_a open containing a and $g, f \in A(X)$ such that $\varphi = g/f$ on U_a and $f \neq 0$ on U_a . In particular we can assume that $U_a = D(f)$. Then we have that

$$\varphi = \frac{g}{f}$$

on $a \in U \cap D(f)$ nonempty. Thus we have that

$$[(U, \varphi)]_{\mathcal{O}_{X,a}} = \left[\left(D(f), \frac{g}{f} \right) \right]_{\mathcal{O}_{X,a}}$$

So this is the image of

$$\left[\frac{g}{f} \right]_{A(X)_{I_X(a)}}$$

To check injectivity, if

$$\left[\left(D(f), \frac{g}{f} \right) \right]_{\mathcal{O}_{X,a}} = \left[\left(X, \frac{0}{1} \right) \right]_{\mathcal{O}_{X,a}}$$

Then there is $U \subseteq D(f)$ containing A such that

$$\frac{g}{f} \Big|_U = 0 \in \mathcal{O}_X(U)$$

Since the distinguished opens form a basis for the Zariski topology, there is h such that $D(h) \subseteq U$. In particular, we must have $h \in A(X) \setminus I_X(a)$. Then

$$\frac{g}{f} \Big|_{D(h)} = \frac{0}{1} \in \mathcal{O}_X(D(h)) = A(X)_h$$

Therefore by definition, there is m such that

$$h^m(g \cdot 1 - f \cdot 0) = h^m g = 0 \in A(X)$$

Since $h \notin I_X(a)$,

$$\left[\frac{g}{f} \right]_{A(X)_{I_X(a)}} = \left[\frac{0}{1} \right]_{A(X)_{I_X(a)}}$$

□

Corollary 2.13

$\mathcal{O}_{X,a}$ is a local ring and the unique maximal ideal is given by

$$I_X(a)A(X)_{I_X(a)} = \left\{ \frac{g}{f} : f \notin I_X(a), g \in I_X(a) \right\} / \sim = \left\{ \frac{g}{f} : f(a) \neq 0, f(a) = 0 \right\} / \sim$$

where $\sim$ is as for localizations.

Proof. $I_X(a)$ is prime. □

Definition 2.10

The **local ring** of X at a is $\mathcal{O}_{X,a}$ and its maximal ideal is denoted

$$I_a := I_X(a)A(X)_{I_X(a)} = \left\{ \frac{g}{f} : f(a) \neq 0, f(a) = 0 \right\}_{/\sim}$$

2.3 Ringed Spaces and Morphisms

Now that we have a characterization of nice functions on affine varieties, we can see morphisms between them as (continuous) maps which preserve those functions. That is, we want a function to have the property that whenever φ is a regular function on U , then $\varphi \circ f$ is a regular function on $f^{-1}(U)$.

$$X \xrightarrow{f} Y$$

$$\mathcal{O}_X(f^{-1}(U)) \xleftarrow{f^*} \mathcal{O}_Y(U)$$

Definition 2.11

A **ringed space** is a topological space with a sheaf of rings $\mathcal{O}_X$ on X . The sheaf is called the **structure sheaf**.

In the case that X is an affine variety we always take the structure sheaf to be the sheaf of regular functions. Any open subset of X is also a ringed space using the restricted sheaf $\mathcal{O}|_U$. For general rings, the composition $\varphi \circ f$ does not necessarily make sense. As a result, we adopt the convention from here until the introduction of schemes that for any ringed space, $\mathcal{F}(U)$ is always a subset of the set of functions from $U \rightarrow K$.

Definition 2.12

Let X, Y be ringed spaces and $f : X \rightarrow Y$ any map. Then for any $\varphi : U \rightarrow K$ for $U \subseteq Y$ open, the map $\varphi \circ f : f^{-1}(U) \rightarrow K$ is called the **pullback** of φ by f and is denoted $f^*\varphi$. If f is continuous and for any $U \subseteq Y$ open, $\varphi \in \mathcal{O}_Y(U)$, we have $f^*\varphi \in \mathcal{O}_X(f^{-1}(U))$, then f is called a **morphism** of ringed spaces. f is an isomorphism of ringed spaces if it is bijective and f^{-1} is also a morphism. We say that a map $f : X \rightarrow Y$ continuous for X, Y affine varieties is a **morphism** of affine varieties if it is a morphism as a ringed space.

In the case that f is a morphism of ringed spaces, then $f^* : \mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(f^{-1}(U))$ is a K -algebra morphism for each U .

Proposition 2.14

The identity map on a ringed space X is a morphism, and if $f : X \rightarrow Y$, $g : Y \rightarrow Z$ are morphisms of ringed spaces, then $g \circ f : X \rightarrow Z$ is a morphism as well.

This tells us that the ringed spaces form a category.

Proposition 2.15

If $f : X \rightarrow Y$ is a morphism of ringed spaces and $U \subseteq X, V \subseteq Y$ are open with $V \subseteq f(U)$, then $f|_U : U \rightarrow V$ is a morphism of ringed spaces.

In particular, the inclusion map $U \rightarrow X$ is a restriction of the identity, so a morphism.

Lemma 2.16

Let $f : X \rightarrow Y$ be a map of ringed spaces, and let $\{U_i\}_{i \in I}$ be an open cover of X such that all the restrictions $f|_{U_i} : U_i \rightarrow Y$ are morphisms of ringed spaces. Then f is a morphism.

Proof. f is continuous since it is continuous on an open cover. Now, let $V \subseteq Y$ be open, and let $\varphi \in \mathcal{O}_Y(V)$. Then for each i ,

$$(f^*\varphi)|_{U_i \cap f^{-1}(V)} = (f|_{U_i \cap f^{-1}(V)})^* \varphi \in \mathcal{O}_X(U_i \cap f^{-1}(V))$$

by assumption. So the sets $\{U_i \cap f^{-1}(V)\}_{i \in I}$ are an open cover of $f^{-1}(V)$, on which the sections $f^*\varphi|_{U_i \cap f^{-1}(V)}$ agree on intersections. So by the gluing property there is a unique map which agrees with the restrictions, which must be $f^*\varphi$. So $f^*\varphi$ is a section in $\mathcal{O}_X(f^{-1}(V))$, and therefore f is a morphism. $\square$

2.4 Morphisms of Affine Varieties

We have thus defined morphisms of ringed spaces, and defined morphisms of affine varieties to be morphisms as ringed spaces. But because we have studied the structure sheaf on affine varieties, we can make more concrete statements about their morphisms.

Proposition 2.17

If $U \subseteq X$ is an open subset of an affine variety and $Y \subseteq \mathbb{A}^n$ an affine variety. Then the morphisms $f : U \rightarrow Y$ are all the maps of the form

$$f(x) = (\varphi_1(x), \dots, \varphi_n(x))$$

where $\varphi_i \in \mathcal{O}_X(U)$ for each i and $f(U) \subseteq Y$.

Proof. Let $f : U \rightarrow Y$ be a morphism. The coordinate projection map $\pi_i : x \mapsto x_i$ on $\mathbb{A}^n$ is a regular function (hence on Y by restriction), so for each i we have that

$$f^* \pi_i(x) = \pi_i(f(x)) \in \mathcal{O}_X(U)$$

by the definition of a morphism. So every morphism is of the above form.

Now, let (φ_i) be regular functions in $\mathcal{O}_X(U)$. We need to show that

$$f = (\varphi_1, \dots, \varphi_n)$$

is a morphism whenever $f(U) \subseteq Y$. Let $Z \subseteq Y$ be closed, so that $Z = V_Y(g_1, \dots, g_m)$ for $g_j \in A(Y)$. Then it follows that

$$f^{-1}(Z) = \{x \in X : \forall j, g_j(\varphi_1, \dots, \varphi_n) = 0\}$$

Since g_j is a polynomial and φ_i lie in $\mathcal{O}_X(U)$, which is a K -algebra, we have that

$$g_j(\varphi_1, \dots, \varphi_n) = f^* g_j$$

is also a regular function, and

$$f^{-1}(Z) = V(f^* g_1) \cap \dots \cap V(f^* g_m)$$

is closed in U (each $V_X(f^* g_i)$ is closed since it is the zero locus of a regular function). Now, let $\psi \in \mathcal{O}_Y(W)$ be a regular function. It follows that locally it is a quotient of polynomials, so

$$f^* \psi = \frac{g(\varphi_1, \dots, \varphi_n)}{h(\varphi_1, \dots, \varphi_n)}$$

is locally a quotient of polynomials, after expanding. Thus $f^* \psi$ is regular, so f is a morphism. $\square$

Example 2.8

Let $X = V(xy - 1) \in \mathbb{A}^2$ and $Y = \mathbb{A}^1$. Then the map $(x, y) \mapsto x : X \rightarrow Y$ is a morphism. If we let $U = \mathbb{A}^1 \setminus \{0\} \subseteq Y$, then this map is surjective, and the map $x \mapsto (x, 1/x)$ is also a morphism which is its inverse. Thus $X \cong U$ as ringed spaces.

Corollary 2.18

Let X, Y be affine varieties. Then there is a 1-1 correspondence between morphisms $X \rightarrow Y$ and K -algebra homomorphisms $A(Y) \rightarrow A(X)$.

Note the contravariance, since we are looking at the pullback.

Proof. Let $f : X \rightarrow Y$ be a morphism. Then by definition of the pullback, $f^* : \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(X)$ is a K -algebra homomorphism:

$$f^*(\varphi + \psi) = (\varphi + \psi)(f(\cdot)) = f^* \varphi + f^* \psi$$

and so on. These rings are precisely $A(Y), A(X)$.

On the other hand, let $g : A(Y) \rightarrow A(X)$. Let $y_1, \dots, y_n$ be the coordinate functions on $Y \subseteq \mathbb{A}^n$, and define $\varphi_i = gy_i \in A(X)$. Then let

$$f = (\varphi_1, \dots, \varphi_n)$$

which is a priori a set-theoretic map from $X \rightarrow \mathbb{A}^n$. To see that this is a morphism, for $h \in K[y_1, \dots, y_n]$,

$$(f^*h)(x) = h(\varphi_1(x), \dots, \varphi_n(x)) = h(gy_1(x), \dots, gy_n(x)) = g(h(y_1, \dots, y_n))(x)$$

where the last equality follows because K -algebra homomorphisms commute with polynomials. Now, for any $h \in I(Y)$, and $x \in X$, we have

$$h(f(x)) = f^*h(x) = (g(h(y)))(x)$$

since $h \in I(Y)$, $h = 0 \in A(Y)$, so $h(f(x)) = 0$. Thus $f(X) \subseteq V(I(Y)) = Y$, so the map is indeed into Y . To see that it is a morphism, we just note that the coordinate functions φ_i are regular, so by the previous theorem f is a morphism. Then we finally see that $f^* = g$, so we have constructed the morphism corresponding to g . $\square$

Corollary 2.19

Two affine varieties are isomorphic if and only if their coordinate rings are isomorphic.

Example 2.9

Let X be the rational normal curve $V(y - x^2, z - x^3) \subseteq \mathbb{A}^3$. Then

$$A(X) = \frac{K[x, y, z]}{\langle y - x^2, z - x^3 \rangle} = K[x]$$

so $X \cong \mathbb{A}^1$.

Proposition 2.20

The automorphism group of $\mathbb{A}^1$ is

$$\text{Aut}(\mathbb{A}^1) = \{f(x) = ax + b, a \in K^*, b \in K\}$$

Proof. The K -algebra homomorphisms of $K[x]$ are those of the form

$$\psi_p : f \mapsto f(p(x))$$

But $\deg f \circ p \geq \deg f \deg p$, and when f is a constant function $f \circ p$ is also constant. Thus if p has degree higher than 1 there are no degree 1 polynomials in the image, and if p is constant then so is ψ_p . Thus the only isomorphisms are of the form $\psi_p, p = ax + b$ where $a \neq 0$. From

correspondence with K -algebra homomorphisms we know that the corresponding morphism is

$$f = (p(x_1), \dots, p(x_n))$$

but we work over $n = 1$, so this is just $f = p$. $\square$

Example 2.10

Let $X = V(x^2 - y^3) \subseteq \mathbb{A}^2$. Then $f : \mathbb{A}^1 \rightarrow X$ given by

$$t \mapsto (t^3, t^2)$$

is bijective and a morphism (since its coordinates are regular). But the pullback map $f^* : K[x, y]/\langle x^2 - y^3 \rangle \rightarrow K[t]$ maps

$$[x] \mapsto [t^3], [y] \mapsto [t^2]$$

so there are no degree 1 elements in the image, hence this is not an isomorphism. So it is not enough to be a bijective morphism to be an isomorphism.

We can use this correspondence and the categorical interpretation of the product to give a categorical characterization of the coordinate ring of products.

Proposition 2.21

If X, Y are affine varieties then the projection map

$$\pi_X : (x_1, \dots, x_n, y_1, \dots, y_m) \mapsto (x_1, \dots, x_n)$$

is a morphism, and similarly for Y .

Proposition 2.22: Universal Property of Products

Let X, Y be affine varieties and π_X, π_Y be projections from $X \times Y \rightarrow X, Y$. If Z is an affine variety and $f_X : Z \rightarrow X, f_Y : Z \rightarrow Y$ are maps then there is a unique morphism $Z \rightarrow X \times Y$ that makes the diagram commute:

$$\begin{array}{ccccc} Z & & & & \\ & \searrow f_Y & & \searrow & \\ & & X \times Y & \xrightarrow{\pi_Y} & Y \\ & \swarrow f_X & \downarrow \pi_X & & \\ & & X & & \end{array}$$

$\exists! f$ (dashed arrow from Z to $X \times Y$)

Proof. The only set theoretic map that satisfies this is $f = (f_X, f_Y)$ by looking at the projections of f . This is a morphism since the coordinates of f_X, f_Y are regular, so so are the coordinates of f . $\square$

By passing through the contravariant correspondence, we get the universal property of tensor products on the coordinate rings.

$$\begin{array}{ccc}
 \begin{array}{c}
 Z \xrightarrow{f_Y} Y \\
 \searrow \exists! f \quad \downarrow \pi_X \\
 X \times Y \xrightarrow{\pi_Y} Y \\
 \downarrow f_X \\
 X
 \end{array}
 & \Longleftrightarrow &
 \begin{array}{c}
 A(Z) \xleftarrow{g_Y} A(Y) \\
 \searrow \exists! g \quad \uparrow \pi_X^* \\
 A(X) \otimes_K A(Y) \xleftarrow{\pi_Y^*} A(Y) \\
 \uparrow \pi_X^* \\
 A(X)
 \end{array}
 \end{array}$$

We have seen that every affine variety may be associated with its coordinate ring $A(X)$, which is a finitely generated, reduced K -algebra. Moreover, if we restrict to equivalence classes, then this map injects the equivalence classes of affine varieties into the finitely generated reduced K -algebras. We see the reverse of this correspondence now.

Let R be a f.g. reduced K -algebra. Then let $b_1, \dots, b_n$ generate R . This gives a surjective K -algebra homomorphism $g : K[x_1, \dots, x_n] \rightarrow R$ defined by

$$g(x_i) = b_i$$

or equivalently $f \mapsto f(b_1, \dots, b_n) \in R$. By the isomorphism theorem, we have

$$R \cong K[x_1, \dots, x_n] / \ker g$$

Set $I = \ker g$. Since R is reduced, I is radical. So if we take $X = V(I) \subseteq \mathbb{A}^n$, then

$$A(X) = K[x_1, \dots, x_n] / I(X) = K[x_1, \dots, x_n] / I(V(I)) = K[x_1, \dots, x_n] / I \cong R$$

This is not a unique representation of R , but any two representations are isomorphic as affine varieties. Thus we see that the contravariant functor from affine varieties to finitely generated reduced K -algebras is an equivalence of categories, meaning that it is bijective on equivalence classes of objects such that $\text{Mor}(X, Y) \cong \text{Mor}(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$. Moreover, since from a categorical perspective, affine varieties are simply specific ringed spaces, we can extend our definitions:

Remark 2.1

From now, an affine variety is *any* ringed space isomorphic to an affine variety (in the old sense) as a ringed space. We write $A(X) = \mathcal{O}_X(X)$, where $\mathcal{O}_X$ is the structure sheaf.

This allows us to see certain open subsets of affine varieties as affine varieties (though they are not subvarieties in the old sense).

Proposition 2.23

Let X be an affine variety and let $f \in A(X)$. Then $D(f)$ is an affine variety with coordinate ring

$$A(D(f)) = \mathcal{O}_X(D(f)) = A(X)_f$$

Proof. Write

$$Y = \{(x, t) \in X \times \mathbb{A}^1 : tf(x) - 1 = 0\} \subseteq X \times \mathbb{A}^1$$

There is a projection morphism $g : Y \rightarrow D(f)$ mapping $(x, t) \mapsto x$, since $f(x)$ is always nonzero for elements of Y . The inverse is given by

$$x \mapsto \left(x, \frac{1}{f(x)}\right)$$

$\frac{1}{f}$ is an element of $\mathcal{O}_X(D(f))$, so this is a morphism and it is the inverse of g . So $D(f)$ is an affine variety as a sub ringed space, meaning that

$$A(D(f)) \cong \mathcal{O}_X(D(f))$$

and we know that

$$\mathcal{O}_X(D(f)) \cong A(X)_f$$

□

Example 2.11

$U = \mathbb{A}^2 \setminus \{(0, 0)\}$ is an open subset. Suppose it were an affine variety. Then its coordinate ring would be $\mathcal{O}_{\mathbb{A}^2}(U) = K[x, y]$, and the identity map on $K[x, y]$ induces the isomorphism

$$\text{id} : U \rightarrow \mathbb{A}^2$$

but this is not a bijection, hence not an isomorphism, contradiction. Thus U is not an affine variety at all.

2.5 Varieties

We conclude the development of affine varieties by generalizing our results to spaces that only look locally like an affine variety. For instance, we showed in the last section that $U = \mathbb{A}^2 \setminus \{(0, 0)\}$ is not an affine variety. However, it is the union $D(x) \cup D(y)$, so every point has a neighborhood which is an affine variety. Thus we allow for more general constructions where affine varieties are glued together into larger spaces, analogously to how manifolds are glued copies of Euclidean space.

This will allow us to remedy some issues with the more restrictive definition of affine varieties. For instance, in the Euclidean topology, no affine variety is compact except the finite sets (they are all compact in the Zariski topology, or more accurately quasi-compact since this is not Hausdorff). Moreover, arbitrary open subsets of affine varieties are not varieties, as we saw above.

Definition 2.13

A **prevariety** is a ringed space X which admits a finite open cover by affine varieties. A morphism of a prevariety is a morphism as a ringed space. The regular functions on $U \subseteq X$ open are the elements of $\mathcal{O}_X(U)$. An open subset of a prevariety which is an affine variety is an affine open set.

Example 2.12

$U = \mathbb{A}^2 \setminus \{(0, 0)\}$ is a prevariety covered by $D(x), D(y)$, which are distinguished open subsets, hence affine varieties. More generally, since the distinguished open subsets form a basis, any open subset of an affine variety has a finite cover by distinguished open subsets, so is a prevariety.

Proposition 2.24

Any open subset of a prevariety is a prevariety.

Proof. Let X be a prevariety covered by $(U_i)_{i=1}^n$ affine varieties, and let $Y \subseteq X$ be open. Then $(Y \cap U_i)_{i=1}^n$ is an open cover of Y . Moreover, each $Y \cap U_i$ is an open subset of U_i , so it is an open subset of an affine variety, hence a prevariety. Thus there are $(V_{i,j})_{j=1}^{m_n}$ affine varieties covering $Y \cap U_i$, and taking all $V_{i,j}$, we have a finite open cover of Y by affine varieties. $\square$

Since the point of prevarieties is to allow for patching together affine varieties locally, we should develop a tool to do this. We will begin by treating the case of two prevarieties.

Definition 2.14

Let X_1, X_2 be two prevarieties and let $U_1 \subseteq X_1, U_2 \subseteq X_2$ be open subsets, with an isomorphism of prevarieties $f : U_1 \rightarrow U_2$. Then the prevariety X formed by gluing X_1, X_2 along f is the ringed space $(X_1 \sqcup X_2) / \sim$, where $\sim$ is the equivalence relation $x \sim x$, together with $a \sim f(a), f(a) \sim a$ for $a \in U_1$. This set is endowed with the quotient topology, and the structure sheaf is given by

$$\mathcal{O}_X(U) = \{\varphi : U \rightarrow K : i_1^* \varphi \in \mathcal{O}_{X_1}(i_1^{-1}(U)), i_2^* \varphi \in \mathcal{O}_{X_2}(i_2^{-1}(U))\}$$

where $i_1 : X_1 \rightarrow X, i_2 : X_2 \rightarrow X$ are the injection maps which take x to its equivalence class in X .

Since X_1, X_2 are prevarieties isomorphic to their images, we get a finite cover of the glued space by affine varieties, so the glued space is indeed a prevariety. Also, it is immediate that $\mathcal{O}_X|_{X_i} = \mathcal{O}_{X_i}$ for $i = 1, 2$.

Example 2.13

Let $X_1 = X_2 = \mathbb{A}^1$, with $U_1 = U_2 = \mathbb{A}^1 \setminus \{0\}$. If $f(x) = 1/x$, then the glued space is the space $\mathbb{A}^1$ together with a point at infinity. This is called the **projective line** $\mathbb{P}^1$. In particular when $K = \mathbb{C}$, this is the Riemann sphere. If we consider the morphisms $X_1 \rightarrow X_2 : x \mapsto x$ and $X_2 \rightarrow X_1 : x \mapsto x$, this maps $[x] \mapsto [1/x]$ for $x \in X_1$, and similarly for X_2 . Moreover, these agree on their combined domain, which is the image of U_1 (and U_2 , by the quotient equivalence relation). Thus the glued map is a morphism on $\mathbb{P}^1 \rightarrow \mathbb{P}^1$, which takes $x \mapsto 1/x$ for $x \in K^*$ and $0 \leftrightarrow \infty$.

On the other hand, $f(x) = x$, we get a space that looks like $\mathbb{A}^1$ but with two points at the origin. This is a non-Hausdorff space over $\mathbb{C}$, since we can take a sequence which converges to both origins. The glued morphism as above is the identity on $\mathbb{A}^1 \setminus \{0\}$ but switches the origins. As a result, the set $\{x : g(x) = x\}$ is the equality locus of two morphisms, but is not a closed set in X .

Pathological behavior like the affine line with two origins will eventually be excluded when we move from prevarieties to varieties. The extension to finite collections of prevarieties is straightforward and just requires a few extra compatibility conditions.

Definition 2.15

Let $(X_i)_{i=1}^n$ be a finite set of prevarieties. For all $i \neq j$ suppose there are $U_{ij} \subseteq X_i$ open and an isomorphism $f_{ij} : U_{ij} \rightarrow U_{ji}$ such that whenever i, j, k are distinct,

$$f_{ij}^{-1} = f_{ji}$$

and $f_{ij}^{-1}(U_{jk}) \subseteq U_{ik}$ with

$$f_{jk} \circ f_{ij} = f_{ik}$$

on $f_{ij}^{-1}(U_{jk})$. Then we define the glued prevariety to be $X = \sqcup X_i / \sim$, where the equivalence relation $\sim$ is defined by $x \sim x$ with $a \sim f(a)$, $f(a) \sim a$ for all $a \in U_{ij}$. By the compatibility relations, $\sim$ is transitive and symmetric, so it is an equivalence relation, and the structure sheaf and topology are defined the same as above.

Example 2.14

Let $X_0 = X_1 = X_2 = \mathbb{A}^2$ and let $U_{i,j} = \{x \in U_i : x_j \neq 0\}$, where a point in X_0 is specified by coordinates (x_1, x_2) , a point in X_1 by (x_0, x_2) , and a point in X_2 by (x_0, x_1) . The space formed by gluing along

$$f_{01}(x_1, x_2) = \left(\frac{1}{x_1}, \frac{x_2}{x_1} \right), f_{02}(x_1, x_2) = \left(\frac{1}{x_2}, \frac{x_1}{x_2} \right), f_{12} = \left(\frac{x_0}{x_1}, \frac{1}{x_1} \right)$$

is the **projective plane** $\mathbb{P}^2$.

Since every prevariety X is finitely covered by affine varieties, we can take an open cover U_i

and set $U_{ij} = U_i \cap U_j$ open, with $f_{ij} : U_i \rightarrow U_j$ given by the identity. This recovers X , so this construction gives any prevariety in terms of affine varieties.

Since a prevariety is a ringed space, in particular it is topological. Moreover, it is the finite union of Noetherian spaces, so by homework it is Noetherian. Moreover, we can endow any open or closed subset with a prevariety structure.

Definition 2.16

Let $U \subseteq X$ be an open subset of a prevariety. Then U is a ringed space endowed with the structure sheaf

$$\mathcal{O}_U(V) = \mathcal{O}_X(V)$$

Moreover, If X is covered by finitely many affine varieties $X_1, \dots, X_n$, then $U \cap X_i$ is open in X_i , hence covered by finitely many distinguished open subsets, which are affine varieties. So U is also covered by finitely many affine varieties. Endowed with this structure, U is called an **open subprevariety** of X .

Definition 2.17

Let $Y \subseteq X$ be a closed subset of a prevariety. Then Y is a ringed space endowed with the structure sheaf $\mathcal{O}_Y(V)$ of functions which are locally restrictions of regular functions:

$$\{\varphi : V \rightarrow K : \forall a \in V, \exists U_a \subseteq X \text{ open } \exists \psi \in \mathcal{O}_X(U_a) \text{ s.t. } \varphi = \psi \text{ on } U_a \cap V\}$$

This is called a **closed subprevariety**.

Proposition 2.25

For any affine open set $W \subseteq X$ and closed set $Y \subseteq X$, the identity map on $W \cap Y$ is an isomorphism of ringed spaces between $W \cap Y$ equipped with the sheaf $\mathcal{O}_Y|_W$ (from the closed subprevariety construction) and the equipped with the sheaf $\mathcal{O}_{W \cap Y}$ (treating $W \cap Y$ as an affine subvariety of W).

Proof. The identity is clearly continuous and its own inverse. So we just need to show it is a morphism in both directions. $\square$

Corollary 2.26

If $Y \subseteq X$ is a closed subset of a prevariety, then $(Y, \mathcal{O}_Y)$ as defined above is a prevariety.

Proof. Cover X by finitely many affine varieties X_i . Then $(Y, \mathcal{O}_Y)$ is covered by $Y \cap X_i$ with the sheaf $\mathcal{O}_Y|_{Y \cap X_i}$. By the proposition, this is isomorphic to $(Y \cap X_i, \mathcal{O}_{Y \cap X_i})$ which is an affine variety. So $(Y, \mathcal{O}_Y)$ is covered by finitely many affine varieties. $\square$

The finite union of open or closed subprevarieties is also an open or closed subprevariety, respectively. However, unioning an open and closed subprevariety together does not in general give a prevariety.

Example 2.15

Let $X = \mathbb{A}^1 \times (\mathbb{A}^1 \setminus \{0\})$, $Y = \{(0, 0)\} \subseteq \mathbb{A}^2$, and let $Z = X \cup Y$. X is open and Y closed in $\mathbb{A}^2$, so they are subprevarieties. However, there is no neighborhood of the origin that looks like an affine variety.

To see this, let U be an open neighborhood of the origin in Z , which is therefore of the form $Z \cap V$ for V open in $\mathbb{A}^2$. Then $V \cap D(y)$ is a nonempty open set, hence dense in $\mathbb{A}^2$, and therefore it is also irreducible by irreducibility of $\mathbb{A}^2$. $V \cap D(y)$ is also contained in U so its closure in U is U , thus it is dense in X . Therefore U is also irreducible.

The dimension of U is two, since it lies in $\mathbb{A}^2$ and we have the strict chain

$$\{(0, 0)\} \subsetneq U \cap V(x) \subsetneq U$$

(note that $U \cap V(x)$ is nonempty and open in $V(x) \cong \mathbb{A}^1$, which is irreducible, hence irreducible). Then y is in the coordinate ring of U , so its vanishing locus has codimension 1, implying it has dimension 1. But it is a single point so has dimension 0, contradiction.

Proposition 2.27

Let $Y \subseteq X$ be a closed subprevariety. Then

1. The inclusion $i_Y : Y \rightarrow X$ is a morphism,
2. If $f : Z \rightarrow X$ is a morphism and $f(Z) \subseteq Y$ where Z is a prevariety, then the same set theoretic map is a morphism from $Z \rightarrow Y$.

Proof. 1. The inclusion is closed by definition of the subspace topology. It is a morphism since if $\varphi \in \mathcal{O}_X(U)$, then $i_Y^* \varphi$ is a map from $i_Y^{-1}(U) \rightarrow K$. Since i_Y is the restriction of the identity to Y , for any $a \in i_Y^{-1}(U)$ we have $a \in U \subseteq X$ with $\varphi = \varphi$ on $U \cap i_Y^{-1}(U)$. So i_Y is a morphism.

2. f is clearly still continuous, and if $\varphi \in \mathcal{O}_Y(U)$, then at any point $a \in U$ we have $a \in U_a \subseteq X$ as well as $\psi \in \mathcal{O}_X(U_a)$ with $\varphi = \psi$ on $U \cap U_a$. $f^* \psi \in \mathcal{O}_Z(f^{-1}(U_a))$ and since $f(Z) \subseteq Y$, $f^{-1}(U_a) = f^{-1}(U \cap U_a)$. Then we have $f^* \psi = f^* \varphi \in \mathcal{O}_Z(f^{-1}(U_a \cap U))$. Taking the union over all a , $f^* \varphi$ is regular on an open cover so it is regular. Thus $f^* \varphi \in f^{-1}(U)$, meaning that f is a morphism. $\square$

Proposition 2.28

The inverse image of an open (resp. closed) subprevariety under a morphism is an open (closed) subprevariety.

Proof. Straightforward by continuity. $\square$

Example 2.16

Let $f : \mathbb{A}^2 \rightarrow \mathbb{A}^2$ be $(x, y) \mapsto (xy, y)$. Then $f(\mathbb{A}^2)$ is the set $Z = \mathbb{A}^1 \times (\mathbb{A}^1 \setminus \{0\}) \cup \{(0, 0)\}$ that we investigated earlier, which is neither an open nor closed subprevariety. So this property is not preserved under forward images.

The union of a closed and open subset is a **locally closed** subset, and a **constructible set** is a finite union of locally closed sets. We can show that the forward image of a constructible set under a morphism is constructible.

Finally, we study products of prevarieties, because this will allow us to state the condition that ensures a variety avoids the pathological behavior of some prevarieties. If X, Y are prevarieties, then we want to make sense of the product $X \times Y$. Set theoretically this is no problem, but we need to endow it with a ringed space structure.

The only way that we can currently build up prevarieties is by gluing them together. One way that we can do this is to pick affine covers U_i of X and V_j of Y , and glue together the sets $U_i \cap V_j$, which we do know have prevariety structures as affine varieties. However, this is not a priori independent of the chosen covers. We will solve this by defining the product via a universal property, so it is unique up to unique isomorphism.

Definition 2.18

Let X, Y be prevarieties. A **product of prevarieties** is a prevariety P with morphisms $\pi_X : P \rightarrow X, \pi_Y : P \rightarrow Y$ such that for any prevariety Z with morphisms $f_X : Z \rightarrow X, f_Y : Z \rightarrow Y$ there is a unique morphism $f : Z \rightarrow P$ such that the following commutes:

$$\begin{array}{ccccc}
 Z & & & & \\
 & \searrow \exists! f & & \searrow f_Y & \\
 & & P & \xrightarrow{\pi_Y} & Y \\
 & \searrow f_X & \downarrow \pi_X & & \\
 & & X & &
 \end{array}$$

In particular, when $P = X \times Y$ and Z is assumed to be affine, we showed that this holds.

Proposition 2.29

If X, Y are prevarieties then there is a product P of X and Y , which is unique up to unique isomorphism.

Proof. First we show uniqueness. Let P be a product and P' another product. Applying the universal property of P to the projections of P' , and vice versa, we get morphisms $f : P \rightarrow P', g : P' \rightarrow P$ such that

$$\begin{array}{ccc} P & \xrightarrow{\pi_Y} & Y \\ \pi_X \downarrow & \swarrow g & \uparrow \pi'_Y \\ X & \xleftarrow{\pi'_X} & P' \end{array}$$

This implies the following diagram:

$$\begin{array}{ccc} P & \xrightarrow{\pi_Y} & Y \\ \pi_X \downarrow & \swarrow g \circ f & \uparrow \pi'_Y \\ X & \xleftarrow{\pi'_X} & P' \end{array}$$

By the universal property for P , $\text{id}_P = g \circ f$, and by the same for P' , $f \circ g = \text{id}_{P'}$. So P, P' are isomorphic by a unique isomorphism.

Now, for existence, we let P be constructed as above, where it is glued across affine varieties $U_i \times V_j$. To check that this satisfies the universal property, we have that each $U_i \times V_j$ offers a projection morphism $U_i \times V_j \rightarrow U_i \subseteq X$ and one to $V_j \subseteq Y$. They agree on intersections, so they glue into a morphism $\pi_X : P \rightarrow X, \pi_Y : P \rightarrow Y$. Now, let Z be a prevariety and $f_X : Z \rightarrow X, f_Y : Z \rightarrow Y$ be morphisms. We want to show that

$$f = (f_X, f_Y) : Z \rightarrow P$$

is a morphism. It is clearly unique since this is the only possible way the diagram can commute. It suffices to check this on an open cover. Consider the open sets $f^{-1}(U_i \times V_j)$. These are open subprevarieties, so we can cover them by affine open subsets Z_k . On each Z_k , $f_X|_{Z_k} : Z_k \rightarrow U_i$ is a morphism as the composition of the open embedding morphism with a morphism, and similarly for $f_Y|_{Z_k}$. Then since we are consider maps from affine varieties, the universal property for products of affine varieties applies, so $f|_{Z_k} = (f_X|_{Z_k}, f_Y|_{Z_k})$ is a morphism from $Z_k \rightarrow U_i \times V_j$. By the gluing construction of prevarieties, the embedding $U_i \times V_j \rightarrow P$ is a morphism. We can then glue these into a morphism $f : Z \rightarrow P$. So this is a morphism and the universal property holds. $\square$

Proposition 2.30

If $X' \subseteq X, Y' \subseteq Y$ are closed subprevarieties then $X' \times Y'$ with the product structure is isomorphic to $X' \times Y'$ with the closed prevariety structure inherited from $X \times Y$ with the product structure.

Finally, we conclude this section by defining a variety to be a prevariety which is sufficiently nice to avoid pathological topological spaces like the affine line with two origins. In this case, the set equipped with the complex topology is not Hausdorff. We cannot directly require that varieties are Hausdorff prevarieties, as this is too strong of a requirement (we cannot find disjoint open neighborhoods). However, it is a result that a topological space X is Hausdorff if and only if its diagonal is closed in the product topology of $X \times X$. Since our products do not carry the product topology, the following is a useful definition even though being Hausdorff is not:

Definition 2.19

A prevariety X is called a **variety** if its diagonal

$$\Delta_X = \{(x, x) : x \in X\} \subseteq X \times X$$

is closed under the Zariski topology.

Lemma 2.31

Affine varieties are varieties.

Proof. If $X \subseteq \mathbb{A}^n$ then we can write the diagonal as the relative vanishing locus

$$\Delta_X = V_{X \times X}(x_1 - y_1, x_2 - y_2, \dots, x_n - y_n) \subseteq X \times X$$

so Δ_X is closed. □

Proposition 2.32

Let $f_1 : Y_1 \rightarrow X_1, f_2 : Y_2 \rightarrow X_2$ be morphisms (of prevarieties). Then $(f_1, f_2) : Y_1 \times Y_2 \rightarrow X_1 \times X_2$ is a morphism.

Proof. The maps $\pi_1 : Y_1 \times Y_2 \rightarrow Y_1, \pi_2 : Y_1 \times Y_2 \rightarrow Y_2$ are morphisms. Let $g_1 = f_1 \circ \pi_1, g_2 = f_2 \circ \pi_2$ are morphisms. Then there is a unique morphism $f = (F_1, F_2)$ such that $F_1 = g_1, F_2 = g_2$ by the universal property of products, which implies that (f_1, f_2) is a morphism. □

Lemma 2.33

Open and closed subprevarieties of varieties are also varieties.

Proof. Consider the inclusion map $i_{Y \times Y} : Y \times Y \rightarrow X \times X$. If we can show that this is a morphism, then it will follow by continuity that the diagonal is closed, since $\Delta_Y = i_{Y \times Y}^{-1}(\Delta_X)$.

To show this, we note that the open/closed embedding $i_Y : Y \rightarrow X$ is a morphism. Then by the proposition above, $i_{Y \times Y} = (i_Y, i_Y)$ (where the inclusions work on each coordinate separately) is a morphism. $\square$

Definition 2.20

A variety of a pure dimension 1 is called a **curve** and variety of pure dimension 2 is a **surface**. If $Y \subseteq X$ is a subvariety, X is pure dimensional, and Y has pure dimension $\dim X - 1$, then Y is a **hypersurface** in X .

Lemma 2.34

Let $f, g : X \rightarrow Y$ be morphisms of prevarieties, with Y a variety. Then:

1. the graph

$$\Gamma_f = \{(x, f(x)) : x \in X\} \subseteq X \times Y$$

is closed in $X \times Y$,

2. the set

$$\{x \in X : f(x) = g(x)\}$$

is closed in X .

Proof. 1. The graph is the preimage of Δ_Y closed under the map

$$(\text{id}, f)$$

which is a morphism by the proposition above.

2. This is the preimage of Δ_Y under the map

$$(f, g)$$

which is a morphism by the proposition above. $\square$

Proposition 2.35

If X, Y are varieties then $X \times Y$ is a variety.

Proof. The diagonal is given by

$$\Delta_{X \times Y} = \{(x, y), (x, y) : x \in X, y \in Y\} \subseteq X \times Y \times X \times Y$$

This is isomorphic to

$$\{(x, x), (y, y)\} \subseteq X \times X \times Y \times Y$$

This is the set $(\Delta_X \times Y \times Y) \cap (X \times X \times \Delta_Y)$ which are both closed, so this is closed. $\square$

Chapter 3

Projective Varieties

3.1 Construction and Homogeneous Polynomials

In the last chapter we introduced the projective line $\mathbb{P}^1$, which we saw was constructed from $\mathbb{A}^1$ by adding “points at infinity.” If we do this for each $\mathbb{A}^n$, it turns out that a large class of varieties lie in some $\mathbb{P}^n$, and that such varieties have beneficial properties. We begin by describing the construction and topology of projective spaces and varieties. We can do this using a gluing procedure as before, but it is more instructive to build this up by hand, transfer results from affine space, and then finally reconnect this with the theory of varieties by imposing a ringed space structure.

Definition 3.1

Projective space of dimension n is defined as a set by the space of all 1-dimensional linear subspaces of K^{n+1} .

Note that every one dimensional linear subspace can be specified by giving a nonzero vector which generates it:

$$\ell = \langle (x_0, \dots, x_n) \rangle$$

Moreover, two vectors which generate the same subspace agree up to a homogeneous rescaling of their coordinates by $\lambda \neq 0$. Thus we can see $\mathbb{P}^n$ as

$$\mathbb{P}^n = K^{n+1} \setminus \{0\} / \sim$$

where $\sim$ is the equivalence relation of agreeing up to homogeneous rescaling by $\lambda \neq 0$.

Definition 3.2

The **homogeneous coordinates** for an equivalence class $[(x_0, \dots, x_n)] \in \mathbb{P}^n$ for $(x_0, \dots, x_n) \in K^{n+1} \setminus \{0\}$ are

$$(x_0 : \dots : x_n)$$

Homogeneous coordinates are defined up to a constant scaling by $\lambda \neq 0$ and are never all zero. We need to connect affine space with projective space, which proceeds as follows:

Consider the map $f : \mathbb{A}^n \rightarrow \mathbb{P}^n$ which takes $(x_1, \dots, x_n) \mapsto (1 : x_0 : \dots : x_n)$. This is injective since the first coordinate prevents scaling. The image is the set

$$U_0 = \text{im } f = \{(1 : x_1 : \dots : x_n)\} = \{(x_0 : x_1 : \dots : x_n) : x_0 \neq 0\}$$

There is also an inverse

$$f^{-1} : U_0 \rightarrow \mathbb{A}^n, \quad (x_0 : x_1 : \dots : x_n) = \left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0} \right)$$

Thus, as a set, we can think of $\mathbb{A}^n$ as a subset of $\mathbb{P}^n$. This is called the **affine part**. Meanwhile, the complement is the set of points in $\mathbb{P}^n$ where $x_0 = 0$, and the other coordinates are also not all zero, defined up to scaling. So

$$\mathbb{P}^n \setminus \mathbb{A}^n = \mathbb{P}^{n-1}$$

where we think of $\mathbb{P}^{n-1}$ as the space of points at infinity in $\mathbb{P}^n$.

Example 3.1

Let $K = \mathbb{C}$. The affine coordinate to homogeneous coordinate map $\pi : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{P}_{\mathbb{C}}^n$ is surjective. If we restrict to $S^n = \{(x_0, \dots, x_n) : |x_0|^2 + \dots + |x_n|^2 = 1\}$, then S is compact and $\pi|_S$ is surjective, so $\mathbb{P}_{\mathbb{C}}^n$ is compact.

We would like to make sense of solutions to algebraic conditions in $\mathbb{P}^n$. Observe that since homogeneous coordinates are only defined up to scaling, polynomials in their coordinates may not be well defined: if $f(x, y) = x^2 - y$ then $(1 : 1) = (2 : 2)$ but

$$f(1, 1) = 0, f(2, 2) = 2$$

so we cannot necessarily detect the vanishing locus directly. However, if all terms are of the same degree, then they will scale together. While the value is still not well defined, we can detect when the function is exactly zero.

Definition 3.3

A polynomial $f \in K[x_0, \dots, x_n]$ is called **homogeneous** of degree d if all its monomials are of all degree d .

As above, we will prefer to formulate this for non-polynomial rings. The graded ring structure generalizes the notion of degree.

Definition 3.4

A **graded ring** is a ring R with abelian additive subgroups $R_1, R_2, \dots$ such that

$$R = \bigoplus_{d \in \mathbb{N}} R_d$$

so that each element of R is a finite sum of elements of R_d , with a unique decomposition

$$f = \sum_{d \in \mathbb{N}} f_d, \quad \text{finitely many nonzero}$$

and whenever $f \in R_d, g \in R_e, fg \in R_{d+e}$. For $f \neq 0 \in R$, the **degree** of f is

$$\deg(f) = \max \{d : f_d \neq 0\}$$

The nonzero elements of each R_d are called **homogeneous elements**. A graded ring where $\lambda f \in R_d$ whenever $\lambda \in K, f \in R_d$, and R is a K -algebra is called a **graded K -algebra**.

Clearly the polynomials, where R_d denote those polynomials where all monomials are of degree d , form a graded ring.

Proposition 3.1

If $f, g \in R$ are nonzero in a graded ring then

$$\deg(fg) = \deg f + \deg g$$

Proposition 3.2

If R is a nonzero graded ring, then 1 is homogeneous of degree 0.

Proof. There is some R_d which is nonzero, so let $x \in R_d$ be nonzero. Then the homogeneous decomposition of $x \cdot 1$ is just

$$x \cdot 1 = x \in R_d$$

so $(x \cdot 1)_d = x$ and zero for every other component. Thus $\deg(x \cdot 1) = \deg x$ so $\deg 1 = 0$. $\square$

Corollary 3.3

The graded ring structure on $K[x_0, \dots, x_n]$ is uniquely determined by the values $\deg(x_0), \dots, \deg(x_n)$.

Definition 3.5

An ideal of a graded ring is called a **homogeneous ideal** if it can be generated by homogeneous elements (not necessarily of the same degree).

Lemma 3.4

Let J, J_1, J_2 be ideals in a graded ring R .

(a) J is homogeneous if and only if

$$f = \sum_d f_d \in J \implies f_d \in J \forall d$$

or equivalently if $J = \bigoplus_d (J \cap R_d)$.

(b) If J_1, J_2 are homogeneous, then $J_1 + J_2, J_1 J_2, J_1 \cap J_2, \sqrt{J_1}$ are homogeneous.

(c) If J is homogeneous then R/J is a graded ring with decomposition

$$R/J = \bigoplus_d R_d/J \cap R_d$$

Proof. We first observe that we have the convolution

$$(fg)_d = \sum_{d_1+d_2=d} f_{d_1} g_{d_2}$$

(a) ($\implies$) Assume J is homogeneous generated by the elements h^i . Then $f \in J$ is a finite sum

$$f = \sum_{i \in I} g^i h^i$$

where g^i are not necessarily homogeneous. But we have that the homogeneous decomposition of f is

$$f_d = \sum_{i \in I} g_{d-\deg(h^i)}^i h^i$$

and only finitely many are nonzero, so $f_d \in J$.

($\impliedby$) Assume R is a graded ring such that $f \in J \implies f_d \in J$. Then we want to say that $J = \langle f_d : f \in J, d \in \mathbb{N} \rangle$ is generated by all homogeneous elements. Indeed, each f_d is contained in J by assumption, and each f can be written as a finite sum of its homogeneous elements. So J is homogeneous.

(b) For the first two, let $J_1 = \langle \{h^{(i)} : i \in I_1\} \rangle, J_2 = \langle \{g^{(j)} : j \in I_2\} \rangle$. Then

$$\begin{aligned} J_1 + J_2 &= \left\langle \left\{ h^{(i)} : i \in I_1 \right\} \cup \left\{ g^{(j)} : j \in I_2 \right\} \right\rangle \\ J_1 J_2 &= \left\langle \left\{ h^{(i)} g^{(j)} : i \in I_1, j \in I_2 \right\} \right\rangle \end{aligned}$$

which are all homogeneous. For $J_1 \cap J_2$, we have that

$$f \in J_1 \cap J_2 \implies \begin{cases} f \in J_1 \\ f \in J_2 \end{cases} \implies \forall d, \begin{cases} f_d \in J_1 \\ f_d \in J_2 \end{cases} \implies \forall d, f_d \in J_1 \cap J_2$$

so $J_1 \cap J_2$ is homogeneous by part (a). For $\sqrt{J}$, we use the condition from (a). If $f \in \sqrt{J}$, write

$$f = f_0 + \dots + f_d$$

and induct on d . If $f^m \in J$, then

$$(f^m)_{md} = f_d^m$$

which is in J by part (a). So $f_d \in \sqrt{J}$, meaning $f - f_d \in \sqrt{J}$ as well. Inducting, we find that all f_d lie in $\sqrt{J}$. So $\sqrt{J}$ is homogeneous.

- (c) We need to show that $R_d \cap J$ is a subgroup of R_d , and that there exists a unique homogeneous decomposition in R/J . For the first, we restrict the projection $R \rightarrow R/J$ to R_d to get $\varphi : R_d \rightarrow R/J$ a group homomorphism with kernel $R_d \cap J$. Then

$$R_d / R_d \cap J \cong \text{im}(\varphi_d) \subseteq R_d$$

which is a subgroup. The existence of a homogeneous decomposition is clear since if

$$f = \sum_d f_d \in R$$

then

$$\bar{f} = \sum_d \bar{f}_d \in R/J$$

and uniqueness follows since If

$$\sum_d \bar{f}_d = \sum_d \bar{g}_d \in R/J$$

then

$$\Delta = \sum_d f_d - g_d \in J$$

By part (a), we then have

$$\Delta_d = f_d - g_d \in J \implies \bar{f}_d = \bar{g}_d \in R_d / R_d \cap J \quad \square$$

We will use homogeneous ideals to define formulate projective subvarieties and a projective version of the Nullstellensatz.

Definition 3.6

Let $S \subseteq K[x_0, \dots, x_n]$ be a set of homogeneous polynomials. Then the (projective) **vanishing locus** of S is

$$V_p(S) = V(S) = \{x \in \mathbb{P}^n : f(x) = 0 \forall f \in S\}$$

Sets of the form $V(S)$ are called **projective varieties**. If J is a homogeneous ideal in $K[x_0, \dots, x_n]$ then

$$V_p(J) = V(J) = \{x \in \mathbb{P}^n : f(x) = 0 \forall f \in J \text{ homogeneous}\}$$

Equivalently, if S is the set of homogeneous polynomials in J , then $J = \langle S \rangle$ and $V(J) = V(S)$. If $X \subseteq \mathbb{P}^n$ then we define the ideal of X to be

$$I_p(X) = I(X) = \langle f \in K[x_0, \dots, x_n] : f(x) = 0 \forall x \in X \rangle$$

When we mix the definitions, the affine vanishing locus and ideal are denoted V_a, I_a .

Example 3.2

$\emptyset = V_p(1)$ and $\mathbb{P}^n = V_p(0)$ are both projective varieties.

Definition 3.7

If $f_1, \dots, f_r$ are homogeneous linear polynomials then $V_p(f_1, \dots, f_r)$ is called a **linear subspace** of $\mathbb{P}^n$.

Proposition 3.5

A point $\{a\} \in \mathbb{P}^n$ is a projective variety.

Proof.

□

Example 3.3

Consider $f = x_1^2 - x_2^2 - x_0^2$. In $\mathbb{A}^3$, the real part of $V_a(f)$ looks like a cone. If we intersect with $\{x_0 = 1\}$ and consider which linear subspaces pass through $V_a(f) \cap \{x_0 = 1\}$, the real part looks like a hyperbola. In projective space, we add two more linear subspaces which are those contained in $V_a(f) \cap \{x_0 = 0\}$. These connect the hyperbola so that the real part of $V_p(f)$ is a circle.

This shows that V_a, V_p behave similarly, and many of the results for $V_a(f)$ should be able to be carried over $V_p(f)$.

Definition 3.8

An affine variety $X \subseteq \mathbb{A}^{n+1}$ is called a **cone** if $0 \in X$ and $\lambda x \in X$ whenever $\lambda \in K, x \in X$. Equivalently, the first condition can be taken to be $X \neq \emptyset$. If X is a cone then the **projectivization** of X is the set

$$\mathbb{P}(X) = \pi(X \setminus \{0\}) \subseteq \mathbb{P}^n$$

where $\pi : \mathbb{A}^{n+1} \rightarrow \mathbb{P}^n$ is defined by

$$(x_0, \dots, x_n) \mapsto (x_0 : \dots : x_n)$$

If $X \subseteq \mathbb{P}^n$ is a projective variety then the cone over X is

$$C(X) = \pi^{-1}(X) \cup \{0\}$$

Proposition 3.6

Let $S \subseteq K[x_0, \dots, x_n]$ be a set of homogeneous nonconstant polynomials. Then $V_a(S)$ is a cone.

Proof. $0 \in V_a(S)$ since every homogeneous nonconstant polynomial vanishes at zero. If $x \in V_a(S)$ then for any $f \in S$ of degree d ,

$$f(\lambda x) = \lambda^d f(x) = 0$$

so $\lambda x \in V_a(S)$. □

Proposition 3.7

If $X \subseteq \mathbb{A}^{n+1}$ is a cone then $I(X) = I_a(X)$ is a homogeneous ideal in $K[x_0, \dots, x_n]$.

Proof. From before, we need to show that when $f \in I(X), f_d \in I(X)$. For any $x \in X$, we have

$$0 = f(\lambda x) = \sum_{d \in \mathbb{N}} \lambda^d f_d(x)$$

This is a polynomial in λ which vanishes for all λ , so using the fact that K is algebraically closed, $f_d(x) = 0$ for all d . This follows for all x , so $f_d \in I(X)$ for all d , so $I(X)$ is homogeneous. □

Lemma 3.8

If S is a set of homogeneous nonconstant polynomials then

$$\mathbb{P}(V_a(S)) = V_p(S), \quad C(V_p(S)) = V_a(S)$$

Theorem 3.9

The maps $\mathbb{P}, C$ define a bijection between cones in $\mathbb{A}^{n+1}$ and projective varieties in $\mathbb{P}^n$.

Proof. First we show that this is a well defined bijection. If $X \subseteq \mathbb{A}^{n+1}$ is a cone then by the Nullstellensatz,

$$X = V_a(I_a(X))$$

Define $J = I_a(X)$ and let S be its homogeneous elements. Since J is homogeneous, we have

$$X = V_a(J) = V_a(S)$$

Then the projectivization satisfies

$$\mathbb{P}(X) = \mathbb{P}(V_a(J)) = V_p(J)$$

In the other direction, if Y is a projective variety then let $Y = V_p(J)$ with homogeneous elements S . If J contains constant polynomials then $J = K[x_0, \dots, x_n]$ and $Y = \emptyset$, and $C(Y) = \{0\}$ is a cone. Otherwise, we showed $V_a(S)$ is a cone, and

$$C(Y) = C(V_p(J)) = C(V_p(S)) = V_a(S)$$

To show the bijection, let X be a cone. X is nonempty so $I_a(X)$ does not contain the constants. Then

$$C(\mathbb{P}(X)) = C(\mathbb{P}(V_a(I_a(X)))) = C(V_p(I_a(X))) = V_a(I_a(X)) = X$$

and for Y a projective variety

$$\mathbb{P}(C(Y)) = \pi(\pi^{-1}(Y)) = Y \quad \square$$

Now we would like to formulate a projective Nullstellensatz. The naive statement is unfortunately not true, as the edge cases in the previous results have shown: if

$$X = V_p(K[x_0, \dots, x_n]) = \emptyset$$

then

$$C(X) = \{0\} \implies I_a(C(X)) = \langle x_0, \dots, x_n \rangle$$

which shows that the ideal $I_0 = \langle x_0, \dots, x_n \rangle$ is somehow related to our edge cases from before. Indeed, $V_p(\langle x_0, \dots, x_n \rangle) = \pi(V_a(\langle x_0, \dots, x_n \rangle) \setminus \{0\}) = \pi(\{0\} \setminus \{0\}) = \emptyset$, so

$$I_p(V_p(\langle x_0, \dots, x_n \rangle)) = I_p(\emptyset) = K[x_0, \dots, x_n] \neq \sqrt{\langle x_0, \dots, x_n \rangle}$$

Definition 3.9

The **irrelevant ideal** is the ideal $I_0 = \langle x_0, \dots, x_n \rangle$.

Theorem 3.10: Projective Nullstellensatz

- (a) If X is a projective variety then $V_p(I_p(X))$.
- (b) If $J \subseteq K[x_0, \dots, x_n]$ is a homogeneous ideal and $\sqrt{J}$ is not the irrelevant ideal then $I_p(V_p(J)) = \sqrt{J}$.

This defines an inclusion reversing bijection between projective varieties and radical homogeneous ideals besides the irrelevant ideal.

Proof. (a), and the inclusion reversing part of the statement are the same as in the affine case. Also, the proof of $I_p(V_p(J)) \supseteq \sqrt{J}$ is the same as the affine case.

For the other inclusion, assume J is homogeneous and $\sqrt{J} \neq I_0$. If J contains any constant polynomials then $J = K[x_0, \dots, x_n]$, and then we have

$$I_p(V_p(K[x_0, \dots, x_n])) = I_p(\emptyset) = K[x_0, \dots, x_n] = \sqrt{J}$$

Otherwise, write $X = V_a(J) \subseteq \mathbb{A}^{n+1}$, $Y = V_p(J) \subseteq \mathbb{P}^n$. Since J does not contain the constants, we know from before that

$$\mathbb{P}(X) = Y, C(Y) = X$$

By the affine Nullstellensatz, we have

$$I_a(X) = \sqrt{J} \neq I_0 = I_a(\{0\})$$

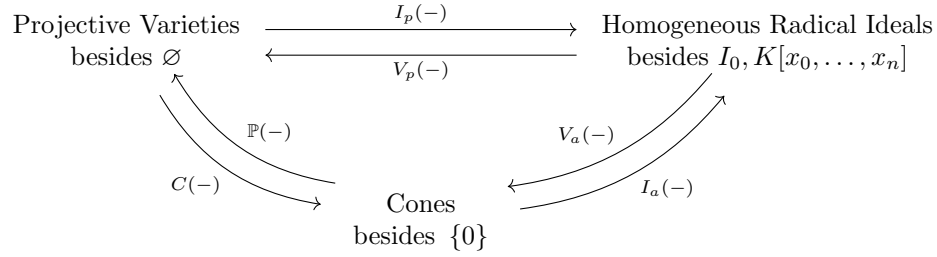
so $X \neq \{0\}$, which implies that $Y \neq \emptyset$ (since we know cones biject with projective varieties, and the cone $\{0\}$ bijects with $\emptyset$). As a result, every constant polynomial in $I_p(Y)$ is the zero polynomial. Moreover, the homogeneous polynomials vanishing on Y (which generate $I_p(Y)$) are all either degree > 0 (in which case they vanish at zero), or the zero polynomial. So such polynomials vanish not only on $X \setminus \{0\}$ (since X is a cone), but also 0 itself, and thus all of X . Thus we have the equalities

$$\begin{aligned} I_p(Y) &= \langle \{f \in K[x_0, \dots, x_n] \text{ homog.} : f|_Y \equiv 0\} \rangle \\ &= \langle \{f \in K[x_0, \dots, x_n] \text{ homog.} : f|_{X \setminus \{0\}} \equiv 0\} \rangle \\ &= \langle \{f \in K[x_0, \dots, x_n] \text{ homog.} : f|_X \equiv 0\} \rangle \end{aligned}$$

X is a cone, so the homogeneous polynomials generate $I_a(X)$, and we conclude that $I_p(Y) = I_a(X) = \sqrt{J}$ by the affine Nullstellensatz. $\square$

As a result, we basically have shown a bijection between projective varieties, cones, and homogeneous radical ideals. However, in each case, we had to exclude some exceptional cases: the cone $\{0\}$, the empty set in $\mathbb{P}^n$, and the ideals $I_0, K[x_0, \dots, x_n]$. This is summarized

in the following diagram:



Proposition 3.11

(a) If S_1, S_2 are sets of homogeneous polynomials in $K[x_0, \dots, x_n]$ then

$$V_p(S_1) \cup V_p(S_2) = V_p(S_1 S_2)$$

and for any collection (S_i) of such sets,

$$\bigcap_{i \in I} V_p(S_i) = V_p\left(\bigcup_{i \in I} S_i\right)$$

(b) If J_1, J_2 are homogeneous ideals of $K[x_0, \dots, x_n]$ then

$$\begin{aligned} V_p(J_1) \cup V_p(J_2) &= V_p(J_1 J_2) = V_p(J_1 \cap J_2) \\ V_p(J_1) \cap V_p(J_2) &= V_p(J_1 + J_2) \end{aligned}$$

(c) If $X_1, X_2 \subseteq \mathbb{P}^n$ are projective varieties then

$$\begin{aligned} I_p(X_1 \cup X_2) &= I_p(X_1) \cap I_p(X_2) \\ I_p(X_1 \cap X_2) &= \begin{cases} \sqrt{I_p(X_1) + I_p(X_2)}, & X_1 \cap X_2 \neq \emptyset \\ K[x_0, \dots, x_n], & X_1 \cap X_2 = \emptyset \end{cases} \end{aligned}$$

3.2 Projective Subvarieties

We now build the coordinate ring of a variety and consider subvarieties.

Definition 3.10

If $Y \subseteq \mathbb{P}^n$ is a variety then its **(projective) coordinate ring** is

$$S(Y) = K[x_0, \dots, x_n] / I_p(Y)$$

By a previous lemma,

$$S(Y) = \bigoplus_{d=0}^{\infty} K_d[x_0, \dots, x_n] / K_d[x_0, \dots, x_n] \cap I_p(Y)$$

is a graded ring, where $K_d[x_0, \dots, x_n]$ denotes the homogeneous polynomials of degree d .

Example 3.4

If Y is the linear subspace $V(x_0 + x_1 + x_2) \subseteq \mathbb{P}^2$, then

$$S(Y) = K[x_0, x_1, x_2] / \langle x_0 + x_1 + x_2 \rangle \cong K[x_0, x_1] = S(\mathbb{P}^1)$$

We will see later that, in analogy with the affine case, this means they are isomorphic as ringed spaces, (hence as projective varieties and varieties).

Recall that, as before, these elements do not make sense as full functions on Y . However, any element of $f \in S(Y)$ with a homogeneous representative does have a well defined vanishing locus on Y . The condition of having a homogeneous representative is equivalent to being homogeneous in the quotient graded ring structure.

Definition 3.11

Let $Y \subseteq \mathbb{P}^n$ be a projective variety. If $J \subseteq S(Y)$ is a homogeneous ideal then its **(relative) vanishing locus** is

$$V_Y(J) = V_{Y,p}(J) = \{x \in Y : f(x) = 0 \forall f \text{ homog.} \in J\}$$

Sets of this form are called **projective subvarieties**. If $X \subseteq Y$ is a subset then its **(relative) vanishing ideal** is

$$I_Y(X) = I_{Y,p}(X) = \langle f \in S(Y) \text{ homog.} : f(x) = 0 \forall x \in X \rangle$$

Example 3.5

For $Y = V(x_0 + x_1 + x_2) \in \mathbb{P}^2$, we have

$$V_Y(-x_0^2 + x_1^2 + x_2^2) = V(x_0 + x_1 + x_2, -(x_1 + x_2)^2 + x_1^2 + x_2^2) = V(x_0 + x_1 + x_2, -2x_1x_2)$$

If K is not of characteristic 2 then we have $x_1 = 0$ or $x_2 = 0$, from which we deduce that

$$V_Y(-x_0^2 + x_1^2 + x_2^2) = \{(1 : 0 : -1), (1 : -1 : 0)\}$$

There is a relative version of the projective Nullstellensatz as well. Also, it can be useful to know that every projective subvariety $X \subseteq Y \subseteq \mathbb{P}^n$ can be written as the vanishing locus of finitely many homogeneous polynomials in $S(Y)$ of the same degree, since

$$V_p(f) = \emptyset \cup V_p(f) = V_p(\langle x_0^d, \dots, x_n^d \rangle) \cup V_p(f) = V_p(\langle x_0^d f, \dots, x_n^d f \rangle)$$

By the properties of the relative vanishing locus, which are the same as the nonrelative projective versions, we see that finite unions and arbitrary intersections of subvarieties are subvarieties.

Definition 3.12

The **Zariski topology** on Y is the topology where the closed sets are the subvarieties.

This is the same as the relative topology on Y . We want to be able to make a similar statement about $\mathbb{A}^n$, namely that since $\mathbb{A}^n \subseteq \mathbb{P}^n$ as a set, we should also have that the Zariski topology on $\mathbb{A}^n$ is the relative topology as well. Algebraically, this will need to involve some relationship between polynomials in $x_1, \dots, x_n$ and homogeneous polynomials in $x_0, \dots, x_n$.

Definition 3.13

If $f \in K[x_0, \dots, x_n]$ is a homogeneous polynomial then its **dehomogenization** is the (not necessarily homogeneous polynomial) in $K[x_1, \dots, x_n]$ obtained by setting $x_0 = 1$. This is a ring homomorphism as $(f + g)^i = f^i + g^i$, $(fg)^i = f^i g^i$, and since it is surjective, applying this to ideals gives an ideal. Thus we define

$$J^i = \{f^i : f \in J\}$$

for J a homogeneous ideal, which is an ideal in $K[x_1, \dots, x_n]$.

Definition 3.14

If $f \in K[x_1, \dots, x_n]$ is nonzero of degree d , then its **homogenization** is the polynomial

$$f^h = \sum_{k=0}^d x_0^{d-k} f_k$$

where as before, f_k is the degree k homogeneous part.

Since f^h is defined relative to the degree, we have $(fg)^h = f^h g^h$ but $(f + g)^h \neq f^h + g^h$. Therefore to homogenize an ideal, we need to take the generated ideal:

$$J^h = \langle f^h : f \neq 0 \in J \rangle$$

Proposition 3.12

If S is a set of homogeneous polynomials in $K[x_0, \dots, x_n]$ then

$$\langle S \rangle^i = \langle S^i \rangle$$

where $S^i = \{f^i : f \in S\}$.

Proof. Any element $f \in \langle S \rangle$ can be written

$$f = \sum_{n=1}^k s_n$$

where $s_n \in S$. Then $f^i \in \langle S \rangle^i$ and

$$f^i = \sum_{n=1}^k s_n^i \in \langle S^i \rangle$$

In the other direction, if $g \in \langle S^i \rangle$ then

$$g = \sum_{n=1}^k s_n^i \implies g = \left(\sum_{n=1}^k s_n \right)^i \in \langle S \rangle^i \quad \square$$

Example 3.6

It is not true that if $S \subseteq K[x_1, \dots, x_n]$ is a set of nonzero polynomials, then $\langle S \rangle^h = \langle S^h \rangle$. For a counterexample, consider $S = \{x_1\}$, then $S^h = S$ and $\langle S^h \rangle = \langle x_1 \rangle \subseteq K[x_0, \dots, x_n]$. But

$$\langle S \rangle^h = \langle f^h : f \in \langle S \rangle \rangle$$

In particular, $x_1 + x_1x_2 \in \langle S \rangle$, but its homogenization is $x_0x_1 + x_1x_2$ which is not in $\langle x_1 \rangle$.

Homogenization and dehomogenization gives the connection that we need, since it relates the behavior of polynomials under the restriction $x_0 \neq 0$, which restricts $\mathbb{P}^n$ to $\mathbb{A}^n$ under our identification. That is, if F is the map

$$(x_1, \dots, x_n) \rightarrow (1 : x_1 : \dots : x_n)$$

then for homogeneous $f \in K[x_0, \dots, x_n]$,

$$f(F(x_1, \dots, x_n)) = f^i(x_1, \dots, x_n)$$

and for $g \in K[x_1, \dots, x_n]$ nonzero,

$$g^h(F(x_1, \dots, x_n)) = g(x_1, \dots, x_n)$$

Theorem 3.13

The subspace topology on $\mathbb{A}^n \subseteq \mathbb{P}^n$ is the Zariski topology on $\mathbb{A}^n$.

Proof. Observe that if $X = V_p(J) \cap \mathbb{A}^n$ is closed in the subspace topology, then by the remarks above, $X = V_a(J^i)$ is closed in the Zariski topology. On the other hand, if $X = V_a(J)$ is Zariski closed then $X = V_p(J^h) \cap \mathbb{A}^n$ is closed in the subspace topology. $\square$

Now that we have found an embedding of affine space in projective space, we can transfer many of the results we found before.

Proposition 3.14

$$\dim \mathbb{P}^n = n.$$

Proof. We just showed the subspace

$$U_0 = \{x_0 = 0\}$$

is isomorphic to $\mathbb{A}^n$. This is also true for each $U_i = \{x_i = 0\}$, and collectively the U_i cover $\mathbb{P}^n$. So $\mathbb{P}^n$ is covered open irreducible sets of dimension n , and thus $\dim \mathbb{P}^n = n$. $\square$

Proposition 3.15

If X is a projective variety, then X is irreducible if and only if $S(X)$ is an integral domain.

Proposition 3.16

If X, Y are nonempty projective varieties in $\mathbb{P}^n$, and $\dim X + \dim Y \geq n$, then $X \cap Y \neq \emptyset$.

We have the intuition that an affine variety can be compactified into a projective variety by adding points “at infinity.” This notion is formalized by taking the projective closure of a variety. If an affine variety has $X = V_a(J)$, then we know also that $X = V_p(J^h) \cap \mathbb{A}^n$. This gives $V_p(J^h)$ as a candidate for the closure.

Proposition 3.17

Let $J \subseteq K[x_1, \dots, x_n]$ be an ideal, and let $X = V_a(J)$. Then the closure of X in $\mathbb{P}^n$ (via the embedding $\mathbb{A}^n \hookrightarrow U_0$) is

$$\overline{X} = V_p(J^h)$$

and if $J = \langle f \rangle$ is nonzero and principal then

$$\overline{X} = V_p(f^h)$$

Proof. We know that $\overline{X} \subseteq V_p(J^h)$ since the latter is a closed set containing X . In the other direction, let $Y = V_p(J')$ be a closed set in $\mathbb{P}^n$ containing X . We want to show that $V_p(J^h) \subseteq Y$, which is equivalent via the Nullstellensatz to saying that $J' \subseteq \sqrt{J^h}$ (assume J' is radical).

J' is homogeneous, so it is generated by its homogeneous elements. Let $g \in J'$ be homogeneous. Note that by dividing out by the highest power of x_0 dividing g , we have $g = x_0^d f^h$ for some $f \in K[x_1, \dots, x_n]$. This is zero on $X \subseteq \mathbb{P}^n$, and since $X \subseteq \mathbb{A}^n$ as well, we have that $(f^h)^i = f$ is zero on $X \subseteq \mathbb{A}^n$. Thus $f \in I_a(X) = I_a(V_a(J)) = \sqrt{J}$. Then $f^m \in J$ for some

J , meaning $(f^m)^h \in J^h$. Homogenization commutes with multiplication so $(f^h)^m \in J^h$, meaning $f^h \in \sqrt{J^h}$ and thus $g \in \sqrt{J^h}$.

To show the result when J is principal, we have

$$\begin{aligned} J = \langle fg : g \in K[x_1, \dots, x_n] \rangle &\implies J^h = \langle (fg)^h : g \in K[x_1, \dots, x_n] \rangle \\ &= \langle f^h g^h : g \in K[x_1, \dots, x_n] \rangle = \langle f^h \rangle \end{aligned} \quad \square$$

Finally, we use this result to prove that, as in the affine case, the ideal of a projective hypersurface is principal.

Definition 3.15

A projective variety $X \subseteq \mathbb{P}^n$ is a **projective hypersurface** if it has pure dimension $n - 1$.

Proposition 3.18

If $X \subseteq \mathbb{P}^n$ is a hypersurface then $I_p(X)$ is principal.

Proof. Say X has irreducible decomposition $X = X_0 \cup \dots \cup X_m$. First assume that no component is contained in $V_p(x_0) = \mathbb{P}^n \setminus \mathbb{A}^n$, the points at infinity. Then

$$X \cap \mathbb{A}^n = \bigcup_{n=1}^{\infty} X_n \cap \mathbb{A}^n$$

Each $X_n \cap \mathbb{A}^n$ is a nonempty set open in the relative topology of X_n . Thus

$$\overline{X \cap \mathbb{A}^n} \supseteq \bigcup_{n=1}^{\infty} X_n = X$$

and clearly the other inclusion is true, too, so the projective closure of $X \cap \mathbb{A}^n$ is X . $X \cap \mathbb{A}^n$ is an affine hypersurface so $X \cap \mathbb{A}^n = V_a(f)$ for some nonzero f , and then by the previous proposition,

$$X = V_p(f^h)$$

In the general case, if $X = X' \cup X_n$ and $X_n \subseteq V_p(x_0)$, then by irreducibility and dimension, $X_n = V_p(x_0)$. Then we have $X' = V_p(f)$ for some f , and

$$X = V_p(x_0 f) \quad \square$$

We make a few remarks on this. First, we can assume that f itself is homogeneous by taking its unique factorization. Second, as we remarked above, it is not true in general that $\langle f, g \rangle^h = \langle f^h, g^h \rangle$, or for more generators. However, this is true if we have a Gröbner basis, which is the basis for computer algebra.

3.3 Projective Varieties as Ringed Spaces

We now impose a ringed space structure on projective varieties and want to show that they are varieties. In particular, they should be compatible with affine space. To start, we want to define regular functions. Recall that for affine varieties, regular functions are locally quotients f/g . For projective space, we need to make a restriction, as even a quotient of two homogeneous polynomials is not well defined (it scales the numerator and denominator differently).

Definition 3.16

Let $U \subseteq X$ be open in X a projective variety. Then $\varphi : U \rightarrow K$ is a **regular function** if it can be written locally as

$$\varphi = \frac{g}{f}$$

for $g, f \in S(X)$ homogeneous of the same degree. The set of regular functions on U is denoted $\mathcal{O}_X(U)$.

Since this is a local condition, we therefore have a sheaf, making $(X, \mathcal{O}_X)$ into a ringed space. Actually, $\mathcal{O}_X(U)$ is also a K -algebra, as

$$\frac{f_1}{g_1} + \frac{f_2}{g_2} = \frac{f_1 g_2 + f_2 g_1}{g_1 g_2}$$

and both the numerator and denominator have degree $d_1 d_2$. Now, we will show that this is a prevariety by using the open cover $\mathbb{P}^n = U_0 \cup \dots \cup U_n$.

Proposition 3.19

Let $X \subseteq \mathbb{P}^n$ be a projective variety, and define

$$U_i = \{(x_0 : \dots : x_n) \in X : x_i \neq 0\}$$

Then each U_i is an affine variety, and in particular X is a prevariety.

Proof. By symmetry we can just show $i = 0$. Say $X = V_p(J)$ for J homogeneous. Define $Y = V_a(J^i)$. We claim that $Y \cong U_0$, in particular via the isomorphism $F : Y \rightarrow U_0$

$$(x_1, \dots, x_n) \mapsto (1 : x_1 : \dots : x_n)$$

with inverse

$$(x_0 : x_1 : \dots : x_n) \mapsto \left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0} \right)$$

These are well defined maps of sets, and they are continuous since

$$\begin{aligned} F^{-1}(V_p(J') \cap U_0) &= V_{a,Y}((J')^i) \\ F(V_a(\tilde{J})) &= V_p(\tilde{J}^h) \cap U_0 \end{aligned}$$

To see that they pull back regular functions, if $\varphi(x_0, \dots, x_n) = g/f$ locally on U_0 , for $f, g \in S(X)_d$, then

$$F^*\varphi = \frac{g^i}{f^i}$$

which is regular on Y . In the other direction, if $\psi(x_1, \dots, x_n) = g/f$ locally on Y , then

$$(F^{-1})^*\psi = \frac{g\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right) x_0^e}{f\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right) x_0^e}$$

where $e = \max(\deg g, \deg f)$. The numerator and denominator are homogeneous polynomials of the same degree. So F, F^{-1} are morphisms, and we conclude. $\square$

Lemma 3.20

Let $X \subseteq \mathbb{P}^n$ be a projective variety, $f_0, \dots, f_m \in S(X)$ be homogeneous of the same degree. Then letting $U = X \setminus V_p(f_0, \dots, f_m)$, the map $f : U \rightarrow \mathbb{P}^m$ given by

$$x \mapsto (f_0(x) : \dots : f_m(x))$$

is a morphism.

Proof. This is well defined as a map of sets since the homogeneous functions scale together, and they cannot all vanish simultaneously since we are away from their vanishing locus. By the gluing property of morphisms, it suffices to check this on an open cover. Set $V_i = \{x_i \neq 0\} \subseteq \mathbb{P}^m$ and $U_i = f^{-1}(V_i) = \{f_i(x) \neq 0\} \subseteq U$. These are open and cover U . We know each V_i is isomorphic to $\mathbb{A}^n$, so we get the following diagram:

$$\begin{array}{ccc} U_0 & \xrightarrow{f} & V_0 \\ & \searrow & \uparrow \cong \\ & \left(\frac{f_1}{f_0}, \dots, \frac{f_m}{f_0}\right) & \downarrow \\ & & \mathbb{A}^n \end{array}$$

The map from $U_0 \rightarrow \mathbb{A}^n$ is a map from a prevariety to an affine variety with coordinates given by quotients of polynomials. We previously showed that rational coordinate maps between affine varieties were morphisms, the same is true if we only assume the domain is a prevariety. Thus f is a morphism on each U_i , and we conclude that f is a morphism. $\square$

Example 3.7

For any $A \in \mathrm{GL}_{n+1}(K)$, the map $x \mapsto Ax$ is an automorphism of $\mathbb{P}^n$ with inverse $x \mapsto A^{-1}x$, since each coordinate is homogeneous of degree 1.

Moreover it is true that every projective automorphism of $\mathbb{P}^n$ is of this form, and they are unique up to scaling. If $\mathrm{PGL}_{n+1}(K)$ is $\mathrm{GL}_{n+1}(K)$ quotiented by the identification $\lambda I_{n+1} \sim I_{n+1}$ for all $\lambda \in K$, then

$$\mathrm{Aut}(\mathbb{P}^n) \cong \mathrm{PGL}_{n+1}(K)$$

We can describe another class of morphisms which are geometrically seen as projecting from a point. To begin, we project from the affine origin point $a = (1, 0, \dots, 0)$.

Proposition 3.21

The map $f : \mathbb{P}^n \setminus \{a\} \rightarrow \mathbb{P}^{n-1}$ defined by

$$(x_0 : \dots : x_n) \mapsto (x_1 : \dots : x_n)$$

is a morphism.

Proof. The coordinates are polynomials which do not simultaneously vanish. $\square$

We can view this as a projection by fixing $x = (x_0 : \dots : x_n) \in \mathbb{P}^n \setminus \{a\}$. Then the line through a, x is parameterized by

$$\begin{aligned} L_{ax} &= \{sa + tx : (s : t) \in \mathbb{P}^1\} = \{(s + tx_0 : tx_1 : \dots : tx_n) : (s : t) \in \mathbb{P}^1\} \\ &= \{(s : tx_1 : \dots : tx_n) : (s : t) \in \mathbb{P}^1\} \end{aligned}$$

Then the point $(x_1 : \dots : x_n)$, which is the output of this mapping, is the unique intersection point of L_{ax} with the line at infinity $L = \{x_0 = 0\}$. In this way, we can view f as a projection from a point onto a linear subspace. Up to translation and a linear transformation, which are isomorphisms, any point and linear subspace can be moved to a, L , so this describes projection from any point onto any linear subspace not containing it. In general a line through a, b looks like

$$L_{ab} = \{sa + tb : (s : t) \in \mathbb{P}^1\}$$

We note that the map f as above does not extend to a morphism on all of $\mathbb{P}^n$. However, if we restrict to some subvarieties containing a , an extension is sometimes possible.

Example 3.8

Consider the subvariety $X = V_p(x_0x_2 - x_1^2)$, which contains $a = (1 : 0 : 0)$. Define $f : X \rightarrow \mathbb{P}^1$ by

$$x = (x_0 : x_1 : x_2) \mapsto \begin{cases} (x_1 : x_2), & x \neq a \\ (x_0 : x_1), & x \neq (0 : 0 : 1) \end{cases}$$

We observe that away from $a, (0 : 0 : 1)$, x_0, x_1 are both nonzero on X , so that on the overlap we have

$$x_0x_2 - x_1^2 = 0 \implies \frac{x_2}{x_1} = \frac{x_1}{x_0}$$

which says that the two homogeneous coordinates are the same. Thus f locally has a rational representation by homogeneous polynomials of the same degree, and thus it is a morphism. Thus the map above has an extension to a . With some more work, we can show that it is actually an isomorphism, so that $X \cong \mathbb{P}^1$.

Proposition 3.22

Let $X \subseteq \mathbb{P}^2$ be any projective conic over a field of $\text{char}(K) \neq 2$. Then $X \cong \mathbb{P}^1$.

Proof. Over such fields, the affine part of any projective conic is isomorphic to either $V_a(x_2 - x_1^2)$ (the standard parabola) or $V(x_1x_2 - 1)$ (the standard hyperbola). After taking the projective closure, any projective conic is thus isomorphic to either $V_p(x_2x_0 - x_1^2)$ or $V_p(x_1x_2 - x_0^2)$, but of course up to change of coordinates these are the same. So every projective conic is the one we just looked at, and we showed that was isomorphic to $\mathbb{P}^1$. $\square$

Now, we want to prove that projective varieties are varieties. We thus need to understand the diagonal of $\mathbb{P}^n \times \mathbb{P}^n$ (since any closed subprevariety of a variety is a subvariety). We will do this by embedding $\mathbb{P}^n \times \mathbb{P}^n$ in a larger space (and will generalize by taking n, m).

Definition 3.17

Let $n, m \in \mathbb{N}$ and $N = (n+1)(m+1) - 1$. Then the **Segre embedding** of $\mathbb{P}^n \times \mathbb{P}^m$ is the embedding $f : \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^N$ given by

$$((x_i), (y_j)) = (z_{ij} = x_i y_j)_{0 \leq i \leq n, 0 \leq j \leq m}$$

Note that this scales correctly so it is a well defined map.

Proposition 3.23

If f is the Segre embedding, then $X = f(\mathbb{P}^n \times \mathbb{P}^m)$ is the projective variety

$$X = V_p(z_{ij}z_{kl} - z_{il}z_{kj}, 0 \leq i \leq n, 0 \leq j \leq m)$$

and f is isomorphic with its image, so in particular it is a closed embedding.

Proof. Notice that points of X clearly lie in the vanishing locus since for such points,

$$z_{ij}z_{kl} - z_{il}z_{kj} = x_i y_j x_k y_l - x_i y_l x_k y_j = 0$$

Now, we pass to an affine chart. If $z = (z_{ij})$ satisfies the equations, then there is some nonzero coordinate, and without loss of generality we assume $z_{00} \neq 0$. Then z lies in the affine chart $U_{00} = \{z_{00} \neq 0\} \subseteq \mathbb{P}^N$. Then we can assume $z_{00} = 1$. So for any i, j we have

$$z_{ij} = z_{i0}z_{0j}$$

so we note that this is the image of

$$f((z_{i0})_i, (z_{0j})_j)$$

Thus we have described X .

Now, we want to show f is an isomorphism. First, notice that $f^{-1}(U_{00}) = U_0^{\mathbb{P}^n} \times U_0^{\mathbb{P}^m}$, so in these coordinates we have

$$f(x_1, \dots, x_n, y_1, \dots, y_m) = (x_1, \dots, x_n, y_1, \dots, y_m, (x_i y_j)_{i,j})$$

So f is injective since we can read off its coordinates, and its inverse is given by the projection onto the first $n + m$ coordinates. This is true on all patches U_{ij} , and since any two points lie in some patch containing both, we see that f is globally injective. f, f^{-1} locally have polynomial coordinates so they are morphisms and we are done. $\square$

Corollary 3.24

Every projective variety is a variety.

Proof. As before we just need to show that $\mathbb{P}^n$ is separated. The diagonal is given by

$$\Delta_{\mathbb{P}^n} = \{((x_0 : \dots : x_n), (y_0 : \dots : y_n) : x_i y_j - x_j y_i = 0 \forall i, j)\}$$

To see that this is true, note that the algebraic condition is equivalent to all 2×2 minors of the matrix

$$\begin{bmatrix} x_0 & \dots & x_n \\ y_0 & \dots & y_n \end{bmatrix}$$

vanishing, which means the rows are linearly dependent. Since they are nonzero, we have that $x = y$ up to scalar multiples, so that it lies in the diagonal. The reverse inclusion is obvious.

In Segre coordinates, this becomes

$$\Delta_{\mathbb{P}^n} = f^{-1}(V_p(z_{ij} - z_{ji}))$$

so this is the preimage of a closed set in $\mathbb{P}^N$, which is therefore closed. $\square$

We remark that this finally tells us that the product of projective varieties in a projective variety. If $X \subseteq \mathbb{P}^n, Y \subseteq \mathbb{P}^m$, then $X \times Y$ lies in $\mathbb{P}^n \times \mathbb{P}^m$ via a closed embedding (the product of two embeddings), and this lies in $\mathbb{P}^N$ via another closed embedding. So $X \times Y$ is isomorphic to a closed subset of $\mathbb{P}^N$ and is a projective variety.

Now, we will develop the notion of completeness. This generalizes the observation that every projective variety over $\mathbb{C}$ is compact in the Euclidean topology. Compactness itself is not useful since every prevariety is Zariski compact. However, we will see that the image under a morphism of a complete set is complete, analogous to compact sets under continuous maps.

Definition 3.18

A map $f : X \rightarrow Y$ between topological spaces is a **closed map** if $f(V) \subseteq Y$ is closed for every $V \subseteq X$ closed.

Theorem 3.25

The projection map $\pi : \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^m$ is a closed map.

Proof. Fix $Z \subseteq \mathbb{P}^n \times \mathbb{P}^m$ closed. This is also closed under the Segre embedding, so it is $V_p(f_1, \dots, f_r)$ for each f_k homogeneous in z_{ij} (equivalently, bihomogeneous in x_i, y_j). We remarked earlier that we can take these to be homogeneous of the same degree d . Fix $a \in \mathbb{P}^m$ and set $g_i(\cdot) = f_i(\cdot, a)$. Then:

$$\begin{aligned} a \notin \pi(Z) &\iff V_p(g_1, \dots, g_r) = \emptyset \\ &\iff \sqrt{\langle g_1, \dots, g_r \rangle} = K[x_0, \dots, x_n] \text{ or } \langle x_0, \dots, x_n \rangle \end{aligned}$$

In either case, for each i we have $x_i^{k_i} \in \langle g_1, \dots, g_r \rangle$ for some k_i . In particular, for any monomial of degree $k \geq k_0 + \dots + k_n$, some factor x_i must be raised to a power at least k_i . Thus for such k , $K[x_0, \dots, x_n]_k = \langle g_1, \dots, g_r \rangle_k$; that is, the degree k homogeneous part of $\langle g_1, \dots, g_r \rangle$ is all of the degree k homogeneous polynomials. We also observe that $k \geq d$ since each g_i is homogeneous of degree d , $\langle g_1, \dots, g_r \rangle$ contains no nonzero polynomials of degree less than d .

Now, every element of $\langle g_1, \dots, g_r \rangle$ is of the form

$$h_1 g_1 + \dots + h_r g_r, \quad h_i \in K[x_0, \dots, x_n]$$

so the map F_k which takes a tuple $(h_1, \dots, h_r)$ to the corresponding element is surjective from $(K[x_0, \dots, x_n]_{k-d})^r \rightarrow K[x_0, \dots, x_n]_k$. This map is K -linear, so if it is surjective then its rank is $\dim K[x_0, \dots, x_n]_k$. This is the number of partitions of k into $n+1$ intervals (to see this, note that any homogeneous monomial $x_0^{k_0} \dots x_n^{k_n}$ satisfies $k_0 + \dots + k_n = k$ which is a partition). By stars and bars, it has dimension

$$\binom{n+k}{n} = \binom{n+k}{k}$$

By linear algebra, this implies that one of the minors of size $\binom{n+k}{k}$ is nonzero. The minors are homogeneous polynomial equations in the coefficients g_i , which are correspondingly homogeneous polynomials in the coordinates of a . So $a \notin \pi(Z)$ if and only if a lies outside the collective projective vanishing locus of polynomial conditions, and therefore $\pi(Z)$ is a closed set. $\square$

Corollary 3.26

$\pi : \mathbb{P}^n \times Y \rightarrow Y$ is closed for any variety Y .

Proof. First we take the case where Y is affine in $\mathbb{A}^m$. Then for any $Z \subseteq \mathbb{P}^n \times Y$ closed, Z can be embedded in $\mathbb{P}^n \times \mathbb{A}^m \subseteq \mathbb{P}^n \times \mathbb{P}^m$. Let $\bar{Z}$ be the closure of Z in $\mathbb{P}^n \times \mathbb{P}^m$. By the previous theorem, the projection $\pi' : \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^m$ is closed, so

$$\pi'(\bar{Z}) \subseteq \mathbb{P}^m$$

is closed. Then returning to $\mathbb{P}^n \times Y$, since Z is closed in $\mathbb{P}^n \times Y$ we have

$$\pi(Z) = \pi'(\bar{Z} \cap (\mathbb{P}^n \times Y)) = \pi'(\bar{Z}) \cap Y$$

which is closed in Y .

Now, in the general case, we can cover any variety Y by affine open sets Y_i . Then any closed set $Z \subseteq \mathbb{P}^n \times Y$ satisfies

$$\pi(Z) \cap Y_i = \pi(Z \cap (\mathbb{P}^n \times Y_i))$$

which is closed in Y_i by the affine case. So $\pi(Z)$ is closed when checking on an open cover, which means it is closed in Y . $\square$

We are now ready to state the definition of completeness.

Definition 3.19

A variety X is called a **complete variety** if the projection map $\pi : X \times Y \rightarrow Y$ is closed for any variety Y .

The previous corollary shows that $\mathbb{P}^n$ is complete, which generalizes the fact that $\mathbb{P}_{\mathbb{C}}^n$ is compact in the Euclidean topology.

Example 3.9

We show that $\mathbb{A}^1$ is not complete. Informally, this is because it is “missing” the points at infinity in its projective closure. For instance, consider the closed hyperbola $V(x_1x_2 - 1) \subseteq \mathbb{A}^1 \times \mathbb{A}^1$. Taking the projection, its image is $\mathbb{A}^1 \setminus \{0\}$, which is not closed. If we take the projective closure, then we have $V_p(x_1x_2 - x_0^2) \subseteq \mathbb{P}^1 \times \mathbb{A}^1$ which projects to $\mathbb{A}^1$.

Proposition 3.27

A closed subvariety of a complete subvariety is complete.

Proof. Let $X' \subseteq X$ be a closed subvariety of a complete subvariety. Then for Y a variety, we have the following diagram:

$$\begin{array}{ccc} X' \times Y & \hookrightarrow & X \times Y \\ \pi' \downarrow & & \downarrow \pi \\ Y & \longleftarrow \cong \longrightarrow & Y \end{array}$$

π is a closed map since X is complete, and the inclusion $X' \times Y \hookrightarrow X \times Y$ is a closed embedding. Finally, the isomorphism is obviously closed, so π' is the composition of closed maps, hence closed $\square$

In particular, every projective variety is complete. We note that there exist complete varieties which are not projective, but they are very hard to construct and not encountered in practice.

Corollary 3.28

If $f : X \rightarrow Y$ is a morphism of varieties and X is complete then $f(X) \subseteq Y$ is a complete closed subvariety. Also, f is a closed map.

Proof. X is complete so the projection $\pi : X \times Y \rightarrow Y$ is a closed map. Since Y is a variety, the graph $\Gamma_f = \{(x, f(x)) : x \in X\} \subseteq X \times Y$ is closed. Therefore $f(X) = \pi(\Gamma_f)$ is closed in Y .

To see that $f(X)$ is complete, let Y' be any other variety, and let $Z \subseteq f(X) \times Y'$ be closed. We lift to $X \times Y'$ via the morphism $\psi = (f, \text{id}_{Y'})$. Thus we have the following diagram:

$$\begin{array}{ccc} \psi^{-1}(Z) & \subseteq & X \times Y' \\ \downarrow \psi & & \downarrow \psi \\ Z & \subseteq & f(X) \times Y' \\ \downarrow \pi' & & \downarrow \pi' \\ \pi'(Z) & \subseteq & Y' \end{array}$$

Since ψ is surjective, we get

$$\pi'(Z) = \pi'(\psi(\psi^{-1}(Z))) = \pi(\psi^{-1}(Z))$$

$\psi^{-1}(Z)$ is closed since ψ is a morphism, hence continuous, and π is a closed map because X is complete. So π' is closed and thus $f(X)$ is complete.

To see that f is a closed map, we know that for any $V \subseteq X$ closed, V itself is complete. So $f(V)$ is the image of the composition $f \circ i$ with i the closed embedding morphism. This itself is a morphism so by the first part, $f(V)$ is closed. $\square$

Example 3.10: Exercise

Generalize the construction of the affine line with two origins by gluing two copies of $\mathbb{P}^1$ along $\mathbb{P}^1 \setminus \{0\}$ to get a projective line with two origins X . Show that the image of the embedding $\mathbb{P}^1 \rightarrow X$ is not closed in X .

Recall that in complex analysis, we know that if a holomorphic function from the Riemann sphere $\mathbb{P}^1_{\mathbb{C}} \rightarrow \mathbb{C}$ is bounded, then it is constant. We have a similar analogue in the algebraic projective setting.

Corollary 3.29

Let X be a connected complete variety. Then the set of regular functions is the constant functions $\mathcal{O}_X(X) = K$.

Proof. Let $\varphi \in \mathcal{O}_X(X)$. Then $\varphi : X \rightarrow \mathbb{A}^1$ is a morphism, and by embedding $\mathbb{A}^1 \subseteq \mathbb{P}^1$, we get a map $\varphi : X \rightarrow \mathbb{P}^1$ with $\infty \notin \varphi(X)$. By the above, $\varphi(X)$ is a closed subset of $\mathbb{P}^1$, so it is relatively closed in $\mathbb{A}^1$, meaning it is a finite set or all of $\mathbb{A}^1$. But in the second case, $\mathbb{A}^1$ is not closed in $\mathbb{P}^1$ so this cannot be the case. Since X is connected, the image is connected, and the only connected finite set is a single point. So φ is constant. $\square$

We saw that when $f \in K[x_0, \dots, x_n]$ is degree 1 and homogeneous, $X = V_p(f)$ is a linear hypersurface. In fact, we can show that this is isomorphic to the particular linear hypersurface $V_p(x_0)$. We show this using the Veronese embedding.

Definition 3.20

Let $n, d > 0$ be integers and let $f_0, \dots, f_N$ be the set of degree d monomials in $K[x_0, \dots, x_n]$ (where $N = \binom{n+d}{d} - 1$ by a stars and bars argument). Then the **Veronese embedding** of degree d is the map

$$F = F_{n,d} : \mathbb{P}^n \rightarrow \mathbb{P}^N, x \mapsto (f_0(x) : \dots : f_N(x))$$

Note that the f_j do not all vanish simultaneously since the monomials $x_0^d, \dots, x_n^d$ do not all vanish simultaneously. This is a map with polynomial coordinates that do not simultaneously vanish, so this is a morphism. Since $\mathbb{P}^n$ is a complete variety and $\mathbb{P}^N$ a variety, the image $X = F(\mathbb{P}^n) \subseteq \mathbb{P}^N$ is a complete closed subvariety, and F is a closed map.

Proposition 3.30

The Veronese embedding is an isomorphism onto its image (and therefore a closed embedding).

Proof. We can check the inverse morphism on an open cover. Let U_i be the open set $\{x_i \neq 0\}$, hence $x_i^d \neq 0$. Setting $x_i = 1$, we get affine coordinates $x_0, \dots, x_{i-1}, x_{i+1}, \dots, x_n$ on U_i . On U_i , the inverse morphism has coordinates (for $j \neq i$)

$$x_j = \frac{x_j x_i^{d-1}}{x_i^d}$$

which is the quotient of coordinate functions where the denominator does not vanish, so F is an isomorphism. $\square$

This embeds $\mathbb{P}^n$ into a bigger space where the degree d conditions are translated into linear conditions.

Example 3.11

For $n = 1$, we have

$$F_{1,d} : \mathbb{P}^1 \rightarrow \mathbb{P}^d, (x_0 : x_1) \mapsto (x_0^d : x_0^{d-1}x_1 : \cdots : x_0x_1^{d-1} : x_1^d)$$

which is called the rational normal curve.

Proposition 3.31

The image of the Veronese embedding is the vanishing locus of the equations

$$X_I \cdot X_J = X_K \cdot X_L$$

where I, J, K, L are multiindices, X_α is the monomial $x_0^{\alpha_0} \cdots x_n^{\alpha_n}$, and $I + J = K + L$ as multiindices.

The above is a quadratic condition, so we see that any projective variety is isomorphic to one (possibly in higher dimension) cut out by quadratic conditions.

Corollary 3.32

Let $X \subseteq \mathbb{P}^n$ be a projective variety and $f \in S(X)$ be homogeneous and nonconstant. Then $X \setminus V(f)$ is an affine variety.

Proof. First, in the case $X = \mathbb{P}^n$ and f is a linear polynomial $f(x) = a_0x_0 + \cdots + a_nx_n$, we can extend $(a_0, \dots, a_n)$ to a basis of K^{n+1} , and write

$$A = \begin{bmatrix} a_0 & \cdots & a_n \\ * & \cdots & * \\ \vdots & \ddots & \vdots \\ * & \cdots & * \end{bmatrix}$$

so that

$$V_p(f) = \{x : (Ax)_0 = 0\} \cong \{x : x_0 = 0\}$$

where the isomorphism follows since A is a projective automorphism. So $\mathbb{P}^n \setminus V_p(f) \cong \mathbb{P}^n \setminus V_p(x_0) \cong \mathbb{A}^1$ is affine. In the general case, where f is of degree d , the Veronese embedding $F_{n,d}$ takes f to a linear polynomial $g = F_{n,d} \circ f$. Then $X \setminus V(f)$ is a closed subset of $\mathbb{P}^n \setminus V(f)$, which is isomorphic to $Y \setminus V(g)$, which is a closed subset of $\mathbb{P}^N \setminus V(g)$, which we just showed is affine. So we are done. $\square$

3.4 Grassmannians

We now offer an introduction to a generalization of projective space which will later enable us to perform blowup modifications on projective varieties – essentially, to “zoom in,” for instance by looking at the tangent space.

Definition 3.21

For $n > 0$ and $0 \leq k \leq n$, the **Grassmannian** $\text{Gr}(k, n)$ is the set of k -dimensional linear subspaces of K^{n+1} .

By definition, this generalizes projective space as $\text{Gr}(1, n) = \mathbb{P}^{n-1}$. We will show that the Grassmannian can be embedded as a projective variety inside $\mathbb{P}^N$ for some large value of N . To do this we will need to be able to write down coordinates which are unique up to scaling. If we take any $L \in \text{Gr}(k, n)$ and pick a basis $v_1, \dots, v_k$ of L , then the matrix

$$\begin{bmatrix} v_{1,1} & \cdots & v_{k,1} \\ \vdots & \ddots & \vdots \\ v_{1,n} & \cdots & v_{k,n} \end{bmatrix}$$

is unique up to column operations, which correspond to right multiplication by elements of $\text{GL}_k(K)$. For any maximal $k \times k$ minor of this matrix, right multiplication by an element $S \in \text{GL}_k(K)$ just serves to scale the minor by $\det(S) \neq 0$. Thus the maximal minors of this matrix are unique up to homogeneous scaling. Moreover they are not all zero since this matrix has rank k .

Definition 3.22

For a linear subspace $L \in \text{Gr}(k, n)$, the vector p_L of all $k \times k$ maximal minors are called the **Plücker coordinates** of L .

We can more generally write this without respect to a basis using the language of tensor products and alternating maps.

Definition 3.23

A multilinear map $f : V^k \rightarrow W$ for two K -vector spaces V, W and $k \in \mathbb{N}$ is called an **alternating map** if $f(v_1, \dots, v_k) = 0$ whenever $v_i = v_j$ for distinct i, j .

We remark that by linearity, and expanding out the quantity

$$f(v_1, \dots, v_i + v_j, \dots, v_i + v_j, \dots, v_k)$$

we recover the more familiar property that switching two arguments reverses the sign of f . More generally,

Proposition 3.33

If $f : V^k \rightarrow W$ is alternating and $\sigma \in S_k$ is a permutation, then

$$f(v_{\sigma(1)}, \dots, v_{\sigma(k)}) = \text{sign}(\sigma) f(v_1, \dots, v_k)$$

Definition 3.24

A k -fold **alternating tensor product** of a K -vector space V is a K -vector space T together with a k -fold alternating map $\tau : V^k \rightarrow T$ such that for any k -fold alternating map $f : V^k \rightarrow W$ there is a unique linear map $g : T \rightarrow W$ such that $f = g \circ \tau$:

$$\begin{array}{ccc} & & T \\ & \nearrow \tau & \downarrow \exists! g \\ V^k & \xrightarrow{f} & W \end{array}$$

As usual, alternating tensor products are unique up to unique isomorphism. Thus such a product is also denoted $T = \bigwedge^k V$ and $\tau(v_1, \dots, v_k) = v_1 \wedge \dots \wedge v_k$.

Proposition 3.34

For any V, k , an alternating tensor product exists.

Proof. We take T to be the k -fold tensor product $V \otimes \dots \otimes V$, quotiented by the linear subspace L which is generated by elements of the form $v_1 \wedge \dots \wedge v_k$ where $v_i = v_j$ for some $i \neq j$. $\square$

As with alternating maps, we have that

$$v_{\sigma(1)} \wedge \dots \wedge v_{\sigma(k)} = \text{sign}(\sigma) v_1 \wedge \dots \wedge v_k$$

Moreover, a basis for $\bigwedge^k V$ is given by vectors of the form

$$e_{i_1} \wedge \dots \wedge e_{i_k}$$

where each e_{i_j} is chosen from a basis $\{e_1, \dots, e_n\}$ of V , and each is distinct. Since permutation only performs scalar multiplication, we can also assume that $i_1 < \dots < i_k$, which then gives that

$$\dim \bigwedge^k V = \binom{n}{k}$$

Finally, we note that $\bigwedge^0 V \cong K$, $\bigwedge^1 V \cong V$, and by the dimension formula, $\dim \bigwedge^n K^n = 1$, so it is isomorphic to K . In particular, the determinant gives a canonical isomorphism.

Now, select vectors $v_1, \dots, v_k \in K^n$ for $0 \leq k \leq n$, and suppose they have coordinates $v_i = \sum_j a_{ij} e_j$ in some basis $\{e_j\}$. Let us determine how to write the coordinates of $v_1 \wedge \dots \wedge v_k$ in terms of the $e_{i_1} \wedge \dots \wedge e_{i_k}$. By multilinearity and regrouping so every term has increasing indices, we have

$$\begin{aligned} v_1 \wedge \dots \wedge v_k &= \sum_{i_1, \dots, i_k} a_{1i_1} \dots a_{ki_k} e_{i_1} \wedge \dots \wedge e_{i_k} \\ &= \sum_{i_1 < \dots < i_k} \left(\sum_{\sigma \in S_k} \text{sign}(\sigma) a_{1i_{\sigma(1)}} \dots a_{ki_{\sigma(k)}} \right) e_{i_1} \wedge \dots \wedge e_{i_k} \end{aligned}$$

This coefficient is precisely the Leibniz determinant formula for the $k \times k$ submatrix with rows $i_1, \dots, i_k$ selected from the following matrix:

$$\begin{bmatrix} a_{11} & \cdots & a_{k1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{kn} \end{bmatrix}$$

Thus we see that the coefficients of $v_1 \wedge \cdots \wedge v_k$ are precisely the maximal minors of the above matrix, which are the same as the Plücker coordinates we found earlier.

Lemma 3.35

$v_1 \wedge \cdots \wedge v_k = 0 \in \bigwedge^k V$ if and only if $v_1, \dots, v_k$ are linearly dependent in V .

Proof. $v_1, \dots, v_k$ are linearly dependent if and only if all $k \times k$ maximal minors are zero, which is equivalent to all coordinates of $v_1 \wedge \cdots \wedge v_k$ being zero, hence equivalent to $v_1 \wedge \cdots \wedge v_k$ being zero. $\square$

Lemma 3.36

If $v_1, \dots, v_k$ and $w_1, \dots, w_k$ are two sets of linearly independent vectors in K^n , then $v_1 \wedge \cdots \wedge v_k, w_1 \wedge \cdots \wedge w_k$ are linearly dependent in $\bigwedge^k V$ if and only if they span the same k dimensional linear subspaces of V .

Proof. We remark that by the previous lemma, since both sets are linearly independent, they define k dimensional subspaces and their wedge products are nonzero.

($\implies$) If

$$v_1 \wedge \cdots \wedge v_k = \lambda w_1 \wedge \cdots \wedge w_k$$

for some $\lambda \in K$, then taking the wedge product against w_i , we have

$$w_i \wedge v_1 \wedge \cdots \wedge v_k = 0$$

so $\{v_1, \dots, v_k, w_i\}$ is linearly dependent. Since $\{v_1, \dots, v_k\}$ is linearly independent, w_i lies in its span, which is true for all i , so

$$\text{span}(w_1, \dots, w_k) \subseteq \text{span}(v_1, \dots, v_k)$$

and the reverse is true as well.

($\impliedby$) If

$$\text{span}(w_1, \dots, w_k) \subseteq \text{span}(v_1, \dots, v_k)$$

then there is a matrix $B \in \text{GL}_k(K)$ which takes

$$w_i = \sum_j b_{ij} v_j$$

from which it follows that

$$w_1 \wedge \cdots \wedge w_k = \sum_{j_1, \dots, j_k} b_{1j_1} \cdots b_{kj_k} v_1 \wedge \cdots \wedge v_k = \det B v_1 \wedge \cdots \wedge v_k$$

by the same regrouping argument as above. $\det B \neq 0$, so we are done. $\square$

Definition 3.25

Let $0 \leq k \leq n$. The **Plücker embedding** is the map

$$f : \text{Gr}(k, n) \rightarrow \mathbb{P}^{\binom{n}{k}-1}$$

which takes $L \in \text{Gr}(k, n)$ spanned by $v_1, \dots, v_k$ to the coordinates of $v_1 \wedge \cdots \wedge v_k$ with respect to some basis $\{e_j\}$.

By the lemma above, these are independent of the choices of the v_i and the e_j up to homogeneous scaling. Also, since $v_1, \dots, v_k$ are linearly independent, they are not all zero. Finally, the lemma above also tells us that this map is injective.

Example 3.12

For $\text{Gr}(1, n)$, we have

$$f(\text{span}(a_1 e_1 + \cdots + a_n e_n)) = (a_1 : \cdots : a_n) \in \mathbb{P}^{n-1}$$

so $\text{Gr}(1, n) = \mathbb{P}^{n-1}$ as we expected.

Thus we have seen that we have a somewhat natural embedding of the Grassmannian into projective space, but this is only useful if we can show that it is open or closed. We will show that it is a closed subset of $\mathbb{P}^N$, which means the Plücker embedding defines the structure of a projective variety on $\text{Gr}(k, n)$. Here, we note that the image of a linear subspace under the Plücker embedding is given by the pure tensor

$$v_1 \wedge \cdots \wedge v_k$$

Conversely, we know that every such nonzero pure tensor has $v_1, \dots, v_k$ linearly independent, so their span is the preimage of this pure tensor. Therefore the Grassmannian, embedded in $\mathbb{P}^N$, is the set of pure tensors in $\bigwedge^k V$ (while $\bigwedge^k V$ in general consists of linear combinations thereof).

Lemma 3.37

Let $\omega \in \bigwedge^k K^n$ be nonzero with $k < n$, and consider the map

$$f : K^n \rightarrow \bigwedge^{k+1} K^n, v \mapsto v \wedge \omega$$

Then f has rank at least $n - k$, with equality if and only if ω is a pure tensor.

Proof. Pick a basis $v_1, \dots, v_r$ for $\ker f$ (where for dimension reasons we have $r = n - \text{rank } f$). Extend this to a basis $v_1, \dots, v_r$ of K^n . Then write

$$\omega = \sum_{i_1 < \dots < i_k} a_{i_1, \dots, i_k} v_{i_1} \wedge \dots \wedge v_{i_k}$$

Since the v_i , $1 \leq i \leq r$, are in the kernel of f , we have that

$$0 = v_i \wedge \omega = \sum_{i_1 < \dots < i_k} a_{i_1, \dots, i_k} v_i \wedge v_{i_1} \wedge \dots \wedge v_{i_k}$$

For each term, if $i \in \{i_1, \dots, i_k\}$ then the corresponding tensor for that term is zero. If $i \notin \{i_1, \dots, i_k\}$ then the tensor is nonzero, so since the LHS is zero, by linear independence we have that the coefficients $a_{i_1, \dots, i_k}$ are zero $i \notin \{i_1, \dots, i_k\}$. This is true for all $i \leq r$ so the only nonzero coefficients are those such that $\{1, \dots, r\} \subseteq \{i_1, \dots, i_k\}$. Since ω is nonzero there must be at least one nonzero coefficient, so we have $k \geq r$, which implies that $\text{rank}(f) \geq n - k$. Moreover, if there is equality then only the coefficient $a_{1, \dots, k}$ can be nonzero so this is a pure tensor.

Conversely, if ω is a pure tensor $\omega_1 \wedge \dots \wedge \omega_k$ then each ω_j is in the kernel of f (and they are linearly independent), so we have

$$\dim \ker(f) \geq k \implies \text{rank}(f) \leq n - k$$

so we have both inequalities and thus $\text{rank}(f) = n - k$. □

Corollary 3.38

Under the Plücker embedding, $\text{Gr}(k, n)$ is a closed subset of $\mathbb{P}^{\binom{n}{k}-1}$.

Proof. The statement is true for $n = k$ as the Grassmannian is a single point. So assume $n < k$, as we need this for the previous result. As we discussed above, our previous results have shown that the image of the Plücker embedding is the set of all (nonzero) pure tensors in $\bigwedge^k K^n$ (up to equivalence classes by homogeneous scaling).

Then, the previous lemma says that this condition is equivalent to the map $f : v \mapsto v \wedge \omega$ having rank $n - k$. Since the rank is always at least $n - k$, it is enough to have the condition $\text{rank}(f) \leq n - k$. This is the condition that all $n - k + 1$ minors of the matrix of f vanish. This is a polynomial in the entries of the matrix of f , which in turn can be written as polynomials in the Plücker coordinates of ω . Therefore $\text{Gr}(k, n)$ can be defined by polynomial vanishing conditions, and is a closed subset. □

The Plücker embedding indirectly gives an affine cover of $\text{Gr}(k, n)$, by considering the affine covers of $\mathbb{P}^{\binom{n}{k}-1}$. We can explicitly understand what these patches look like.

Proposition 3.39

$\text{Gr}(k, n)$ admits an open cover by spaces isomorphic to $\mathbb{A}^{(n-k) \times k}$.

Proof. Without loss of generality, we will consider the patch

$$U_0 = \left(\mathbb{P}^{\binom{n}{k}-1} \setminus V_p(x_0) \right) \cap \text{Gr}(k, n)$$

This is the subset of $\text{Gr}(k, n)$ where the coordinate of

$$e_1 \wedge \cdots \wedge e_k$$

is nonzero. In other words, based on our descriptions of the elements of $\text{Gr}(k, n)$ as the column spans of matrices with the correct maximal minors, we know that any element of U_0 is represented as the column span of a block matrix of the form

$$\begin{bmatrix} A \\ B \end{bmatrix}$$

where A is $k \times k$ and B is $(n-k) \times k$. The coordinate of $e_1 \wedge \cdots \wedge e_k$ is the determinant of A , so A is invertible. In particular, since the column span is invariant under right multiplication by elements of $\text{GL}_k(K)$, we know that this is the same element as the one represented by

$$\begin{bmatrix} I_k \\ BA^{-1} \end{bmatrix}$$

Since the I_k block gives a nonzero coordinate, the matrix B may be chosen freely. Thus, U_0 is parameterized by

$$U_0 = \left\{ \text{ColumnSpan} \begin{bmatrix} I_k \\ C \end{bmatrix} : C \in \mathbb{A}^{(n-k) \times k} \right\}$$

Moreover, the maximal minors of

$$\begin{bmatrix} I_k \\ C \end{bmatrix}$$

are polynomials in the entries of C , so the map

$$f : \mathbb{A}^{(n-k) \times k} \rightarrow U_0, C \mapsto \begin{bmatrix} I_k \\ C \end{bmatrix}$$

is a morphism. Moreover, the minor of

$$\begin{bmatrix} I_k \\ C \end{bmatrix}$$

given by choosing all the rows of I_k except the i th, and the j th row of C , is exactly C_{ij} (up to sign, and we can permute the rows based on the parities of i, j to make sure this works out). Thus the map is also injective, and the inverse map is a projection since the C_{ij} are contained as Plücker coordinates of the block matrix. Therefore U_0 is isomorphic to $\mathbb{A}^{(n-k) \times k}$, and $\text{Gr}(k, n)$ is covered by copies of $\mathbb{A}^{(n-k) \times k}$. $\square$

Corollary 3.40

$\text{Gr}(k, n)$ is irreducible and of dimension $(n - k)k$.

Proof. Any two of the patches has nonempty intersection, and in particular there is a matrix which lies in all the patches. Each patch is irreducible of dimension $(n - k)k$, so for topological reasons it follows that $\text{Gr}(k, n)$ is of dimension $(n - k)k$ and is irreducible as well. $\square$

Proposition 3.41

For $0 \leq k \leq n$, $\text{Gr}(k, n) \cong \text{Gr}(n - k, n)$.

Proof. The set theoretic map $L \mapsto L^\perp$ is a bijection between $\text{Gr}(k, n)$ and $\text{Gr}(n - k, n)$. Also, the inverse map is also $\tilde{L} \mapsto \tilde{L}^\perp$, so it suffices to show that the forward map is a morphism. We can show this on an affine patch, so consider the patch where L is parameterized by

$$\begin{bmatrix} I_k \\ C \end{bmatrix}$$

Note that we have

$$\begin{bmatrix} -C & I_{n-k} \end{bmatrix} \begin{bmatrix} I_k \\ C \end{bmatrix} = 0$$

as a matrix, so the row span of $\begin{bmatrix} -C \\ I_{n-k} \end{bmatrix}$ is orthogonal to L . Since the row span is linearly independent (as there is a nonzero coordinate given by the I_{n-k} minor), we get that the row span lies in $L^\perp$, and for dimension reasons $L^\perp$ is precisely this row span. Moreover, the Plücker coordinates of $L^\perp$ are the maximal minors of $\begin{bmatrix} -C & I_{n-k} \end{bmatrix}$ which are polynomials in the entries of C , so this is a morphism and the desired isomorphism is found. $\square$

Chapter 4

Birational Geometry

4.1 Birational Maps

In birational geometry we wish to study the ways that varieties are related to each other, in particular those which look similar on a dense open set. For instance, we know that $\mathbb{P}^n \times \mathbb{P}^m$ is not isomorphic to $\mathbb{P}^{n+m}$, but they both contain a dense open subset $\mathbb{A}^n \times \mathbb{A}^m \cong \mathbb{A}^{n+m}$. Or, we know that $V(y^2 - x^3)$ is not isomorphic to $\mathbb{A}^1$ itself, but removing the origin we do have

$$V(y^2 - x^3) \setminus \{(0, 0)\} \cong \mathbb{A}^1 \setminus \{0\}$$

via the map $t \mapsto (t^2, t^3)$, $(x, y) \mapsto y/x$. We will first describe this kind of classification and then make a construction that allows us to study these types of relationships.

Definition 4.1

A **rational map** between irreducible varieties X, Y is a morphism $f : U \rightarrow Y$ where U is a nonempty open subset of X . Such a rational map is denoted $f : X \dashrightarrow Y$. Two rational maps $f_1, f_2 : X \dashrightarrow Y$ defined on U_1, U_2 are equivalent if there is U nonempty contained in $U_1 \cap U_2$ where f_1, f_2 agree as morphisms.

We note that if two such maps are equivalent, then since U is dense in $U_1 \cap U_2$, and the locus where $f_1 = f_2$ agree is closed in $U_1 \cap U_2$, it always suffices to check that $f_1|_{U_1 \cap U_2} = f_2|_{U_1 \cap U_2}$.

Unfortunately, as defined, two rational maps need not have a well defined composition. For instance, if $f : \mathbb{A}^1 \rightarrow \mathbb{A}^1, x \mapsto 0$ and $g : \mathbb{A}^1 \dashrightarrow \mathbb{A}^1, x \mapsto 1/x$, then g can never be defined on the image of f , so we cannot compose $g \circ f$.

Definition 4.2

A rational map $f : X \dashrightarrow Y$ with X, Y irreducible is **dominant** if $\text{im } f$ is dense in Y .

Note that if $f : X \dashrightarrow Y$ is dominant and $g : Y \dashrightarrow Z$ is a rational map defined on $V \subseteq Y$, then $g \circ f : X \dashrightarrow Z$ is defined on $f^{-1}(V)$, which is nonempty since f is dominant.

Definition 4.3

A rational map $f : X \dashrightarrow Y$ with X, Y irreducible is a **birational map** if f is dominant and there is $g : Y \dashrightarrow X$ dominant with $g \circ f = \text{id}_X, f \circ g = \text{id}_Y$ under equivalence of rational maps. In this case X, Y are said to be birational.

If we restrict our attention to dominant maps, then composition is always well defined and the composition is also dominant. Thus, the category of irreducible varieties with morphisms given by dominant rational maps forms a category $\text{Var}_K^{\text{dom}}$. Birationality gives us a simple condition to check that two varieties have isomorphic subsets.

Lemma 4.1

Two irreducible varieties X, Y are birational if and only if they contain isomorphic nonempty open subsets.

Proof. ($\Leftarrow$) If $U \subseteq X, V \subseteq Y$ are isomorphic via $f : U \rightarrow V, f^{-1} : V \rightarrow U$, then f, f^{-1} are both rational maps by definition. They are also dominant as the images U, V are dense by assumption. Hence X, Y are birational.

($\Rightarrow$) If $f : X \dashrightarrow Y$ is defined on $U \subseteq X$ and $g = f^{-1} : Y \dashrightarrow X$ is defined on $V \subseteq Y$, then both compositions are defined on

$$f^{-1}(g^{-1}(U)), g^{-1}(f^{-1}(V))$$

respectively, so these two sets are isomorphic (and nonempty open, hence dense). $\square$

In particular, this means that birational irreducible varieties are the same dimension.

Definition 4.4

A **rational function** on X an irreducible variety is a rational map $X \dashrightarrow \mathbb{A}^1 = K$. The set of all rational functions on X (up to equivalence) is denoted $K(X)$ and called the **function field** of X .

A few remarks on this definition:

- In particular, a rational function can be seen as a regular function on the nonempty open subset on which it is defined.
- Moreover, if $\varphi_1, \varphi_2 \in K(X)$, then on the intersection of their domains (which is nonempty), $\varphi_1 + \varphi_2$ and $\varphi_1 \varphi_2$ are both well defined. If φ is not zero everywhere, then we can also define an inverse on $\varphi^{-1}(K^*)$. Thus $K(X)$ forms a field, which justifies calling it the function field.

- If $U \subseteq X$ is a nonempty open subset of an irreducible variety, then any rational function on U is also a rational function on X , and any rational function on X restricts to a rational function on U . Moreover, because rational functions only need to agree on a nonempty open subset to be equal, this restriction is injective. Thus, $K(U) \cong K(X)$.
- As a consequence of the above, birational irreducible varieties have isomorphic function fields.
- In the sense of sheaves, $K(X)$ is the stalk $\mathcal{O}_X(X)$. From our study of affine stalks, if X is affine then $K(X) \cong A(X)_{(0)}$, which is precisely the fraction field of $A(X)$.
- $K(X)$ is a finitely generated field extension of K , and more generally $X \mapsto K(X)$ is a contravariant equivalence of classes from $\text{Var}_K^{\text{dom}}$ to finitely generated field extensions of K .

4.2 Blowing Up

Now, we will study a particular class of rational maps that will give a richer analysis of varieties. We will begin by making this definition on affine varieties and then extend to the general case by gluing.

Definition 4.5

Let $X \subseteq \mathbb{A}^n$ be an affine variety and let $f_1, \dots, f_r \in A(X)$. Setting $U = X \setminus V_X(f_1, \dots, f_r)$, the map

$$U \ni x \mapsto f(x) = (f_1(x) : \dots : f_r(x)) \in \mathbb{P}^{r-1}$$

is a well defined morphism. Let

$$\Gamma_f = \{(x, f(x)) : x \in U\} \subseteq X \times \mathbb{P}^{r-1}$$

be the graph of f . Then the **blowup** of X at $f_1, \dots, f_r$ is $\tilde{X} = \overline{\Gamma_f}$, the closure of Γ_f in $X \times \mathbb{P}^{r-1}$.

Note that there is an obvious projection $\tilde{X} \rightarrow X$ by projecting on the first factor. In this case, U is isomorphic with its preimage, so $\tilde{X}$ is birational with X so long as X is irreducible and the f_i are not all zero. In other words, above U , $\tilde{X}$ looks like Γ_f . On the other hand, the point of the blowup is that it replaces the remaining vanishing locus by a set with richer structure. This is $\tilde{X} \setminus U$ (identified via the projection morphism), which we call the **exceptional set** of $\tilde{X}$.

Definition 4.6

If $Y \subseteq X$ is a closed subvariety then the restrictions $f_1, \dots, f_r \in A(Y)$ define a blowup $\tilde{Y} \subseteq Y \times \mathbb{P}^{r-1} \subseteq X \times \mathbb{P}^{r-1}$. Since Y is closed, $\tilde{Y}$ is closed in $\tilde{X}$, and this is called the **strict transform** of Y in $\tilde{X}$.

This gives us a way to think about the blowup of subvarieties in a way that they remain embedded, and allows us to track where subvarieties get moved around to via the blowup, which we will see gives useful geometric intuition. We also remark that the strict transform is the closure of $Y \cap U$ in $\tilde{X}$. Also, if X has irreducible decomposition $X_1 \cup \dots \cup X_n$, then $\tilde{X} = \tilde{X}_1 \cup \dots \cup \tilde{X}_n$.

Example 4.1

If $r = 1$, then $\tilde{X}$ lies in $X \times \mathbb{P}^0 = X$. Thus, if f is nonzero and X is irreducible, $\tilde{X} = X$. More generally, $\tilde{X}$ becomes the closure of U , and if $f \equiv 0$ then this is the empty set (otherwise it is the union of the irreducible components which intersect U).

Of course, working with closures is unwieldy, so now we work to figure out what these blowups look like.

Lemma 4.2

Let X be an affine variety and $f_1, \dots, f_r \in A(X)$. Then $\tilde{X}$ lies in

$$\tilde{X} \subseteq \{(x, y) \in X \times \mathbb{P}^{r-1} : y_i f_j(x) = y_j f_i(x) \forall i, j\}$$

In particular, the RHS is a closed subvariety of $X \times \mathbb{P}^{r-1}$.

Proof. The equality obviously holds on U , since here we have $y = f(x)$. Since this is an equality of homogeneous polynomials of the same degree on a variety, this is true everywhere on $\tilde{X}$. $\square$

Example 4.2

For the blowup of $\mathbb{A}^n$ at the coordinate functions $x_1, \dots, x_n$, we actually have equality

$$\tilde{\mathbb{A}}^n = Z := \{(x, y) \in X \times \mathbb{P}^{r-1} : y_i f_j(x) = y_j f_i(x) \forall i, j\}$$

To see this, we will first do this formally. Consider the coordinate patch $U_1 = \{(x, y) \in \tilde{\mathbb{A}}^n : y_1 = 1\} \subseteq Z$. On this patch, the previous lemma tells us that for fixed $x_1, y_2, \dots, y_j \in K$, we must have $x_j = x_1 y_j$. Therefore there is a map

$$\mathbb{A}^n \rightarrow U_1 : (x_1, y_2, \dots, y_j) \mapsto (x_1, x_1 y_2, \dots, x_1 y_n) \times (y_1, \dots, y_j)$$

This is an isomorphism, so Z is covered by the irreducible affine patches U_j . These all intersect, so Z is irreducible of dimension n . But it contains the closed subvariety $\tilde{\mathbb{A}}^n$, which is also of dimension n , so we have $\tilde{\mathbb{A}}^n = Z$.

Thinking about this geometrically, this tells us that after picking any $y \in \mathbb{P}^{n-1}$, we get a curve in Z parameterized by x_1 , which is defined including when $x_1 = 0$. This can be seen as the strict transform of the line $\{x_1 \cdot (1, y_2, \dots, y_n) : x_1 \in \mathbb{A}^1\}$ in $\tilde{\mathbb{A}}^n$.

We note that in particular, the point $(0, \dots, 0) \in \mathbb{A}^n$ has a preimage isomorphic to $\mathbb{P}^{r-1}$, which is seen to intersect each of these curves at different points. Thus, we can see the exceptional set as parameterizing the different directions in which we may approach the origin.

The description of the above is very geometric, and the notion of a blowup is really to study the geometry of X at a ‘zoomed-in’ level. We will show that the blowup only depends on an ideal, and not the specific functions (which is more geometric, though it is not 1-1 with subvarieties since we will see that taking the radical *does* change the blowup).

Lemma 4.3

Let X be an affine variety and let $f_1, \dots, f_r, f'_1, \dots, f'_s \in A(X)$ be two sets of polynomials. Then if $\langle f_1, \dots, f_r \rangle = \langle f'_1, \dots, f'_s \rangle$, there is an isomorphism F so that the following commutes:

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{F \cong} & \tilde{X}' \\ & \searrow \pi & \swarrow \pi' \\ & X & \end{array}$$

In this sense, the blowup is only determined by the ideal generated by the functions.

Proof. If the ideals coincide, then there are $g_{i,k}, h_{j,k} \in A(X)$ so that

$$f_i = \sum_k g_{i,k} f'_k, \quad f'_j = \sum_k h_{j,k} f_k$$

It follows that

$$f_i = \sum_j g_{i,j} \sum_k h_{j,k} f_k$$

We then construct F via the definition

$$\tilde{X} \ni (x, y) \mapsto \left(x, \left(\sum_k h_{1,k}(x) y_k : \dots : \sum_k h_{s,k}(x) y_k \right) \right) \in \tilde{X}'$$

Then this clearly commutes with projections to the first coordinate. To see that it is well defined, we note that on U , we have $y_j = f_j(x)$, so we have

$$y_i = \sum_j g_{i,j} \sum_k h_{j,k} y_k$$

and therefore this holds on all of $\tilde{X}$. So in particular, if

$$\sum_k h_{j,k} y_k = 0$$

for all k simultaneously at some point (x, y) , then it follows that y is zero, which is impossible. Therefore, our map F is a well defined morphism into $X \times \mathbb{P}^{r-1}$. To see that its image lies in $\tilde{X}'$, we note that by construction, on U this restricts to

$$(x, y) \mapsto \left(x, \left(\sum_k h_{1,k}(x) f_k(x) : \cdots : \sum_k h_{s,k}(x) f_k(x) \right) \right) = (x, y')$$

So $F(U)$ lies in $\tilde{X}'$, and therefore its closure $F(\tilde{X})$ does as well. Taking the reverse construction gives the inverse morphism. $\square$

Definition 4.7

If X is an affine variety and $J \subseteq A(X)$ is an ideal then the blowup of X at J is the blowup of X at any finite set of generators of J (defined up to isomorphism). If $Y \subseteq X$ is a closed affine subvariety then the blowup of X at Y is the blowup of X at $I(Y)$.

If Y is just a point a , then we can take the blowup of an affine neighborhood U containing a , and glue the blowup of U to $X \setminus \{a\}$ along $U \setminus \{a\}$.

We note that it is not the case that the blowup at J depends only on $\sqrt{J}$. For instance, the blowup of $\mathbb{A}^2$ at $\langle x^2, y \rangle$ is distinct from the blowup at $\langle x, y \rangle$ (the first effective parameterizes parabola approaches).

Definition 4.8

Let X be a variety covered by the affine patches U_i . Let $Y \subseteq X$ be a closed subvariety. Then the blowup of X at Y is the blowup of each U_i at $Y \cap U_i$, glued along the blowups of $U_i \cap U_j$ at $Y \cap U_i \cap U_j$.

We remark briefly that so far, we have only defined the general blowup of a variety at a subvariety, and not along an ideal. We will not make this definition in this course, as it requires gluing ideals together using a sheaf of ideals.

However, we *can* make this definition for projective varieties (recall that this covers most examples in practice) by making the same definition as above: for $f_1, \dots, f_r \in S(X)$ homogeneous of the same degree, the blowup is the closure of

$$\{(x, f(x)) : x \in X\} \subseteq X \times \mathbb{P}^{r-1}$$

which is a projective variety.

This local study of varieties gives us a way to look at tangents to varieties.

Proposition 4.4

Let X be a variety, let $a \in X$ be a point, and let $\tilde{X}$ be the blowup of X at a . Then the exceptional set $\pi^{-1}(a)$ is a projective variety.

Proof. We remarked that for blowing up at a point, this can be seen to be given by taking an affine neighborhood U , and then we have

$$\tilde{X} = (X \setminus \{a\}) \cup_{U \setminus \{a\}} \tilde{U}$$

where the union denotes gluing along $U \setminus \{a\}$. Thus the exceptional set is a closed set contained in the blowup $\tilde{U} \subseteq U \times \mathbb{P}^{n-1}$, and is therefore a projective subvariety. $\square$

Actually, the exceptional set is contained in $\{a\} \times \mathbb{P}^{n-1}$. Since it is a projective variety we can then take an affine cone, lying in $\mathbb{A}^n$ (here $\mathbb{A}^n$ is the same space U lies in, hence n is the local codimension of X at a).

Definition 4.9

Let X be a variety, let $a \in X$ be a point, and let $\tilde{X}$ be the blowup of X at a . Then the **tangent cone** of X at a is the affine cone $C_a X = C(\pi^{-1}(a)) \subseteq \mathbb{A}^n$ over the exceptional set.

In the case that $a = 0$ and X is affine, we can canonically embed $C_0 X$ into the same affine space that X lies in.

Example 4.3

Let $X_1 = V(x_2 + x_1^2) \subseteq \mathbb{A}^2$ (over $\mathbb{R}$, a downward parabola). Then the blowup of X_1 at the origin is the strict transform inside the blowup of $\mathbb{A}^2$. In particular, the exceptional set of $\tilde{X}_1$ is the intersection of $\tilde{X}_1$ with the exceptional set of $\tilde{\mathbb{A}}^2$. Remembering that this is a copy of $\mathbb{P}^1$, our intersection will be some number of points, and then the cone over these points will be precisely the lines going through them. This gives geometric meaning to the tangent cone.

Example 4.4

Let $X_2 = V(x_2^2 - x_1^2 - x_1^3) \subseteq \mathbb{A}^2$ and consider the blowup at the origin. On the blowup of $\mathbb{A}^2$ we have

$$x_i y_j = x_j y_i$$

Therefore away from the exceptional set, we can fix a particular scaling of the homogeneous coordinates, hence giving $\lambda \neq 0$ so that $\lambda x = y$. On the graph of X_2 we have

$$x_2^2 - x_1^2 - x_1^3 = 0$$

So therefore for this choice of λ we have

$$\lambda^2(x_2^2 - x_1^2 - x_1^3) = 0 \implies y_2^2 - y_1^2 - y_1^2 x_1 = 0$$

We choose λ pointwise away from the exceptional set, but the RHS equality holds on the entire blowup away from the exceptional set, and hence on the exceptional set as well. In particular, on the exceptional set we have $x_1 = 0$ so we get

$$y_2^2 - y_1^2 = 0 \implies \pi^{-1}(0) = \{(0, (1 : 1)), (0, (1 : -1))\}$$

Observe that the λ argument had the effect of discarding all but the lowest order terms. Intuitively this is because the higher order terms are negligible, for instance when working over $\mathbb{C}$.

Proposition 4.5

Let $X = V(J) \subseteq \mathbb{A}^n$ be an affine variety containing the origin. Then the blowup of X at the origin is

$$V(\{f^{in} : f \in J\})$$

where f^{in} is the lowest order terms of f .

Note that it does not suffice to take the initial terms of a generating set, except in the case $J = \langle f \rangle$ is principal.

Proof. □

For $\mathbb{A}^n$, we saw that the blowup was irreducible of dimension n , and the exceptional set $\mathbb{P}^{n-1}$ was a subvariety of dimension $n - 1$. This relationship holds in general.

Proposition 4.6

Let X be affine and $f_1, \dots, f_r \in A(X)$, and let $\tilde{X}$ be the blowup at $f_1, \dots, f_r$. Then every component of the exceptional set has codimension 1 in $\tilde{X}$.

In this case the exceptional set is also called the exceptional hypersurface.

Proof. It suffices to prove this on each affine cover $\tilde{X} \cap (X \times U_i)$, where $U_i = \{y_i \neq 0\}$. On this set, the condition

$$f_i y_j = f_j y_i$$

implies that whenever $f_i = 0$, all the f_j vanish simultaneously. Therefore on this cover, $\pi^{-1}(V(f_1, \dots, f_r)) \cap (X \times U_i) = \pi^{-1}(f_i) \cap (X \times U_i)$. Whenever $\tilde{X} \cap (X \times U_i)$ is nonempty, f_i cannot vanish on it, since then $\tilde{X} \cap (X \times U_i)$ would be contained in the empty set, and so would the closure, which is precisely $\tilde{X}$. But in this case the open set U which is the complement of the blowup set would be empty, but this implies that X itself is empty as well. Thus $\tilde{X} \cap (X \times U_i)$ is cut out by a single equation, and thus on this set each component of the exceptional set has codimension 1, and the full statement then follows by taking the cover. □

Since taking the tangent cone increases the dimension by 1, we have the following statement we informally mentioned earlier:

Proposition 4.7

If X is a variety and $a \in X$ then the tangent cone $C_a X$ has dimension $\text{codim}_X \{a\}$.

Recall that the codimension about a point is the maximum of the dimensions of the irreducible components containing that point.

Proof. We can assume every irreducible component contains a , and that X is not $\{a\}$ itself. Up to taking an affine cover, we can also assume X is affine. Say X has irreducible decomposition $X = X_1 \cup \dots \cup X_n$. Then $\dim C_a X$ is the maximum of $\dim C_a X_i$. Here, the exceptional set of the blowup of X_i is of pure dimension $\dim X_i - 1$, so its cone has dimension $\dim C_a X_i = \dim X_i$. Taking maxima, we get $\dim C_a X = \text{codim}_X \{a\}$. $\square$

Note in the above we had to assume that the exceptional set was nonempty after intersecting with each X_i . This is false if we blow up a point, but true for $\dim X_i \geq 1$, but of course no irreducible component can be only $\{a\}$ as long as X is not $\{a\}$.

As we mentioned earlier, the exceptional set of the blowup allows one to identify the direction from which the zero locus is approached. This gives a way in which we can take an equivalent of directional limits of indeterminate forms. That is, for a birational map $f : X \dashrightarrow \mathbb{P}^{r-1}$ with X affine, in general the maximal domain of $f = (f_1 : \dots : f_r)$ is $X \setminus V(f_1, \dots, f_r)$, as on the zero locus the form is indeterminate. This can be resolved by blowing up X along this locus and then extending $f : \tilde{X} \rightarrow \mathbb{P}^{r-1}$. The extension happens by projecting onto the projective coordinate, since away from the exceptional set the projective coordinate is always $f(x)$.

As a specific example, $\mathbb{P}^1 \times \mathbb{P}^1$ is birational with $\mathbb{P}^2$ since they share open subsets, hence there is some birational mapping between them. Resolving the indeterminacies of this map gives an isomorphism between blowups.

Proposition 4.8

$\mathbb{P}^1 \times \mathbb{P}^1$ blown up in one point is isomorphic to $\mathbb{P}^2$ blown up in two points.

Proof. By the Segre embedding,

$$\mathbb{P}^1 \times \mathbb{P}^1 \cong \{(x_0 : x_1 : x_2 : x_3) \subseteq \mathbb{P}^3 : x_0 x_3 = x_1 x_2\}$$

This defines a hypersurface, and we saw before that projecting from a point (in particular, $(0 : 0 : 0 : 1)$, where the projection just forgets the last coordinate) gives a rational map to $\mathbb{P}^2$. However, this is not a priori defined at the point $(0 : 0 : 0 : 1)$. After blowing up, we get a birational map $\tilde{X} \dashrightarrow \mathbb{P}^2$. The inverse birational map has two points of indeterminacies, so blowing up these ends up resolving the projection into an isomorphism. We will not show this formally. $\square$

4.3 Smooth Varieties

As we have seen, blowing up gives us a tool to study varieties locally. In particular, we can formalize notions such as tangents, and we will see now that we can also give meaning to the intuitive notion of singularities such as cusps.

To start, we will begin with defining the tangent space. Recall that we defined the tangent cone, but this space is harder to work with since it is not a linear space.

Definition 4.10

Let X be a variety and $a \in X$. Up to choosing an affine neighborhood and translation, assume X is affine and $a = 0$ (we will show the definition is not dependent on this choice). The **tangent space** of X at a is the set

$$T_a X = V(f_1 : f \in I(X)) \subseteq \mathbb{A}^n$$

where f_1 is the linear terms of f .

Since we assume the origin is in X , every $f \in I(X)$ has zero constant term. For instance, if $X = V(y - x^2 - 2x)$, we can see that the tangent space is

$$T_a X = V(y - 2x)$$

Proposition 4.9

If S is a generating set for $I(X)$ then $T_a X = V(f_1 : f \in S)$.

Proof. Let $f, g \in S$ and $x \in T_a(X)$. Then $(f + g)_1 = f_1 + g_1$ so this vanishes on $T_a X$. If h is any polynomial, then

$$(hf)_1 = hf_1 + f_1 h$$

and since $f(x) = h(x) = 0$, $(hf)_1$ vanishes on $T_a X$. Thus $V(f_1 : f \in S) \subseteq V(f_1 : f \in I(X))$, and the other inclusion follows since the right hand side is cut out by more conditions. $\square$

However, note that as with the blowup, it is important that we take $I(X)$ and not simply any ideal with radical $I(X)$. For instance, $V(x) = V(x^2)$ but the linear term on the left cuts out $V(X)$ while the linear term on the right cuts out $\mathbb{A}^1$. Finally, note that since $C_a X = V(f^{in} : f \in I(X))$, and $f_1 = f^{in}$ whenever f_1 is nonzero, so C_a is cut out by more conditions and therefore $C_a X \subseteq T_a X$. In particular,

$$\dim T_a X \geq \text{codim}_X \{a\}$$

We know $C_a(X)$ is a union of affine lines, so we might be led to the guess that $T_a X$ is in some sense the linear space spanned by $C_a X$. This will turn out not to always be true. For instance, for

$$X = V(x_2^2 - x_1^3) \subseteq \mathbb{A}^2$$

we know by taking initial terms that

$$C_a X = V(x_2^2) = V(x_2)$$

which is itself a linear space, but this has zero linear term so

$$T_a X = V(0) = \mathbb{A}^2$$

However, X here contains a cusp at the origin, which gives us some sense for why this inequality might occur.

Now, we will show that this construction can be treated in a more intrinsic sense.

Lemma 4.10

Let X be an affine variety in $\mathbb{A}^n$ and assume $a = 0 \in X$. Then $I_X(a) = \langle \overline{x_1}, \dots, \overline{x_n} \rangle \subseteq A(X)$. Then

$$\mathrm{Hom}_K(T_a X, K) \cong I(a)/I(a)^2$$

via a natural isomorphism of K vector spaces. In particular, $T_a X$ is isomorphic to the dual space of $I(a)/I(a)^2$.

Effectively, $I(a)^2$ contains the polynomials starting at degree 2.

Proof. We write down the K linear map $\varphi : I(a) \rightarrow \mathrm{Hom}_K(T_a X, K)$ by

$$\overline{f} \mapsto f_1|_{T_a X}$$

This is well defined since if we choose another representative $\overline{f} = \overline{f} + \overline{h}$ for some $h \in I(X)$, then $h_1|_{T_a X} = 0$ so this is well defined. It is also surjective because any element of $\mathrm{Hom}_K(T_a X, K)$ is a linear polynomial in $x_1, \dots, x_n$, and can hence be extended to a map on all of $\mathbb{A}^n$ with no constant term. So by the isomorphism theorem we want to show that $\ker \varphi = I(a)^2$.

($\supseteq$) If $\overline{f}, \overline{g} \in I(a)$ then $(fg)_1 = f(0)g_1 + g(0)f_1 = 0$. So any element of $I(a)^2$ vanishes under φ .

($\subseteq$) If we define $W = \{g_1 : g \in I(X)\}$, then this is of some degree k within $\mathrm{span}(x_1, \dots, x_n)$. Denote this span by V , which we can see as the dual space $(K^n)^*$. Then the zero locus of W is precisely $T_a X$, which is hence of dimension $n - k$. So $(T_a X)^\perp = \{h \in V : h|_{T_a X} = 0\}$ is a linear space of dimension k which contains W , hence $(T_a X)^\perp = W$. Now, if $\overline{f} \in \ker \varphi$, then $f_1 \in (T_a X)^\perp = W$. Therefore by the definition of W , there is some $g \in I(X)$ with $g_1 = f_1$. So $\overline{f} - \overline{g}$ vanishes at the origin and also has zero linear term. Since g vanishes on X , $\overline{f} = \overline{f} - \overline{g}$, and since $f - g$ has no constant or linear term, $\overline{f} - \overline{g} \in I(a)^2$. $\square$

Finally, to remove the dependence on an affine neighborhood, we move to an even more intrinsic description that agrees with our most general definition of a variety. Recall that the local ring $\mathcal{O}_{X,a} = S^{-1}A(X)$ where $S = A(X) \setminus I(a)$, and $\mathcal{O}_{X,a}$ is a local ring with the maximal ideal

$$I_a = S^{-1}I(a) = \left\{ \frac{g}{f} : g, f \in A(X), g(a) = 0, f(a) \neq 0 \right\}$$

Corollary 4.11

In the setting above,

$$I(a)/I(a)^2 \cong S^{-1}I(a)/(S^{-1}I(a))^2$$

Therefore the tangent space of any variety at any point is always isomorphic to the dual of $I(a)/I(a)^2$ (as a vector space).

This justifies why our choice of shifting to the origin and taking affine neighborhoods did not lose any generality.

Proof. We claim that multiplying by an element of S is invertible on $I(a)/I(a)^2$. Under the claim, it follows that localizing is equivalent to multiplication, and hence already contained in $I(a)/I(a)^2$. Therefore localizing does nothing:

$$I(a)/I(a)^2 = S^{-1} \left(I(a)/I(a)^2 \right)$$

It is a theorem from commutative algebra that localization commutes with quotients, so this becomes

$$S^{-1}I(a)/S^{-1}I(a)^2$$

Thus we merely prove the claim. Take $f \in S$, so that $f(a) \neq 0$. Then $\bar{f} \in A(X)/I(a)$ is the class corresponding to the constant term of f , which is nonzero and hence there is $\bar{c} \in A(X)/I(a)$ so that $\bar{f}\bar{c} = 1$ (we can take c to be the constant function $1/f(a)$). Then for any element of $S^{-1}(I(a)/I(a)^2)$, which by definition has the form g/f for $g \in I(a)/I(a)^2$ and $f \in S$ nonzero at a , we get

$$\frac{g}{f} = \frac{cg}{1} \in S^{-1} \left(I(a)/I(a)^2 \right)$$

but $cg \in I(a)/I(a)^2$. Thus as we claimed above, every element of $S^{-1}(I(a)/I(a)^2)$ is already an element of $I(a)/I(a)^2$. $\square$

Example 4.5

For any morphism of varieties $f : X \rightarrow Y$, and a point $a \in X$, this description gives a linear map $T_f : T_a X \rightarrow T_{f(a)} Y$.

As we remarked above, $C_a X \subseteq T_a X$ always, but there is not necessarily equality. However, when there is, we essentially say that X is nice enough to be approximated linearly around a .

Definition 4.11

For X a variety, a point $a \in X$ is called a **smooth point** (or **regular point** or **nonsingular point**) if $T_a X = C_a X$. Otherwise it is a **singular point**, and

Chapter 5

Schemes

In the preceding chapters, we have established a correspondence between affine varieties and reduced finitely generated K -algebras, as well as exploring the geometry of glued affine varieties, particularly projective ones. In modern algebraic geometry, one generalizes varieties further by studying schemes. These allow geometric results to be transferred to settings of more general rings than simply finitely generated K -algebras. In particular, there is no reference explicitly made to a ground field.

5.1 Affine Schemes as Topological Spaces

Analogous to the study of varieties, we will first define the simpler class of *affine* schemes. Initially, we will define them as sets, then give them a topology, then add a ringed space structure. Once we have done this, we will be able to use the ringed space theory from before to define gluing of affine schemes, and look at general schemes as the resulting spaces. Along the way, we will see that we will need to refine our definition of morphisms of ringed spaces, since the rings will no longer be functions to a common base field.

Definition 5.1

Let R be a ring. Then we define the **spectrum** or **affine scheme** of R to be

$$\operatorname{Spec} R = \{\mathfrak{p} : \mathfrak{p} \subseteq R \text{ a prime ideal}\}$$

Example 5.1

If $R = A(X)$ for some affine variety A , then we know from before that points of X correspond to maximal ideals in $A(X)$ via $a \leftrightarrow I_X(a)$. However, this correspondence is with maximal ideals, which are not necessarily prime ideals. In particular, the spectrum is actually composed of a point for each irreducible subvariety of X .

Example 5.2

$\text{Spec } \mathbb{Z}$ has points of the form $\langle p \rangle$, plus the point $\langle 0 \rangle$.

Example 5.3

For any field K , $\text{Spec } K$ is a single point. However, there are rings with single point spectra that are not fields. For instance, $\text{Spec } \mathbb{Z}/2\mathbb{Z}$ has the unique point $\langle 2 \rangle$. Similarly, $\text{Spec } K[x]/\langle x^2 \rangle$ has the unique point $\langle x \rangle$.

We remark that the identification of a ring with its spectrum *as a set* is not one-to-one. That is, let R be a ring and J its nilradical. Then every prime ideal in R contains J , since if $f^m = 0 \in \mathfrak{p}$, then $f \in \mathfrak{p}$. Moreover, prime ideals in R/J biject with prime ideals in R containing J . Thus as sets, $\text{Spec } R = \text{Spec } R/J$. This is also true as topological spaces, so we will see that in order to get a useful classification of rings via schemes, we will need to progress to ringed space structures.

To define the Zariski topology on affine varieties, we had to define vanishing loci of coordinate functions. Since in the case of affine varieties, coordinate functions are quotients of the polynomial ring, they have a natural interpretation as functions to the base field. However, for schemes this is trickier, and in fact as mentioned before, there is not even a canonical base field that it makes sense to map to everywhere, nor an obvious object that should become the functions which we evaluate.

To resolve this, first note that for the case of affine varieties, we have (up to the extra points for irreducible subvarieties) that X “looks like” $\text{Spec } A(X)$. Thus, the functions on $\text{Spec } A(X)$ should be precisely the elements of $A(X)$, and in general we will have elements of R acting on $\text{Spec } R$.

Definition 5.2

Let R be a ring and $\mathfrak{p} \in \text{Spec } R$. Then the **residue field** of $\text{Spec } R$ at $\mathfrak{p}$ is the fraction field

$$K(\mathfrak{p}) = \text{Frac} \left(R/\mathfrak{p} \right)$$

(note that $R/\mathfrak{p}$ is an integral domain). If $f \in R$ then the **value** of f at $\mathfrak{p}$ is the element

$$[f]_1 \in K(\mathfrak{p})$$

Alternately, this is seen as the image of f under the composition of ring homomorphisms $R \rightarrow R/\mathfrak{p} \rightarrow K(\mathfrak{p})$.

In particular, $f(\mathfrak{p}) \iff f \in \mathfrak{p}$.

Example 5.4

If $R = A(X)$ for an affine variety X over K , then for $a \in X$ and $\mathfrak{p} = I_X(a)$, then the quotient $R/\mathfrak{p}$ kills the nonconstant terms when interpreted as a polynomial in $(x - a)$. That is, $[f] = f(a)$. Then $R/\mathfrak{p} \cong K$ is already a field, so $K(\mathfrak{p}) \cong K$, and evaluation agrees with the normal definition. On the other hand, for points of the form $\mathfrak{p} = I_X(Y)$ for a irreducible subvariety, the same logic shows that $K(\mathfrak{p})$ is the set of rational functions $Y \dashrightarrow K$.

Example 5.5

If $R = \mathbb{Z}$ and $f \in \mathbb{Z}$, then for $\mathfrak{p} = \langle p \rangle$ we have $R/\mathfrak{p} \cong \mathbb{Z}/p\mathbb{Z} \cong K(\mathfrak{p})$, and $f(\mathfrak{p})$ is the equivalence class of f in $\mathbb{Z}/p\mathbb{Z}$. If $\mathfrak{p} = \langle 0 \rangle$, then $R/\mathfrak{p} \cong \mathbb{Z}$ and hence $K(\mathfrak{p}) \cong \mathbb{Q}$, with $f(0)$ simply equal to the integer f as an element of $\mathbb{Q}$.

Definition 5.3

Let R be a ring. The **vanishing locus** of a subset $S \subseteq R$ is the set

$$V(S) = \{\mathfrak{p} : f(\mathfrak{p}) = 0 \forall f \in S\} \subseteq \text{Spec } R$$

Correspondingly, for a subset $X \subseteq \text{Spec } R$, the **ideal** of X is the set

$$I(X) = \{f : f(\mathfrak{p}) = 0 \forall \mathfrak{p} \in X\} \subseteq R$$

We remark that $I(X)$ is actually an ideal because we saw that evaluation is the composition of ring homomorphisms. Also, since $f(\mathfrak{p}) = 0 \iff f \in \mathfrak{p}$, we have the alternate characterizations

$$V(S) = \{\mathfrak{p} \in \text{Spec } R : S \subseteq \mathfrak{p}\}$$

and

$$I(X) = \bigcap_{\mathfrak{p} \in X} \mathfrak{p}$$

The following are entirely analogous to the case of varieties.

Proposition 5.1

The following are properties of V :

1. $V(S) = V(\langle S \rangle)$,
2. $V(S_1 S_2) = V(S_1) \cup V(S_2)$,
3. $\bigcap_{i \in J} V(S_i) = V(\bigcup_{i \in J} S_i)$,
4. $V(1) = \emptyset$,
5. $V(\emptyset) = \text{Spec } R$.

Also, V, I both reverse inclusions.

Since $\text{Spec } R$ is a vanishing locus, inclusions are reversed, and finite unions of vanishing loci are vanishing loci, we can define a topology on $\text{Spec } R$ in the analogous way as for varieties.

Definition 5.4

The **Zariski topology** on $\text{Spec } R$ is the topology where the closed sets are those of the form $V(S)$ for some $S \subseteq R$.

Since connectedness, irreducibility, and dimension are topological properties, we do not need to make new definitions here. However, there are some characteristics about these which are different for schemes. In particular, points are not necessarily closed. We can see that

$$\overline{\{\mathfrak{p}\}} = \bigcap_{\mathfrak{p} \in V(S)} V(S) = V\left(\bigcup_{\mathfrak{p} \supseteq S} S\right) = V(\mathfrak{p}) = \{\mathfrak{q} \in \text{Spec } R : \mathfrak{p} \subseteq \mathfrak{q}\}$$

That is, $\mathfrak{p}$ is closed precisely when it is maximal. For affine varieties, this means that points corresponding to genuine points in X are still closed, but those corresponding to irreducible (positive dimensional) subvarieties are not. Instead, these are called “generic points,” and the closure of $\mathfrak{p} = I(Y)$ for $Y \subseteq X$ is the collection of $I(Z)$ for all subvarieties $Z \subseteq Y$.

There is an analogue of the Nullstellensatz for schemes.

Proposition 5.2: Nullstellensatz for Schemes

Let R be a ring. Then:

- (a) For any $X \subseteq \text{Spec } R$ closed, $V(I(X)) = X$,
- (b) For any $J \subseteq R$ an ideal, $I(V(J)) = \sqrt{J}$.

Thus there is a bijection between closed subsets of $\text{Spec } R$ and radical ideals in R .

Proof. As in the case of affine varieties and projective varieties, three implications are simply consequences of the inclusion properties. The only difficult one is $(\subseteq)$ for (b), which follows from the general commutative algebra fact that

$$\bigcap_{J \subseteq \mathfrak{p} \in \text{Spec } R} \mathfrak{p} = \sqrt{J} \quad \square$$

The proofs of the following are analogous to the ones using the affine Nullstellensatz:

Proposition 5.3

Let J_1, J_2 be ideals in R . Then

$$V(J_1) \cup V(J_2) = V(J_1 \cap J_2) = V(J_1 J_2)$$

and

$$V(J_1) \cap V(J_2) = V(J_1 + J_2)$$

If X_1, X_2 are closed subsets of $\text{Spec } R$, then

$$I(X_1 \cup X_2) = I(X_1) \cap I(X_2)$$

and

$$I(X_1 \cap X_2) = \sqrt{I(X_1) + I(X_2)}$$

5.2 Affine Schemes as Ringed Spaces

Now, to understand the functions on $\text{Spec } R$, we proceed as in the affine case and look first at functions on distinguished open sets.

Definition 5.5

Let R in a ring and $f \in R$. Then the **distinguished open set** of f is the set

$$D(f) = \{\mathfrak{p} \in \text{Spec } R : f(\mathfrak{p}) \neq 0\} = \{\mathfrak{p} \in \text{Spec } R : f \notin \mathfrak{p}\}$$

Example 5.6

We saw that for affine varieties, X , if $\Phi : X \hookrightarrow \text{Spec } A(X)$ represents the identification of points in X with closed points in $\text{Spec } R$, then

$$\Phi^{-1}(V(J)) = \{a \in X : m_a \supseteq J\} = V_X(J)$$

so the preimage of the scheme theoretic vanishing locus is precisely the classical one. In particular,

$$\Phi^{-1}(D(f)) = D(f) \subseteq X$$

in the classical sense.

The important fact about distinguished opens before was that they formed a basis. This is still true: if U is open and $U = \text{Spec } R \setminus V(S)$, then

$$U = \text{Spec } R \setminus \bigcap_{f \in S} V(f) = \bigcup_{f \in S} \text{Spec } R \setminus V(f) = \bigcup_{f \in S} D(f)$$

However, note that for affine subsets, $\text{Spec } R$ was Noetherian, so that every open set was a finite union. Now, we lose Noetherianity, so we cannot build every open as a finite union of

distinguished opens. We now develop the analogous partition of unity for scheme theoretic distinguished opens:

Lemma 5.4

Let $D(f) = \bigcup_{i \in I} D(f_i)$. Then there is a finite subindex set $I_0 \subseteq I$, and $r_i \in R$ so that

$$f^m = \sum_{i \in I_0} r_i f_i$$

for some $m \in \mathbb{N}$.

Proof. We first observe that

$$V(f) = \bigcap_{i \in I} V(f_i) = V(\langle f_i : i \in I \rangle)$$

Taking the ideal on both sides, we get

$$\sqrt{\langle f \rangle} = \sqrt{\langle f_i : i \in I \rangle}$$

Thus f lies in the right hand side, which implies the result. $\square$

Now, it is time to look at the ring of regular functions on $\text{Spec } R$. First, we look at the classical case to look at what we would want out of our structure sheaf. In the classical case we have the following properties:

$$\mathcal{O}_{\text{Spec } R}(\text{Spec } R) = R, \quad \mathcal{O}_{\text{Spec } R}(D(f)) = R_f$$

Combined with the sheaf property, this completely determines $\mathcal{O}_{\text{Spec } R}$, since for any U and open cover $D(f_i)$ of U , we have that $\varphi \in \mathcal{O}_{\text{Spec } R}(U)$ is determined by the restrictions $\varphi_i \in \mathcal{O}_{\text{Spec } R}(D(f_i))$, which agree on overlaps as elements of $\mathcal{O}_{\text{Spec } R}(D(f_i f_j)) = R_{f_i f_j}$. So after choosing the open cover, we have determined the form of every regular function on U . However, this is a difficult definition to work with because it is not obvious that this is independent of the choice of distinguished opens. Also, it is not a local condition, which is important to help us work with the sheaves.

Definition 5.6

Let R be a ring and $U \subseteq \text{Spec } R$ open. A **regular function** on U is a family $\varphi = (\varphi_{\mathfrak{p}})_{\mathfrak{p} \in U}$ with $\varphi_{\mathfrak{p}} \in R_{\mathfrak{p}}$ so that for any $\mathfrak{p} \in U$, there is an open set $\mathfrak{p} \in U_{\mathfrak{p}} \subseteq U$ and $f, g \in R$ so that

$$\frac{g}{f} = \varphi_{\mathfrak{q}} \in R_{\mathfrak{q}}$$

for all $\mathfrak{q} \in U_{\mathfrak{p}}$, with $f \notin \mathfrak{q}$.

Letting $\mathcal{O}_{\text{Spec } R}(U)$ be the collection of regular functions above, we can define addition and multiplication analogously, giving a ring structure. More importantly, because this is now a local condition, we will get the gluing property and thus $\mathcal{O}_{\text{Spec } R}$ is a sheaf.

Regular functions now have a natural evaluation. That is, for $\mathfrak{p} \in U$, $\varphi_{\mathfrak{p}}$ is naturally an element of $R_{\mathfrak{p}}$, but by the quotient map is also an element of $R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$, where $\mathfrak{p}R_{\mathfrak{p}}$ is the maximal ideal of the local ring $R_{\mathfrak{p}}$. Then, since localization commutes with quotients, this is the localization of $(R/\mathfrak{p})_{\mathfrak{p}R_{\mathfrak{p}}} = (R/\mathfrak{p})_{\langle 0 \rangle} = K(\mathfrak{p})$. So regular functions have evaluations living in the same space as evaluation of elements of $\text{Spec } R$. However, we will see that because the evaluation map takes $\varphi(\mathfrak{p}) = [\varphi_{\mathfrak{p}}]$ rather than $\varphi_{\mathfrak{p}}$ itself, the values $[\varphi_{\mathfrak{p}}]$ do not in general determine φ .

Recall that previously we had that stalks of regular functions looked like the localizations $\mathcal{O}_{X,a} \cong A(X)_{I(a)}$.

Lemma 5.5

Let R be a ring and $\mathfrak{p} \in \text{Spec } R$. Then $\mathcal{O}_{\text{Spec } R, \mathfrak{p}} \cong R_{\mathfrak{p}}$, explicitly via the isomorphism

$$\Phi : \mathcal{O}_{\text{Spec } R, \mathfrak{p}} \rightarrow R_{\mathfrak{p}}, \quad [(U, \varphi)] \mapsto \varphi_{\mathfrak{p}}$$

Proof. First we write down the inverse map. Any element of $R_{\mathfrak{p}}$ is of the form

$$\frac{g}{f}, \quad f, g \in R, f(\mathfrak{p}) \neq 0$$

This extends to a regular function globally on $D(f)$, so we map this

$$R_{\mathfrak{p}} \xrightarrow{\Psi} \mathcal{O}_{\text{Spec } R, \mathfrak{p}}, \quad \frac{g}{f} \mapsto \left[\left(D(f), \varphi = \left(\frac{g}{f} \right)_{\mathfrak{q} \in D(f)} \right) \right]$$

Then clearly $\Phi \circ \Psi$ is the identity. On the other hand, for $[(U, \varphi)] \in \mathcal{O} - \text{Spec } R, \mathfrak{p}$, there is an open neighborhood $\mathfrak{p} \in U_{\mathfrak{p}} \subseteq U$ where $\varphi_{\mathfrak{q}} = g/f$ for all $\mathfrak{q} \in U_{\mathfrak{p}}$. Then by restricting, we have that

$$[(U, \varphi)] = \left[\left(U_{\mathfrak{p}}, \left(\frac{g}{f} \right)_{\mathfrak{q} \in U_{\mathfrak{p}}} \right) \right] = \Psi \left(\frac{g}{f} \right) = \Psi(\Phi([(U, \varphi)]))$$

so Ψ, Φ are inverses. It is clear that Φ is a ring homomorphism, and Ψ is a ring homomorphism since $\Psi(\varphi\psi)$ is defined on $D(f_1f_2)$. $\square$

Recall that a section of a sheaf is uniquely determined by its germs at all stalks. This is the reason that we a priori defined a regular function to be the collection of its germs, together with the locality condition.

Now we compute the regular function rings on distinguished opens.

Proposition 5.6

Let R be a ring and $f \in R$. Then $\mathcal{O}_{\text{Spec } R}(D(f)) \cong R_f$ via the isomorphism

$$\Phi : R_f \rightarrow \mathcal{O}_{\text{Spec } R}(D(f)), \quad \frac{g}{f^m} \mapsto \varphi = \left(\frac{g}{f^m} \in R_{\mathfrak{p}} \right)_{\mathfrak{p} \in D(f)}$$

In particular, $\mathcal{O}_{\text{Spec } R}(\text{Spec } R) \cong R$ by taking $f = 1$.

This shows that our sheaf satisfies the three properties we suggested above in our intuitive properties desired of the structure sheaf.

Proof. This is clearly a ring homomorphism.

Injectivity: Let $g/f^m \in \ker \Phi$. Then $g/f^m = 0 \in R_{\mathfrak{p}}$ for every $\mathfrak{p} \in D(f)$. That is, there is $h_{\mathfrak{p}} \in R \setminus \mathfrak{p}$ so that $gh_{\mathfrak{p}} = 0 \in R$. We want to show that there is f^r so that $f^r g = 0$.

Define $J = \text{Ann}_R(g)$, which is the set $\{h \in R : hg = 0 \in R\}$. Then we need to show that $f^r \in J$. To show this, we will show that $V(J) \subseteq V(f)$, which implies by the Nullstellensatz that $\sqrt{f} \subseteq \sqrt{J}$, which gives the desired result. To see this, note that by definition,

$$V(J) = \{\mathfrak{p} \in \text{Spec } R : J \subseteq \mathfrak{p}\}$$

By definition, $h_{\mathfrak{p}}$ annihilates g , but also $h_{\mathfrak{p}} \notin \mathfrak{p}$. Thus, J is not contained in $\mathfrak{p}$. In particular, $\mathfrak{p} \notin V(J)$. This is true for all $\mathfrak{p} \in D(f)$, so

$$\text{Spec } R \setminus V(f) \subseteq \text{Spec } R \setminus V(J) \implies V(J) \subseteq V(f)$$

This proves the claim and shows injectivity.

Surjectivity: Let $\varphi \in \mathcal{O}_{\text{Spec } R}(D(f))$. Then at any $\mathfrak{p} \in D(f)$, we have $\mathfrak{p} \in U_{\mathfrak{p}} \subseteq U$ and $\varphi = g_{\mathfrak{p}}/f_{\mathfrak{p}}$ on $U_{\mathfrak{p}}$. But by shrinking to a basis element, we can assume that $U_{\mathfrak{p}}$ is of the form $D(h_{\mathfrak{p}})$, so that $f_{\mathfrak{p}} \notin \mathfrak{q}$ for any $\mathfrak{q} \in D(h_{\mathfrak{p}})$. In other words, $D(h_{\mathfrak{p}}) \subseteq D(f_{\mathfrak{p}})$

We want to show that φ can be written in the form $g_{\mathfrak{p}}/h_{\mathfrak{p}}$. To see this, we have $V(h_{\mathfrak{p}}) \supseteq V(f_{\mathfrak{p}})$ so $\sqrt{\langle h_{\mathfrak{p}} \rangle} \subseteq \sqrt{\langle f_{\mathfrak{p}} \rangle}$. So $h_{\mathfrak{p}}^r = cf_{\mathfrak{p}}$ for some $r \in \mathbb{N}$ and $c \in R$. Therefore we have that on $D(h_{\mathfrak{p}})$,

$$\frac{g_{\mathfrak{p}}}{f_{\mathfrak{p}}} = \frac{cg_{\mathfrak{p}}}{h_{\mathfrak{p}}^r}$$

Thus, redefining $h_{\mathfrak{p}}, g_{\mathfrak{p}}$ as $h_{\mathfrak{p}}^r, cg_{\mathfrak{p}}$, respectively, we have that $D(h_{\mathfrak{p}}^r) = D(h_{\mathfrak{p}})$, so that we now have shown that $D(f)$ is covered by $D(f_{\mathfrak{p}})$, and on each, we have $\varphi = g_{\mathfrak{p}}/f_{\mathfrak{p}}$. Then, by the scheme theoretic partition of unity lemma, we have a finite collection P_0 so that

$$f^m = \sum_{\mathfrak{p} \in P_0} r_{\mathfrak{p}} f_{\mathfrak{p}}$$

Now, on the overlaps $D(f_{\mathfrak{p}}f_{\mathfrak{q}})$, we have

$$\frac{g_{\mathfrak{p}}}{f_{\mathfrak{p}}} = \varphi = \frac{g_{\mathfrak{q}}}{f_{\mathfrak{q}}} \in \mathcal{O}_{\text{Spec } R}(D(f_{\mathfrak{p}}f_{\mathfrak{q}}))$$

Both sides are elements of $R_{\mathfrak{p}\mathfrak{q}}$. So by injectivity,

$$\frac{g_{\mathfrak{p}}}{f_{\mathfrak{p}}} = \frac{g_{\mathfrak{q}}}{f_{\mathfrak{q}}} \in R_{\mathfrak{p}\mathfrak{q}}$$

so there is n so that

$$(f_{\mathfrak{p}}f_{\mathfrak{q}})^n(g_{\mathfrak{p}}f_{\mathfrak{q}} - g_{\mathfrak{q}}f_{\mathfrak{p}}) = 0 \in R$$

The above is a priori true for each $D(f_p f_q)$, but with different n . However, looking only at the finite sum, we can choose n sufficiently large so that this holds for all $p, q \in P_0$ with the same n . Now, we can redefine g_p to be $g_p f_p^n$, f_p to be f_p^{n+1} . Using this, we now have that $D(f)$ is covered by finitely many $D(f_p)$ with

$$\varphi = \frac{g_p}{f_p} \text{ on } D(f_p)$$

and $g_p f_q = g_q f_p \in R$.

Now, applying the partition of unity lemma again we still have a representation

$$f^m = \sum_{p \in Q_0} r_p f_p$$

(though possibly with different r_p , and a smaller subcollection Q_0 , as our f_p have now changed, even though the corresponding distinguished opens have not). Define

$$g = \sum_{p \in Q_0} r_p g_p$$

Then we want to say that

$$\frac{g}{f^m} = \frac{g_q}{f_q} \text{ on } D(f_q)$$

which implies surjectivity. Clearing denominators, this is equivalent to

$$\sum_{p \in Q_0} r_p g_p f_q = g_q = g_q f^m = \sum_{p \in Q_0} r_p f_p g_q$$

But this is true because $g_p f_q = g_q f_p$. □

Example 5.7

We stated previously that as a set, $\text{Spec } K[x]/\langle x^2 \rangle$ is a single point. To see this, let $R = K[x]/\langle x^2 \rangle$. Then $R \cong \mathcal{O}_{\text{Spec } R}(\text{Spec } R)$, so any regular function is of the form

$$\varphi = a + bx, \quad a, b \in K$$

In particular, this ring has nilpotent elements, even though for affine varieties the coordinate ring cannot. In this case, φ is nilpotent if

$$0 = \varphi^n = a^n + na^{n-1}bx$$

Thus this is nilpotent if and only if $a = 0$, and hence when $\varphi \in \langle x \rangle$. Every prime ideal contains the nilradical so $\langle x \rangle$ is in every prime ideal. On the other hand, $R/\langle x \rangle \cong K$ is a field so this is maximal. Letting $\mathfrak{p} = \langle x \rangle$, this shows that $\text{Spec } K[x]/\langle x^2 \rangle = \{\mathfrak{p}\}$ and $K(\mathfrak{p}) = K$. But passing through the evaluation homomorphisms, we have that $\varphi(\mathfrak{p}) = a$, which means that φ is not uniquely determined by its values. Geometrically, $\text{Spec } R$ looks like a single point, but one that has some infinitesimal width that

allows functions to take linear parts as well as values. After making more definitions, one can also view these functions as contractions against the tangent bundle.

Example 5.8

We saw before that $\text{Spec } \mathbb{Z} = \{\langle 0 \rangle, \langle 2 \rangle, \dots\}$. If we look at, say, $D(\langle 6 \rangle)$, we get $\text{Spec } \mathbb{Z} \setminus V(\langle 6 \rangle)$. $V(\langle 6 \rangle)$ is the ideals so that $\langle 6 \rangle \subseteq \langle p \rangle$ (note that $\langle 0 \rangle$ does not contain in $\langle 6 \rangle$), so

$$D(\langle 6 \rangle) = \text{Spec } \mathbb{Z} \setminus \{\langle 2 \rangle, \langle 3 \rangle\}$$

Now, say we define the regular function $\varphi = 5/6 \in \mathcal{O}_{\text{Spec } \mathbb{Z}}(D(\langle 6 \rangle))$. Evaluated at 0, we just interpret this as a fraction in $\mathbb{Q}$. Evaluated at $p \neq 2, 3$, 6 is invertible in $\mathbb{Z}/p\mathbb{Z}$, so $5/6$ becomes $\overline{5}\overline{6}^{-1}$. For instance, $\varphi(\langle 5 \rangle) = \overline{0}$ and $\varphi(\langle 11 \rangle) = \overline{52} = \overline{10}$. That is, $\varphi(p)$ gives the meaning of the fraction $5/6$ within $\mathbb{Z}/p\mathbb{Z}$.

Now, we have endowed $\text{Spec } R$ with the structure of a ringed space. We would like to define morphisms of affine schemes as simply morphisms as ringed spaces. However, the problem is that our definition of morphisms of ringed spaces was that the pullbacks $f^*\varphi$ took regular functions to regular functions. The problem is that $f^*\varphi$ was defined as the set theoretic map $\varphi \mapsto \varphi(f)$, which doesn't make sense here as the regular functions do not map into the same field. Moreover, even if they did, we just saw that a regular function is not determined purely by its values, which is the only thing a set theoretic pullback would be able to detect.

Thus, in the more general setting we now have that the pullback is not directly induced as a set theoretic map, but rather part of the data of a morphism. To make our new definition, we should better understand the set theoretic pullback. If $f : X \rightarrow Y$ is a morphism of varieties, then $f^* : \mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(f^{-1}(V))$ for each open $V \subseteq Y$, and this is compatible with restrictions. Moreover, because this is compatible with restrictions, it induces a well defined map of stalks (which is not dependent on choice of $[(U, \varphi)]$ since we can pass to their intersection). This map becomes $f_a^* : \mathcal{O}_{Y, f(a)} \rightarrow \mathcal{O}_{X, a}$, and moreover it preserves the maximal ideals: the maximal ideals are given by

$$\begin{aligned} I_a &= \{\psi \in \mathcal{O}_{X, a} : \psi(a) = 0\} \\ I_{f(a)} &= \{\varphi \in \mathcal{O}_{Y, f(a)} : \varphi(f(a)) = 0\} \end{aligned}$$

The pullback satisfies

$$(f_a^*)^{-1}(I_a) = \{\varphi \in \mathcal{O}_{Y, f(a)} : f^*\varphi(a) = 0\}$$

which is precisely the same condition, so $(f_a^*)^{-1}(I_a) = I_{f(a)}$. The preservation of maximal ideals under the pullback will be the only new condition we need to impose.

Remark 5.1

From now on, we discard the convention that regular functions of ringed spaces are always of the form

$$\mathcal{F}(U) \subseteq \{\varphi : U \rightarrow K\}$$

Definition 5.7

A **locally ringed space** is a ringed space $(X, \mathcal{O}_X)$ so that each stalk $\mathcal{O}_{X,p}$ is a local ring.

Example 5.9

We saw that for X an affine variety (hence any prevariety, by taking affine opens), the stalks are $\mathcal{O}_{X,a} = A(X)_{I(a)}$ which is local. Similarly, for affine schemes we just showed the stalks are $\mathcal{O}_{\text{Spec } R, \mathfrak{p}} = R_{\mathfrak{p}}$ which is also local.

Definition 5.8

A **morphism** of locally ringed spaces $(X, \mathcal{O}_X), (Y, \mathcal{O}_Y)$ is a continuous map $f : X \rightarrow Y$ together with a collection of ring homomorphisms for each $V \subseteq Y$ open:

$$f_V^* : \mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(f^{-1}(V))$$

such that the pullbacks are compatible with restrictions, so that if $V \subseteq W \subseteq Y$ open, then for $\varphi \in \mathcal{O}_Y(W)$,

$$f_V^*(\varphi|_V) = (f_W^*\varphi)|_{f^{-1}(V)} \in \mathcal{O}_X(f^{-1}(V))$$

This induces a map of stalks $f_{\mathfrak{p}}^* : \mathcal{O}_{Y,f(\mathfrak{p})} \rightarrow \mathcal{O}_{X,\mathfrak{p}}$. We also require that

$$(f_{\mathfrak{p}}^*)^{-1}I_{\mathfrak{p}} = I_{f(\mathfrak{p})}$$

Now, we want to use this definition to see that the correspondence between affine varieties and finitely generated reduced K algebras is properly extended to a correspondence between all rings using schemes. We first use note the following lemma from commutative algebra:

Lemma 5.7

Let $\varphi : S \rightarrow R$ be a ring homomorphism and $\mathfrak{p} \subseteq R$ prime. Then the map

$$\varphi_{\mathfrak{p}} : S_{\varphi^{-1}(\mathfrak{p})} \rightarrow R_{\mathfrak{p}}, \quad \frac{g}{f} \mapsto \frac{\varphi(g)}{\varphi(f)}$$

is a well defined ring homomorphism. Moreover, $\varphi_{\mathfrak{p}}^{-1}(\mathfrak{p} \cdot R_{\mathfrak{p}}) = \varphi^{-1}(\mathfrak{p}) \cdot S_{\varphi^{-1}(\mathfrak{p})}$, which is to say that $\varphi_{\mathfrak{p}}$ preserves the maximal ideals.

Proposition 5.8

For any two rings R, S , there is a bijection between morphisms from $\text{Spec } R \rightarrow \text{Spec } S$ and ring homomorphisms from $S \rightarrow R$.

In particular, isomorphisms between rings correspond to isomorphisms between their affine

schemes, so there is a bijection between affine schemes and rings (up to isomorphism).

Proof. One direction is easy: any morphism f from $\text{Spec } R \rightarrow \text{Spec } S$ comes with the data of a pullback $f_{\text{Spec } S}^*$, which is a ring homomorphism from $\mathcal{O}_{\text{Spec } S}(\text{Spec } S) = S$ to $\mathcal{O}_{\text{Spec } R}(f^{-1}(\text{Spec } S)) = R$. Denote the map $f \mapsto f_{\text{Spec } S}^*$ by Φ .

Naturally, the inverse direction is more difficult since we need to provide the data of the morphism given a ring homomorphism. Let $\varphi : S \rightarrow R$ be a ring homomorphism. We want to construct the inverse $f = \Psi(\varphi) : \text{Spec } R \rightarrow \text{Spec } S$. First, recall that the preimage of a prime ideal is a prime ideal. Thus, as a set map, we will simply send $\mathfrak{p} \in \text{Spec } R$ to $\varphi^{-1}(\mathfrak{p}) \in \text{Spec } S$.

To see that this is continuous, we take a closed set

$$V(J) = \{\mathfrak{q} : J \subseteq \mathfrak{q} \subseteq S \text{ prime ideal}\}$$

The preimage is

$$\begin{aligned} f^{-1}(V(J)) &= \{\mathfrak{p} \subseteq R \text{ prime ideal} : J \subseteq \varphi^{-1}(\mathfrak{p})\} \\ &= \{\mathfrak{p} \subseteq R \text{ prime ideal} : \varphi(J) \subseteq \mathfrak{p}\} = V(\varphi(J)) \end{aligned}$$

which is also closed, so f is continuous.

Now, let $V \subseteq \text{Spec } S$ be open. We need to define $f_V^* : \mathcal{O}_{\text{Spec } S}(V) \rightarrow \mathcal{O}_{\text{Spec } R}(f^{-1}(V))$. Every element of $\mathcal{O}_{\text{Spec } S}(V)$ is a collection $(\sigma_{\mathfrak{q}})_{\mathfrak{q} \in V}$ which is locally given as g/f , and every element of $\mathcal{O}_{\text{Spec } R}(f^{-1}(V))$ is given by a collection $(\eta_{\mathfrak{p}})_{\mathfrak{p} \in f^{-1}(V)}$ which is locally given as $\tilde{g}/\tilde{f}$. By the lemma above, at each $\mathfrak{p}$ there is a well defined ring homomorphism $\varphi_{\mathfrak{p}}$ which applies φ to both numerator and denominator, hence it maps into $\mathcal{O}_{\text{Spec } R}(f^{-1}(V))$.

$$f_V^* : \mathcal{O}_{\text{Spec } S}(V) \rightarrow \mathcal{O}_{\text{Spec } R}(f^{-1}(V)), \quad (\sigma_{\mathfrak{q}})_{\mathfrak{q} \in V} \mapsto (\eta_{\mathfrak{p}} = \varphi_{\mathfrak{p}}(\sigma_{\mathfrak{p}}))_{\mathfrak{p} \in f^{-1}(V)}$$

This is compatible with restrictions because it is a pointwise condition. Moreover, on the stalk $\mathcal{O}_{\text{Spec } S, \mathfrak{p}}$, the induced map is precisely $f_{\mathfrak{p}}^* = \varphi_{\mathfrak{p}}$, which by the lemma above preserves the maximal ideal. Therefore we have constructed a morphism $\text{Spec } S \rightarrow \text{Spec } R$.

One may verify that the two maps we have defined are indeed inverses. $\square$

As with affine varieties, this correspondence between morphisms of affine schemes and ring homomorphisms defines a contravariant equivalence of categories between affine schemes and rings.

Example 5.10

Consider the morphism of affine varieties $f : \mathbb{A}_K^2 \rightarrow \mathbb{A}_K^1, (x, y) \mapsto xy$. Then by the correspondence for affine varieties, we get a homomorphism of K -algebras $f^* : K[t] \rightarrow K[x, y], t \mapsto xy$. By the correspondence for affine schemes, we then get a morphism $F : \text{Spec } K[x, y] \rightarrow \text{Spec } K[t]$ of affine schemes...

To see what this morphism does, first start with the ideals of points $m_{(a,b)} =$

$\langle x - a, y - b \rangle \in \text{Spec } K[x, y]$. Then

$$\begin{aligned} F(m_{(a,b)}) &= (f^*)^{-1}(m_{a,b}) = \{\eta \in K[t] : f^*(\eta) \in m_{(a,b)}\} \\ &= \{\eta \in K[t] : \eta(xy) \in m_{(a,b)}\} = \{\eta \in K[t] : \eta(ab) = 0\} = \langle t - ab \rangle = m_{ab} \end{aligned}$$

Thus we understand the behavior of F on the maximal ideals (which are the points in the plane). For other subvarieties, we get nonmaximal ideals, such as $\langle x \rangle$. This is mapped to

$$F(\langle x \rangle) = \varphi^{-1}\langle x \rangle = \{\eta(t) : \eta(xy) \in \langle x \rangle\}$$

But since $\eta(xy)$ is divisible by x , the only requirement is that $\eta(0) = 0$. Therefore $F(\langle x \rangle) = m_0 = \langle t \rangle$.

Similarly,

$$F(\langle 0 \rangle) = \{\eta(t) : \eta(xy) = 0\}$$

which is only possible if η itself is zero. Thus $F(\langle 0 \rangle) = \langle 0 \rangle$ which effectively says that the generic point for $\mathbb{A}_K^2$ is mapped to the generic point for $\mathbb{A}_K^1$.

Finally,

$$F(\langle x + y \rangle) = \{\eta(t) : \eta(xy) \in \langle x + y \rangle\}$$

In particular, note that $\langle x + y \rangle$ is the ideal of the set $(x, -x) \in \mathbb{A}_K^2$. Thus, $\eta(xy)$ must vanish at all such points, so that $\eta(-x^2)$ vanishes for all $x \in K$. That is, if η is the polynomial

$$a_0 + a_1t + \dots + a_nt^n$$

then

$$a_0 + a_1x^2 + \dots + a_nx^{2n}$$

must vanish as a function on K . This is only possible if this polynomial is the zero polynomial, and hence if η is zero. So

$$F(\langle x + y \rangle) = \langle 0 \rangle$$

as well.

Now, we illustrate the general strategy. If $f : X \rightarrow Y$ is a map of affine varieties and $\varphi = f^* : A(Y) \rightarrow A(X)$ is the ring homomorphism, then $F = \varphi^*$ is computed as follows: For $\mathfrak{p} \in \text{Spec } A(X)$, we take the corresponding irreducible subvariety $Z_{\mathfrak{p}} = V_X(\mathfrak{p}) \subseteq X$. Then, $f(Z_{\mathfrak{p}})$ is irreducible in Y . It may not be closed, so define $Z' = \overline{f(Z_{\mathfrak{p}})} \subseteq Y$, which is now an irreducible subvariety. Finally, we have $F(\mathfrak{p}) = \mathfrak{q} = I_Y(Z')$.

Example 5.11

In the previous example, beginning with $\langle x + y \rangle \in \text{Spec } K[x, y]$, this corresponds to the subvariety

$$\{(x, -x) : x \in K\} \subseteq \mathbb{A}_K^2$$

Under the map $(x, y) \mapsto xy$, we get the set

$$\{-x^2 : x \in K\} \subseteq \mathbb{A}_K^1$$

Since K is algebraically closed, this is all of $\mathbb{A}_K^1$. The ideal corresponding to $\mathbb{A}_K^1$ is the zero ideal, so we see that $F(\langle x + y \rangle) = \langle 0 \rangle$, as before.

Example 5.12

Consider the inclusion map $i : \mathbb{Z} \hookrightarrow \mathbb{Z}[x]$. This defines a map $i^* : \text{Spec } \mathbb{Z}[x] \rightarrow \text{Spec } \mathbb{Z}$ (we call $\text{Spec } R[x]$ the **affine line** over R for general rings R).

This map is visualized in the following diagram:

Indeed, note that $\text{Spec } \mathbb{Z}$ is the points $\langle p \rangle, \langle 0 \rangle$. So $\text{Spec } \mathbb{Z}[x]$ decomposes into the disjoint preimages of each point of $\text{Spec } \mathbb{Z}$ under i^* . For each $\langle p \rangle \in \text{Spec } \mathbb{Z}$, $\langle p \rangle$ is also prime in $\mathbb{Z}[x]$, since the quotient is

$$\mathbb{Z}[x]/\langle p \rangle = \mathbb{Z}/p\mathbb{Z}[x]$$

which is a domain. Similarly, $\mathbb{Z}[x]$ itself is a domain so $\langle 0 \rangle$ is also prime there.

The ideal $\langle 2, x \rangle$ has the quotient

$$\mathbb{Z}[x]/\langle 2, x \rangle = \mathbb{Z}/2\mathbb{Z}$$

which is a domain so this is also prime. By the same logic, so is $\langle p, x \rangle$ for general p and so is $\langle 0, x \rangle = \langle x \rangle$. This gives a section for the affine line where $x = 0$.

Similarly, so is $\langle 2, x + 1 \rangle$, and $\langle p, x + 1 \rangle$.

In general, each fiber over p looks like:

$$(i^*)^{-1}(\langle p \rangle) = \{\langle p \rangle, \langle p, x \rangle, \dots, \langle p, x + p - 1 \rangle\}$$

The point $\langle p \rangle$ is the generic point, in the sense that its closure is the entire fiber. The other points are closed.

On the other hand, as we saw before, $\langle x \rangle$ is in the fiber over 0. $x^2 + 1$ is irreducible over $\mathbb{Z}$, so this also defines a prime ideal in the fiber over 0. Taking the closure, we see that the closure intersects some points in the fibers over primes. For instance,

$$\langle 5, x + 2 \rangle, \langle 5, x + 3 \rangle \in \overline{\langle x^2 + 1 \rangle}$$

This is reflective of the fact that

$$(x + 2)(x + 3) \equiv x^2 + 1 \pmod{5}$$

which can be manipulated to show that $x^2 + 1 \in \langle 5, x + 2 \rangle, \langle 5, x + 3 \rangle$.

Note that we have not yet made a proper definition of an affine subscheme. This was not because it is obvious, but rather the opposite. Note that an affine subvariety is simply a closed subset. But in this case, we saw before that the set structure of a scheme (and even its topological space structure) does not uniquely identify its ringed space structure.

Instead, for $Y \subseteq X$ closed (affine varieties), write $J = I(Y) \subseteq A(X)$ be the radical ideal. Then $A(Y) = A(X)/J$ and the map $Y \rightarrow X$ corresponds to the quotient map $A(X) \rightarrow A(Y)$. We will make this our definition of subschemes, with the exception that we no longer need to demand that J is radical.

Definition 5.9

Let R be a ring and $X = \text{Spec } R$. Then an **affine subscheme** of X is a morphism

$$f : (Y = \text{Spec } R/J) \rightarrow \text{Spec } R$$

which is associated to the quotient map $R \rightarrow R/J$ for J some ideal in R .

We make a few remarks. First, recall that prime ideals in R/J correspond to prime ideals in R containing J (which is $V(J)$). Thus if Y is an affine subscheme associated to the ring R/J , then f is injective and $f(Y) = V(J)$.

Also, by the isomorphism theorem, maps into quotients are in correspondence with surjective maps $R \rightarrow S$, and in this case $Y = \text{Spec } S$.

Thus, we see that ideals are in correspondence with affine subschemes via

$$J \leftrightarrow (\text{Spec } (R/J) \rightarrow \text{Spec } R)$$

Similarly, the set theoretic union and intersection are best replaced with ring theoretic definitions for schemes. Luckily, subschemes preserve everything that we need. In particular, because

$$V(J_1) \cap V(J_2) = V(J_1 + J_2), \quad V(J_1) \cup V(J_2) = V(J_1 \cap J_2)$$

we can make the following definitions:

Definition 5.10

Let R be a ring and consider the subschemes

$$\text{Spec } (R/J_1), \text{Spec } (R/J_2)$$

Then the **intersection** and **union** are defined as

$$\text{Spec } (R/J_1 + J_2), \quad \text{Spec } (R/J_1 \cap J_2)$$

respectively.

Example 5.13

Consider the varieties

$$\operatorname{Spec} K[x, y] / \langle y - x^2 \rangle, \quad \operatorname{Spec} K[x, y] / \langle y \rangle$$

which correspond to the x axis and the parabola $y = x^2$. Their intersection is

$$\operatorname{Spec} K[x, y] / \langle y, y - x^2 \rangle = \operatorname{Spec} K[x] / \langle x^2 \rangle$$

We already saw that this is a double point, which corresponds to the intuitive idea that $y = x^2$ has a double zero at the origin.

We note that we saw every regular function on $\operatorname{Spec} K[x] / \langle x^2 \rangle$ was of the form $a + bx$. To phrase this more generally, we have

$$\dim_K \mathcal{O}_{X_1 \cap X_2}(X_1 \cap X_2) = \dim_K K[x] / \langle x^2 \rangle = 2$$

where the dimension is as a K -algebra. We call this the intersection multiplicity of X_1, X_2 .

In this sense, if X_1, X_2 are curves in $\mathbb{A}_K^2$ which do not share irreducible components, then each X_i can be seen as an affine subscheme, and one can take the scheme theoretic intersection. If we separate these into connected components

$$X_1 \cap X_2 = Z_1 \sqcup \cdots \sqcup Z_m$$

Then $\dim_K \mathcal{O}_{Z_i}(Z_i)$ is the intersection multiplicity at Z_i , which agrees with the multiplicity in the classical sense (for instance, as in Bezout's theorem).

This defines the closed subschemes. For open subsets, as in the case of affine varieties these will not neatly correspond to a subscheme using the inherited structure, but one can put a scheme structure on open subsets to define open subschemes.

Proposition 5.9

Let R be a ring and $f \in R$. Then the ring homomorphism $R \rightarrow R_f$ induces an open embedding morphism

$$\operatorname{Spec} R_f \xrightarrow{i_f} \operatorname{Spec} R$$

and the image is $D(f)$. In particular, $\operatorname{Spec} R_f$ is isomorphic to the set $D(f)$ with the restricted structure sheaf $\mathcal{O}_{\operatorname{Spec} R}|_{D(f)}$.

Proof. Recall that from commutative algebra, the ideals in R_f are the ideals in R which do not contain a power of f . For prime ideals, the prime ideals of R_f are the prime ideals which do not contain f , which is precisely $D(f)$ (as a set). The bijection between these ideals is defined by

$$\mathfrak{p} \mapsto \mathfrak{p} \cdot R_f$$

Thus the map i_f is an injection, and it is continuous with image $D(f)$. To show that it is an open embedding, we show the inverse is continuous. Let $V(J) \subseteq \operatorname{Spec} R_f$ be closed. Then the inverse image under $(i_f)^{-1}$ is just given by applying i_f , and

$$i_f(V(J)) = V(i_f^{-1}(J)) \cap D(f)$$

which is closed in $D(f)$. Thus i_f is a homeomorphism, so we have isomorphism as topological spaces.

To compare their structure sheaves, let $U \subseteq D(f)$ be open. Then we need to show that

$$i_f^* : \mathcal{O}_{D(f)}(U) \rightarrow \mathcal{O}_{\operatorname{Spec} R_f}(U)$$

is an isomorphism, with the identification implicitly via i_f . By sheaf gluing, the regular functions on U are determined by their restrictions on a cover, so we can study this only on a basis of the opens in $D(f)$, which is given by sets of the form $D(fg)$. So showing isomorphism on this basis implies that they glue into an isomorphism on U .

Now, consider some set $U = D(fg)$. This has two structure sheaves: one as a subset of the distinguished open set, given as

$$\mathcal{O}_{D(f)}(D(fg)) = \mathcal{O}_{\operatorname{Spec} R} D(fg) = R_{fg}$$

and one in its identification within $\operatorname{Spec} R_f$, given by

$$\mathcal{O}_{\operatorname{Spec} R_f}(D(fg)) = (R_f)_{fg} = R_{fg}$$

Thus $\mathcal{O}_{D(f)}(U) \cong \mathcal{O}_{\operatorname{Spec} R_f}(U)$, so $\operatorname{Spec} R_f \cong D(f)$ as locally ringed spaces. $\square$

In particular, distinguished opens are affine schemes.

To consider more general opens, and really to be able to consider more general spaces, we now allow objects that locally look like affine schemes, similarly to how prevarieties locally look like an affine variety. However, we drop the assumption of having a finite open cover.

Definition 5.11

A **scheme** is a locally ringed space with an open cover by affine schemes, with morphisms given by morphisms as locally ringed space.

Note that we do not restrict the rings defining the affine scheme, and because there is no finiteness assumption, schemes may not be Noetherian (in fact, affine schemes did not need to be either). Also, even though we do not impose a separatedness condition, we do not call these “preschemes.” Rather, for schemes, we specifically denote the condition that the diagonal is closed by calling the scheme separated.

This immediately implies that for any scheme X and open set $U \subseteq X$, we have a subscheme as follows: First, covering X with affine schemes $(U_i)_i$, $U \cap U_i$ is covered by some distinguished opens $(D_{U_i}(f_{ij}))_j$. Then the collection $(D_{U_i}(f_{ij}))_{ij}$ is an open cover by affine schemes, so the subringed space U is a scheme, which we call an **open subscheme**.

On the other hand, a **closed subscheme** is given by the data of a morphism $i : Y \rightarrow X$ from some other scheme Y , so that there is some open cover of X by affine schemes (U_i) so that

$$i|_{i^{-1}(U_i)} : i^{-1}(U_i) \rightarrow U_i$$

is a morphism which is an affine subscheme. In this case, the image of each restriction is closed in U_i , and is a homeomorphism on the open cover, so this is a closed embedding. However, it can be shown that the condition that if this holds for one open affine cover, then it also holds for every open affine cover.

5.3 Properties of Schemes

The first thing that we would like to do with schemes is translate the theory of varieties into schemes. One obvious way to do this would be to simply take a prevariety with an affine cover, convert the affine cover to an affine scheme by taking the spectrum of the coordinate ring, and gluing back.

That is, for X a prevariety and U_i an affine open cover, with the intersections further covered by affine opens V_{ijk} covering $U_i \cap U_j$, we define

$$U_{i,\text{Sch}} = \text{Spec } A(U_i)$$

and we glue them along

$$V_{ijk,\text{Sch}} = \text{Spec } A(V_{ijk})$$

via the morphisms induced from the maps $A(U_i) \rightarrow A(V_{ijk})$. However, this is not obvious independent of choices. Instead, we will simply directly construct the resulting scheme and show that it matches the intuitive construction.

Definition 5.12

Let X be a prevariety K . Then we define X_{Sch} as a set by the set of all irreducible closed subsets of X . For any closed subset $Z \subseteq X$, we define $\mathcal{Z}$ to be the set of all irreducible closed subsets contained in Z , and use these as the closed subsets in a topology on X_{Sch} . Similarly for $U \subseteq X$ open we denote $\mathcal{U} = X_{\text{Sch}} \setminus \mathcal{Z}$, and define $\mathcal{O}_{X_{\text{Sch}}}(\mathcal{U}) = \mathcal{O}_X(U)$. Since all opens in X_{Sch} arise this way, this completely defines the structure sheaf, with the restrictions given in the inherited way.

Proposition 5.10

If X is a prevariety over K , then $(X_{\text{Sch}}, \mathcal{O}_{X_{\text{Sch}}})$ is a scheme. In addition, for a morphism $f : X \rightarrow Y$ of prevarieties, there is an induced morphism $f_{\text{Sch}} : X_{\text{Sch}} \rightarrow Y_{\text{Sch}}$.

Proof. We do not give the proof here, as it is mostly definition checking. The idea is that for the first claim, we follow the intuitive gluing idea from above. Begin with $U_i \subseteq X$ an open affine variety, and associate this with $\text{Spec } A(U)$. We check that there is a covering

map $\bigcup_{i \in I} \operatorname{Spec} A(U_i) \rightarrow X_{\text{Sch}}$. Since the elements of $\operatorname{Spec} A(U_i)$ correspond to irreducible closed subsets of U_i , taking the closures in X gives an irreducible closed subset of X . This motivates the definition that we have given for X_{Sch} as a set. One checks that the gluing also gives the topological structure. Finally, we can check that under this gluing, we still have $\mathcal{O}_X(U) \cong \mathcal{O}_{X_{\text{Sch}}}(\mathcal{U})$.

For the second claim, it suffices to glue the morphism along affine covers. First, cover Y by affine subsets V_i , and then cover $f^{-1}(V_i)$ by affine subsets U_{ij} . Thus we get $f : U_{ij} \rightarrow V_i$ which induces pullbacks $f^* : \mathcal{O}_Y(V_i) \rightarrow \mathcal{O}_X(U_{ij})$. The spectra $\operatorname{Spec} \mathcal{O}_X(U_{ij}), \operatorname{Spec} \mathcal{O}_Y(V_i)$ lie in $X_{\text{Sch}}, Y_{\text{Sch}}$, and we can show that these maps glue together into a map $X_{\text{Sch}} \rightarrow Y_{\text{Sch}}$. $\square$

In particular, since closed sets in X_{Sch} correspond to closed sets in X , their dimensions are the same.

Example 5.14

Consider the variety $\mathbb{P}^1$. Beginning with the gluing construction, we know $\mathbb{P}^1$ has cover by two copies of $\mathbb{A}^1$, so $\mathbb{P}_{\text{Sch}}^1$ should look like two copies of $\operatorname{Spec} K[x]$ and glue together along the map

$$\operatorname{Spec} K[x]_x \rightarrow \operatorname{Spec} K[w]_w, \quad w \mapsto \frac{1}{x}$$

On the side of schemes, this does the standard gluing for the closed points of $\operatorname{Spec} K[x]$. It also maps the two generic points to each other since the preimage of the zero ideal is the zero ideal. Thus we get a copy of $\mathbb{P}^1$, together with one generic point, in $\mathbb{P}_{\text{Sch}}^1$.

This aligns with the direction construction we gave, since we know $\mathbb{P}^1$ has closed irreducible subsets that are either points or the whole space. So indeed, the topological structure is a collection of closed points, plus a generic point for $\mathbb{P}^1$.

The map from $X \rightarrow X_{\text{Sch}}$ defines a fully faithful functor from prevarieties over K into schemes (which means $\operatorname{hom}(X, Y)$ bijects with $\operatorname{hom}(X_{\text{Sch}}, Y_{\text{Sch}})$ under the functor). This is also phrased as saying that prevarieties are a full subcategory of schemes.

We will now develop some language about properties of schemes that will allow us to classify what this subcategory looks like, or alternatively how to identify when a scheme arises from a (pre)variety. A few properties are clearly necessary: we know that prevarieties are finite gluings of affine varieties, and the coordinate rings of affine varieties are reduced finitely generated K -algebras. Also, the gluing maps $X \rightarrow Y$ correspond to K -algebra homomorphisms $A(Y) \rightarrow A(X)$. Lastly, for varieties, we said that X was separated when the diagonal Δ_X was closed in $X \times X$.

We will begin with definitions of the first three properties to identify schemes arising from prevarieties. Finally, we will define the product of two schemes to define separatedness.

Definition 5.13

Let Y be a scheme. Then a **scheme over Y** is a pair (X, f) of a scheme X with the data of a morphism $f : X \rightarrow Y$. A **morphism of schemes over Y** between $(X_1, f_1), (X_2, f_2)$ is a morphism $g : X_1 \rightarrow X_2$ so that the following commutes:

$$\begin{array}{ccc} X_1 & \xrightarrow{g} & X_2 \\ & \searrow f_1 & \swarrow f_2 \\ & Y & \end{array}$$

If S a ring then a scheme over S is a scheme over $\text{Spec } S$.

Note that if $X = \text{Spec } R$ then a scheme over a ring S is equivalent to the data of a ring homomorphism $S \rightarrow R$.

Definition 5.14

A scheme (X, f) over Y is of **finite type** over Y if there is an open cover (U_i) of Y so that each $f^{-1}(U_i)$ has a finite affine open cover $\text{Spec } R_{ij}$ so that the induced ring homomorphism $S_i \rightarrow R_{ij}$ defines a S_i -algebra structure on R_{ij} which is finitely generated. A scheme X is **reduced** if each $\mathcal{O}_X(U)$ is reduced.

Proposition 5.11

If X is a scheme, then the following are equivalent:

- (a) X is reduced,
- (b) There is an open cover of X by affine schemes $\text{Spec } R_i$ with each R_i reduced,
- (c) For every $\mathfrak{p} \in X$, $\mathcal{O}_{X, \mathfrak{p}}$ is reduced.

For the case of K a field, we know $\text{Spec } K$ is a single pointed space. A scheme over K , then is a morphism $f : X \rightarrow \text{Spec } K$. Of course the set theoretic map has no information, but the pullback $f_U^* : K \rightarrow \mathcal{O}_X(U)$ defines the structure of a K -algebra on each $\mathcal{O}_X(U)$. In particular, if $U = \text{Spec } R$ is an affine open, then R is a K -algebra.

Therefore, if X is of finite type over some field K , then (because there is only one open cover of $\text{Spec } K$) X admits an affine open cover $\text{Spec } R_i$, where each R_i is a finitely generated K -algebra. If X is also reduced then each R_i is also reduced.

Corollary 5.12

If Y is a scheme and K is an algebraically closed field, then Y is equal to X_{Sch} for some prevariety X over K if and only if Y is reduced and of finite type over K . In this case, $\text{hom}_{\text{Schemes}_K}(X_{\text{Sch}}, Y_{\text{Sch}}) \cong \text{hom}_{\text{Prevar}}(X, Y)$ via the map $f \mapsto f_{\text{Sch}}$.

Proof. The forward direction is obvious. In the reverse, we are given a finite cover $U_i = \text{Spec } R_i$ of Y with R_i finitely generated and reduced K -algebras. Each is the coordinate ring $R_i = A(X_i)$ of some affine variety over K . We can then glue these into a prevariety and verify that $X_{\text{Sch}} = Y$ (we do not formally do this here).

One can also verify that f_{Sch} is a morphism of schemes over K . To see that $f \mapsto f|_{\text{Sch}}$ is injective, one checks that there are injective maps making the following commute:

$$\begin{array}{ccc} X & \hookrightarrow & X_{\text{Sch}} \\ \downarrow f & & \downarrow f_{\text{Sch}} \\ Y & \hookrightarrow & Y_{\text{Sch}} \end{array}$$

This then implies that f_{Sch} is determined by f . For surjectivity, given a morphism of schemes over K from X_{Sch} to Y_{Sch} , we mimic the above and take an affine cover, then glue the induced maps of affine varieties together into a map $f : X \rightarrow Y$ of prevarieties, so that $g = f_{\text{Sch}}$. $\square$

Lastly, we need to figure out how to define the separatedness condition for arbitrary schemes. To do this, we need a proper notion of the product of two schemes.

This cannot be done set theoretically by building on the Cartesian product $X \times Y$. However, this does not work, since for instances $\mathbb{A}_K^1 \times \mathbb{A}_K^1 \cong \mathbb{A}_K^2$ as varieties. However, if we take the Cartesian product of sets underlying schemes then we get

$$\begin{aligned} (\mathbb{A}_K^1)_{\text{Sch}} \times (\mathbb{A}_K^1)_{\text{Sch}} &= \{([a], [b]) : (a, b) \in \mathbb{A}_K^2\} \cup \{([a], [\mathbb{A}_K^1]) : a \in \mathbb{A}_K^1\} \\ &\quad \cup \{([\mathbb{A}_K^1], [b]) : b \in \mathbb{A}_K^1\} \cup \{([\mathbb{A}_K^1], [\mathbb{A}_K^1])\} \end{aligned}$$

In particular, this misses most of the generic points for one dimensional varieties in $\mathbb{A}_K^2$.

Another approach is to recall that for affine varieties, $A(X \times Y) = A(X) \otimes_K A(Y)$, so we could try to define

$$\text{Spec } R \otimes \text{Spec } S = \text{Spec}(R \otimes S)$$

However, it is not clear which ring to take the tensor product over (we can always take it over $\mathbb{Z}$, but we may want to take it over a field).

Definition 5.15

Let $X = \text{Spec } R, Y = \text{Spec } S$ be affine schemes over $Z = \text{Spec } A$ (in particular, so that R, S have the structure of A -algebras). Then the **fiber product** of X, Y over Z is

$$\text{Spec } R \times_{\text{Spec } A} \text{Spec } S = \text{Spec}(R \otimes_A S)$$

This gives a commutative diagram using the induced maps $A \rightarrow R, S$, together with

the maps from the tensor products:

$$\begin{array}{ccc}
 R \otimes_A S & \xleftarrow{1 \otimes_A s \leftarrow s} & S \\
 \downarrow r \otimes_A 1 \leftarrow r & & \uparrow \\
 R & \xleftarrow{\quad} & A
 \end{array}$$

Taking the spectrum everywhere, the diagram reverses and we have

$$\begin{array}{ccc}
 X \times_Z Y & \xrightarrow{\pi_X} & Y \\
 \downarrow \pi_Y & & \downarrow f_Y \\
 X & \xrightarrow{f_X} & Z
 \end{array}$$

Example 5.15

If X, Y are affine varieties over K , then $X_{\text{Sch}} = \text{Spec } A(X)$, $Y_{\text{Sch}} = \text{Spec } A(Y)$. Then $X_{\text{Sch}} \times_{\text{Spec } K} Y_{\text{Sch}} = \text{Spec}(A(X) \otimes_K A(Y)) = \text{Spec}(A(X \times Y)) = (X \times Y)_{\text{Sch}}$. Thus the fiber product behaves as we expect for products of affine varieties.

Example 5.16

Let $X = \text{Spec } K$ and $Z = \text{Spec } K[t]$, with X a scheme over Z via the morphism induced from the K -algebra map $t \mapsto 0$. Let $Y = \text{Spec } K[x, y]$ and let it be a scheme over Z via the morphism induced from $t \mapsto xy$.

Then

$$X \times_Z Y = \text{Spec} (K \otimes_{K[t]} K[x, y]) = \text{Spec} (K[t]_{\langle t \rangle} \otimes_{K[t]} K[x, y])$$

A lemma from commutative algebra says that if $\varphi : R \rightarrow S$ is a ring homomorphism and J is an ideal in R , then

$$R/J \otimes_R S \cong S/\langle \varphi(J) \rangle$$

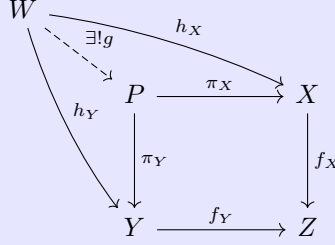
Then we get

$$X \times_Z Y = \text{Spec} (K[x, y]_{\langle xy \rangle})$$

In other words, taking the fiber product against the point X gave the scheme associated to the fiber $\varphi^{-1}(0)$ under $\varphi : t \mapsto xy$.

Definition 5.16

If $f_X : X \rightarrow Z, f_Y : Y \rightarrow Z$ are arbitrary schemes over Z , then a **fiber product** of X, Y over Z is a scheme P together with morphisms of schemes $\pi_X : P \rightarrow X, \pi_Y : P \rightarrow Y$, so that for any other scheme W with morphisms $h_X : W \rightarrow X, h_Y : W \rightarrow Y$ so that $f_X \circ h_X = f_Y \circ h_Y$, then there is a unique $g : W \rightarrow P$ so that the following commutes:



Proposition 5.13

If X, Y, Z are affine schemes, then the fiber product $X \times_Z Y$ as affine schemes that we defined earlier is indeed a fiber product in the general sense. If X, Y, Z are arbitrary schemes, then a fiber product exists and is unique up to unique isomorphism.

Proof. Again we just provide a sketch.

In the affine case, if W is affine as well the whole diagram is the opposite map of the tensor product, after taking spectra. For general W we simply choose a cover by affine schemes and then glue the morphisms, which shows both existence and uniqueness.

For general schemes we do a similar gluing of affine products (first covering Z , then preimages on X, Y , then preimages on P). $\square$

Since we have uniqueness, we denote the fiber product of X, Y over Z by $X \times_Z Y$.

Definition 5.17

Let $f_1 : X_1 \rightarrow Y, f_2 : X_2 \rightarrow Y$ be closed subschemes. Then the **scheme-theoretic intersection** $X_1 \cap X_2$ is defined to be the morphism $X_1 \times_Y X_2 \rightarrow Y$ which is given by $f_1 \circ \pi_{X_1} = f_2 \circ \pi_{X_2}$.

In the case that $U = \text{Spec } R$ is an affine open in Y , then the preimages in X_1, X_2 are of the form $\text{Spec } R/J_1, \text{Spec } R/J_2$ for two ideals $J_1, J_2 \subseteq R$, respectively. Taking the tensor products, we get a preimage in $X_1 \times_Y X_2$:

$$i^{-1}(U) = \text{Spec } R/J_1 \times_{\text{Spec } R} \text{Spec } R/J_2 = \text{Spec } (R/J_1 \otimes_R R/J_2) = \text{Spec } R/J_1 + J_2$$

so this agrees with our previous definition of scheme theoretic intersections for affine subschemes.

Definition 5.18

Let X be a scheme over Y . Then the **diagonal morphism** $\Delta_X : X \rightarrow X \times_Y X$ is the morphism induced by the pair $(\text{id}_X, \text{id}_X)$ from the universal property of $X \times_Y X$. X is **separated** over Y if Δ_X is a closed embedding.

Equivalently, X is separated over Y if Δ_X has an image closed in $X \times_Y X$.

Proposition 5.14

If K is an algebraically closed field then $X \mapsto X_{\text{Sch}}$ is a bijection between varieties over K and separated, reduced schemes of finite type over K . Moreover, this bijection induces a bijection between morphisms from $X \rightarrow Y$ and morphisms from $X_{\text{Sch}} \rightarrow Y_{\text{Sch}}$ as schemes over K .

This allows us to treat varieties as the same as their schemes.

Remark 5.2

We adopt the following conventions: a variety is now a separated reduced scheme of finite type over an algebraically closed field K . A point of a variety is a closed point in X_{Sch} . A morphism of varieties is a morphism over K .

Example 5.17

We now have the definition $\mathbb{A}_K^n = \text{Spec } K[x_1, \dots, x_n]$ when K is algebraically closed.

Example 5.18

To see why we need to require that the morphisms as schemes are morphisms over K , consider the complex conjugation map $\varphi : \mathbb{C}[x_1, \dots, x_n] \rightarrow \mathbb{C}[x_1, \dots, x_n]$ (which conjugates all coefficients). However, this is not a $\mathbb{C}$ -algebra morphism. The ring homomorphism induces a scheme morphism $\mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$, but this morphism is not a morphism of varieties since the diagram to $\text{Spec } \mathbb{C}$ does not commute.

Chapter 6

Sheaves over Schemes

At this point, it would be possible to study how the correspondence between schemes and rings gives geometric meaning to many properties of rings and ring homomorphisms. Instead, however, we will spend the remainder of this course studying sheaves of modules over a sheaf of rings, which will enable the study of, for instance, invariants on varieties.

6.1 Sheaves of Modules

The first motivation for studying sheaves of modules is to observe that there are many objects which may not look like sheaves of rings. For instance, they may not be equipped with multiplication, but still be compatible with multiplication by something else. In particular, we will look at sheaves over affine schemes $X = \operatorname{Spec} R$, with the sheaf elements given by an R -module M .

Example 6.1

If X is a smooth curve (for instance, the Riemann sphere $X = \mathbb{P}_{\mathbb{C}}^1$), then at each point $p \in X$ we have associated the tangent space $T_p X \cong \mathbb{C}^1$. A priori we have defined these to be independent, but we want to be able to make some sort of connection between them. For instance, we might want to look at the tangent sheaf

$$T_X(U) = \{\varphi = (\varphi_p)_{p \in U} : \varphi_p \in T_p U \text{ varies "nicely" with } p\}$$

Of course the definition of varying nicely will have to some precisely stated local definition to properly make this a definition. Each $T_X(U)$ is clearly a $\mathbb{C}$ -module since each $T_p U$ is a vector space. But we could also multiply by different scalars at each point p , rather than scaling globally. For instance, we could multiply by a regular function $\psi \in \mathcal{O}_X(U)$, defining

$$\psi \cdot \varphi = (\psi(p)\varphi_p)_{p \in U} \in T_X(U)$$

Thus $T_X(U)$ can be treated as a $\mathcal{O}_X(U)$ -module.

Definition 6.1

Let X be a scheme. A **(pre-)sheaf of modules** on X is a (pre-)sheaf $\mathcal{F}$ on X so that $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ module for all U open, so that the restrictions are $\mathcal{O}_X$ -module homomorphisms in the sense that

$$\begin{aligned}(\varphi + \psi)|_V &= \varphi|_V + \psi|_V \\ (\lambda\varphi)|_V &= \lambda|_V \varphi|_V\end{aligned}$$

for $\varphi, \psi \in \mathcal{F}(U)$, $\lambda \in \mathcal{O}_X(U)$ and $V \subseteq U \subseteq X$ open. We will occasionally denote this as an $\mathcal{O}_X$ module.

Note that the restriction is not technically a module homomorphism in the usual sense because λ also passes through the restriction map.

Definition 6.2

A **morphism of (pre-)sheaves of modules** on X is a collection $(f_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U))$ so that each f_U is a $\mathcal{O}_X(U)$ module homomorphism which is compatible with restrictions:

$$f_U(\varphi)|_V = f_V(\varphi|_V)$$

for $V \subseteq U \subseteq X$ and $\varphi \in \mathcal{F}(U)$. Such a morphism is denote $f : \mathcal{F} \rightarrow \mathcal{G}$.

Example 6.2

Taking $\mathcal{F} = \mathcal{O}_X$ clearly gives a sheaf of modules on X .

Example 6.3

If $\mathcal{F}, \mathcal{G}$ are two $\mathcal{O}_X$ modules then the **direct sum** $\mathcal{O}_X$ module $\mathcal{F} \oplus \mathcal{G}$ is given by

$$(\mathcal{F} \oplus \mathcal{G})(U) = \mathcal{F}(U) \oplus \mathcal{G}(U)$$

Example 6.4

Let $n \in \mathbb{N}$ and $d \in \mathbb{Z}$. If U is a nonempty open subset of $\mathbb{P}^n$ then we define the set of formal symbols

$$\begin{aligned}(\mathcal{O}_{\mathbb{P}^n}(d))(U) &= \left\{ \frac{g}{f} : f \in K[x_0, \dots, x_n]_e, g \in K[x_0, \dots, x_n]_{e+d} \text{ and } f(p) \neq 0 \forall p \in U \right\} \\ &\subseteq \text{Frac}(K[x_0, \dots, x_n])\end{aligned}$$

and $\mathcal{O}_{\mathbb{P}^n}(d)(\emptyset) = \{0\}$. When taken as sets, $\mathcal{O}_{\mathbb{P}^n}(d)$ is clearly a presheaf of sets with restrictions taken by the identity. To see that it is a sheaf, if there is an open cover U_i of $U \subseteq \mathbb{P}^n$ and $g_i/f_i \in \mathcal{O}_{\mathbb{P}^n}(d)(U_i)$ which are compatible on overlaps, then for

U_i, U_j nonempty, their intersection is nonempty as well. Thus

$$\frac{g_i}{f_i} = \frac{g_j}{f_j} \in \text{Frac}(K[x_0, \dots, x_n])$$

Since the denominator does not vanish at any U_i , it does not vanish on U either, so we have a global section g/f . Thus $\mathcal{O}_{\mathbb{P}^n}(d)$ is a sheaf of sets.

We define an abelian group structure formally via

$$\frac{g_1}{f_1} + \frac{g_2}{f_2} = \frac{g_1 f_2 + g_2 f_1}{f_1 f_2}$$

which still has a numerator of degree d more than the denominator. Note, however, that multiplication

$$\frac{g_1 g_2}{f_1 f_2}$$

does not give a ring structure, since this lands in the space $\mathcal{O}_{\mathbb{P}^n}(2d)$. More generally, there is a multiplication map

$$\mathcal{O}_{\mathbb{P}^n}(d) \times \mathcal{O}_{\mathbb{P}^n}(e) \rightarrow \mathcal{O}_{\mathbb{P}^n}(d+e), \quad \left(\frac{g_1}{f_1}, \frac{g_2}{f_2} \right) \mapsto \frac{g_1 g_2}{f_1 f_2}$$

We remark that as defined here, elements of $\mathcal{O}_{\mathbb{P}^n}(d)(U)$ cannot generally be interpreted as functions, since the numerators and denominators scale differently when $d \neq 0$. However, for $d = 0$, we have the collection of formal symbols

$$\mathcal{O}_{\mathbb{P}^n}(0)(U) = \left\{ \frac{g}{f} : g, f \in K[x_0, \dots, x_n], f(p) \neq 0 \forall p \in U \right\}$$

Since the numerator and denominator are of the same degree, they scale together, and indeed we know from before that each of these can be interpreted as a well defined regular function on U . Moreover, this definition is injective (when looking at the map from the structure sheaf to $\mathcal{O}_{\mathbb{P}^n}(0)(U)$) since if $g/f = 0$ on U then by density, it is zero everywhere and $g = 0$. It is surjective since if $\varphi \in \mathcal{O}_{\mathbb{P}^n}(U)$, then there is an open cover with $\varphi|_{U_i} = g_i/f_i$ and these definitions compatible. Thus, on $U_i \cap U_j$, we have

$$g_i f_j - g_j f_i = 0$$

and assuming that all U_i are nonempty, this implies that

$$\frac{g_i}{f_i} = \frac{g_j}{f_j}$$

as elements of the fraction field of $K[x_0, \dots, x_n]$. Since $K[x_0, \dots, x_n]$ is a UFD, we can assume from the outset that g_i, f_i are coprime up to units, so that there is a global representative g/f . Therefore $\mathcal{O}_{\mathbb{P}^n}(0) = \mathcal{O}_{\mathbb{P}^n}$ recovers the structure sheaf.

Combining this with the multiplication map for $e = 0$, we define a $\mathcal{O}_{\mathbb{P}^n}(U)$ module structure on $\mathcal{O}_{\mathbb{P}^n}(d)(U)$. $\mathcal{O}_{\mathbb{P}^n}(dd)$ is called the **twisting sheaf** of degree d on $\mathbb{P}^n$.

Example 6.5

Let us consider the global sections on $\mathbb{P}^n$. If $g/f \in \mathcal{O}_{\mathbb{P}^n}(d)(\mathbb{P}^n)$, then f must vanish nowhere, so $V(f) = \emptyset$, so by the Nullstellensatz, $f \in K^*$. Therefore $g/f \in K[x_0, \dots, x_n]_d$. So

$$\mathcal{O}_{\mathbb{P}^n}(d)(\mathbb{P}^n) = \begin{cases} K[x_0, \dots, x_n]_d, & d \geq 0 \\ \{0\}, & d < 0 \end{cases}$$

Example 6.6

Let $U_0 = \{x_0 \neq 0\} \subseteq \mathbb{P}^1$. Then

$$\frac{1}{x_0} \in \mathcal{O}_{\mathbb{P}^1}(-1)(U_0)$$

Example 6.7

For fixed $f \in K[x_0, \dots, x_n]_e$, the map $\varphi \mapsto \varphi f$ via the multiplication map defines a morphism of sheaves

$$\mathcal{O}_{\mathbb{P}^n}(d) \rightarrow \mathcal{O}_{\mathbb{P}^n}(d+e)$$

for all d .

Example 6.8

For $d \neq 0$, $\mathcal{O}_{\mathbb{P}^n}(d)$ is not isomorphic to $\mathcal{O}_{\mathbb{P}^n}$. Indeed, the global sections on $\mathbb{P}^n$ are isomorphic to K , but we just saw that $\mathcal{O}_{\mathbb{P}^n}(d)(\mathbb{P}^n)$ is not isomorphic to K for $d \neq 0$. However, if $U_i = \{x_i \neq 0\} \subseteq \mathbb{P}^n$, then the morphism

$$\mathcal{O}_{\mathbb{P}^n} \Big|_{U_i} \rightarrow \mathcal{O}_{\mathbb{P}^n}(d) \Big|_{U_i}$$

given by $\varphi \mapsto \varphi \cdot x_i^d$ is a morphism (actually on all of $\mathbb{P}^n$), with an inverse on U_i given by $\varphi \mapsto \varphi \cdot x_i^{-d}$. Therefore $\mathcal{O}_{\mathbb{P}^n}(U_i) \cong \mathcal{O}_{\mathbb{P}^n}(d)(U_i)$.

Example 6.9

Recall that $\mathbb{P}^n$ can be seen as the set of lines L in K^{n+1} through the origin. We define the **tautological sheaf**

$$\mathcal{F}(U) = \{\varphi : U \rightarrow K^{n+1} \text{ a morphism} : \varphi(L) \in L \forall L \in U \text{ closed points}\}$$

This is a local condition on functions, so it defines an $\mathcal{O}_X$ module.

One can show that $\mathcal{F} \cong \mathcal{O}_{\mathbb{P}^n}(-1)$.

In particular, this shows that there is no global section for the tautological sheaf, meaning there is no way to pick a nicely varying family of vectors that lie in the corresponding lines.

Let $\mathcal{F}$ be a sheaf of modules on X . Then for any section of $\mathcal{F}$ defined on a set containing p , and any germ of a regular function, one can multiply them on their common domain. This gives a $\mathcal{O}_{X,p}$ module structure on $\mathcal{F}_p$ for each stalk $p \in U$. Similarly, for any morphism $f : \mathcal{F} \rightarrow \mathcal{G}$ of $\mathcal{O}_X$ -modules, there are induced $\mathcal{O}_{X,p}$ -module homomorphisms $f_p : \mathcal{F}_p \rightarrow \mathcal{G}_p$.

Proposition 6.1

$f : \mathcal{F} \rightarrow \mathcal{G}$ is an isomorphism of sheaves if and only if $f_p : \mathcal{F}_p \rightarrow \mathcal{G}_p$ is an isomorphism of $\mathcal{O}_{X,p}$ modules for all p .

Definition 6.3

Let $f : X \rightarrow Y$ be a morphism of schemes and $\mathcal{F}$ a sheaf on X . Then the **pushforward** of $\mathcal{F}$ under f is the sheaf $f_*\mathcal{F}$ defined by

$$(f_*\mathcal{F})(U) = \mathcal{F}(f^{-1}(U))$$

This defines a sheaf on Y , and $f_*\mathcal{F}$ is in fact an $\mathcal{O}_Y$ module, since if $\lambda \in \mathcal{O}_Y(U)$ and $\varphi \in \mathcal{F}(f^{-1}(U))$, then they can be multiplied via

$$(f_U^*)(\lambda)\varphi$$

In particular, recall that the data of a morphism of schemes $f : X \rightarrow Y$ was the data of a pullback $f_U^* : \mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(f^{-1}(U))$. But this is precisely the evaluation on $(f_*\mathcal{O}_X)(U)$. Thus the collection of f_U^* gives the data of a morphism of sheaves on Y $f^* : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$.

As a result, we could have alternatively defined a morphism between locally ringed spaces as the data of a continuous map $f : X \rightarrow Y$ and a morphism of sheaves of rings on Y $f^* : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$, so that the maps on stalks preserve the maximal ideals.

Example 6.10

Let Y be a variety and $p \in Y$ a closed point, with $i : \{p\} \rightarrow Y$ the inclusion. The sheaf on $\{p\}$ is

$$\mathcal{O}_{\{p\}}(\{p\}) = K, \mathcal{O}_{\{p\}}(\emptyset) = \{0\}$$

Taking the pushforward along i , we get

$$i_*\mathcal{O}_{\{p\}}(U) = \mathcal{O}_{\{p\}}(i^{-1}(U)) = \begin{cases} K, & p \in U \\ \{0\}, & p \notin U \end{cases}$$

$i_*\mathcal{O}_{\{p\}}$ is called the **skyscraper sheaf** on Y at p , and is sometimes denoted K_p . For $f \in \mathcal{O}_Y(U)$, if $p \notin U$ then the pullback necessarily sends f to 0. Otherwise, it

evaluates f at p .

We now introduce two constructions that are closely related to constructions in commutative algebra.

Definition 6.4

Let $f : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves on a scheme X . Then the **kernel sheaf** $\ker f$ is defined as

$$(\ker f)(U) = \ker(f_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U))$$

Proposition 6.2

The kernel sheaf is indeed a sheaf.

Proof. First, to see that this is a presheaf, notice that if $\varphi \in (\ker f)(U)$, then for $V \subseteq U$,

$$f_V(\varphi|_V) = f_U(\varphi)|_V = 0|_V = 0$$

because restrictions commute with morphisms of sheaves. Thus $\varphi|_V \in (\ker f)(V)$.

To see that we can glue, let $(U_i)_i$ be an open cover of U and let $\varphi_i \in (\ker f)(U_i)$ which agree on the overlaps. Since $\mathcal{F}$ is a sheaf there is a unique φ where $\varphi|_{U_i} = \varphi_i$, so we already have uniqueness. Moreover, let $\psi = f_U(\varphi)$. Then since f commutes with restrictions, we have

$$\psi|_{U_i} = f_{U_i}(\varphi_i) = 0$$

so $\psi = 0$ and thus $\varphi \in \ker f$. So we have unique gluing and thus $\ker f$ is a sheaf. $\square$

Effectively, the idea is that we can verify whether a section is in the kernel by checking that it is zero on all open covers, so this leads to a sheaf.

Trying to do the same with the image runs into some issues:

Definition 6.5

Let $f : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves on a scheme X . Then the **image presheaf** $\operatorname{im}' f$ is defined as

$$(\operatorname{im}' f)(U) = \operatorname{im}(f_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U))$$

Proposition 6.3

$\operatorname{im}' f$ is a presheaf.

Proof. Let $\psi \in (\operatorname{im}' f)(U)$ and write $\psi = f_U(\varphi)$. Then for $V \subseteq U$ open, we have

$$\psi|_V = (f_U(\varphi))|_V = f_V(\varphi|_V) \in \operatorname{im} f_V$$

so $\psi|_V \in (\operatorname{im}' f)(V)$. Thus $\operatorname{im}' f$ is a presheaf. $\square$

However, this is general not a sheaf, effectively because being in the image cannot be checked on an open cover (a point in the image need only be in the image for one set in the cover, not all). Also, if we are given sections ψ_i which agree in the image presheaf, we don't necessarily know that that preimages agree, so we cannot necessarily glue them together.

Example 6.11

As an explicit counterexample which is not a sheaf, take the inclusion morphism of schemes

$$i : \mathbb{A}^1 \rightarrow \mathbb{A}^2, \quad x_1 \mapsto (x_1, 0)$$

Now, we consider two sheaves on the scheme $X = \mathbb{A}^2$: the structure sheaf, and the pushforward sheaf $i_*\mathcal{O}_{\mathbb{A}^1}$. There is a morphism between these given by

$$i^* : \mathcal{O}_{\mathbb{A}^2} \rightarrow i_*\mathcal{O}_{\mathbb{A}^1}$$

Set $f = i^*$.

We want to show the sheaf axiom fails, so we need to find an open cover of an open set with sections that do not glue. We will take

$$U = \mathbb{A}^2 \setminus \{(0, 0)\}$$

with open cover $U_1 = \{x_1 \neq 0\}, U_2 = \{x_2 \neq 0\}$. $\mathcal{O}_{\mathbb{A}^2}$ has structure sheaf for distinguished open subsets given by:

$$\mathcal{O}_{\mathbb{A}^2}(U_1) = K[x_1, x_2]_{x_1}, \quad \mathcal{O}_{\mathbb{A}^2}(U_2) = K[x_1, x_2]_{x_2}, \quad \mathcal{O}_{\mathbb{A}^2}(U_1 \cap U_2) = K[x_1, x_2]_{x_1 x_2}$$

On the union, the regular functions are just the restrictions of polynomials:

$$\mathcal{O}_{\mathbb{A}^2}(U_1 \cup U_2) = K[x_1, x_2]$$

For the pushforward sheaf, we simply compute using preimages

$$i_*\mathcal{O}_{\mathbb{A}^1}(U_1) = \mathcal{O}_{\mathbb{A}^1}(\mathbb{A}^1 \setminus \{0\}) = K[x_1]_{x_1}, \quad i_*\mathcal{O}_{\mathbb{A}^1}(U_2) = \mathcal{O}_{\mathbb{A}^1}(\emptyset) = \{0\}$$

$i_*\mathcal{O}_{\mathbb{A}^1}(U_1 \cap U_2)$ is also the zero ring since $U_1 \cap U_2 \subseteq U_2$. and

$$i_*\mathcal{O}_{\mathbb{A}^1}(U_1 \cup U_2) = \mathcal{O}_{\mathbb{A}^1}(\mathbb{A}^1 \setminus \{0\}) = K[x_1]_{x_1}$$

The pullback is basically given set theoretically, by setting $x_2 = 0$. So the images are calculated as

$$\begin{aligned} (\text{im}' f)(U_1) &= K[x_1]_{x_1} \\ (\text{im}' f)(U_2) &= \{0\} \\ (\text{im}' f)(U_1 \cap U_2) &= \{0\} \\ (\text{im}' f)(U_1 \cup U_2) &= K[x_1] \end{aligned}$$

In particular, $(\text{im}' f)(U_1 \cup U_2) \neq i_* \mathcal{O}_{\mathbb{A}^1}(U_1 \cup U_2)$ but they agree on the other sets. Take

$$\frac{1}{x_1} \in K[x_1]_{x_1} = (\text{im}' f)(U_1), \quad 0 \in (\text{im}' f)(U_2)$$

Since $1/x_1$ has to restrict to zero on the overlap $U_1 \cap U_2$, these sections are compatible, but they do not glue to a global section since no global section restricts to $1/x_1$ on U_1 .

Thus we see that the inability to directly define a sheaf is an actual barrier that we need to overcome. Before resolving this we will see one more example of a presheaf which does not define a sheaf.

Definition 6.6

Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$ modules. We define the **tensor presheaf** $\mathcal{F} \otimes' \mathcal{G}$ by

$$(\mathcal{F} \otimes' \mathcal{G})(U) = \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$$

We can see that this is a presheaf since restriction on each of $\mathcal{F}(U), \mathcal{G}(U)$ is tensored into a restriction map on the product.

Example 6.12

Let $X = \mathbb{P}_1$ and take the twisting sheaves $\mathcal{F} = \mathcal{O}_X(1), \mathcal{G} = \mathcal{O}_X(-1)$. Take the open cover $U_0 = \{x_0 \neq 0\}, U_1 = \{x_1 \neq 0\}$. Take the sections

$$\begin{aligned} x_0 \otimes \frac{1}{x_0} &\in (\mathcal{F} \otimes' \mathcal{G})(U_0) \\ x_1 \otimes \frac{1}{x_1} &\in (\mathcal{F} \otimes' \mathcal{G})(U_1) \end{aligned}$$

We claim these are compatible on the overlap $U_1 \cap U_2 = D(x_0 x_1)$. To see this, note that on the intersections,

$$\frac{x_0}{x_1}, \frac{x_1}{x_0}$$

are both valid regular functions, and their product is equal to 1. So on the overlap,

$$x_0 \otimes \frac{1}{x_0} = \left(\frac{x_0}{x_1} \frac{x_1}{x_0} \right) \left(x_0 \otimes \frac{1}{x_0} \right) = \left(x_0 \frac{x_1}{x_0} \right) \otimes \left(\frac{1}{x_0} \frac{x_0}{x_1} \right) = x_1 \otimes \frac{1}{x_1}$$

So the sections agree when restricted to $U_1 \cap U_2$. However, there is no global section which restricts to these sections. This is because $\mathcal{G}(\mathbb{P}^1) = \mathcal{O}_{\mathbb{P}^1}(-1)(\mathbb{P}^1) = \{0\}$ since the negative twisting sheaves have no global sections. Thus the tensor sheaf also has only zero as a global section.

The one subtlety is that it is not a priori obvious that the sections we have chosen are not zero in the tensor product. To see that they are nonzero, we consider

$$\mathcal{F}(U_0) \otimes_{\mathcal{O}_X(U_0)} \mathcal{G}(U_0)$$

But on each U_0 , we saw before that the twisting sheaves are all isomorphic as

$$\mathcal{O}_{\mathbb{P}^n}(d)(U_i) \cong \mathcal{O}_{\mathbb{P}^n}(U_i) \implies \mathcal{O}_{\mathbb{P}^n}(d)(U_i) \cong K[x_i]$$

and explicitly the isomorphism is by multiplication by x_i^d , which forces

$$\mathcal{F}(U_0) \otimes_{\mathcal{O}_X(U_0)} \mathcal{G}(U_0) = (K[x_0] \cdot x_0) \otimes_{K[x_0]} \left(K[x_0] \cdot \frac{1}{x_0} \right) = K[x_0] \cdot \left(x_0 \otimes \frac{1}{x_0} \right)$$

so $x_0 \otimes \frac{1}{x_0}$ is in fact the generator (as a $K[x_0]$ module) of $\mathcal{F} \otimes' \mathcal{G}(U_0)$, and hence nonzero. So the global sections cannot restrict to our sections, meaning they cannot glue, and thus the tensor presheaf here is not a sheaf.

6.2 Sheafification and Exact Sequences

To avoid this issue, we will expand our attention slightly to sections which only look locally like elements of our presheaf. By changing the condition to a local one, we will immediately upgrade to a sheaf. This procedure can be done for any presheaf.

Definition 6.7

Let $\mathcal{F}'$ be a presheaf on a scheme X . Then the **sheafification** of $\mathcal{F}'$ is the sheaf $\mathcal{F}$ with sections given by

$$\mathcal{F}(U) = \left\{ \begin{array}{l} \varphi = (\varphi_p)_{p \in U} : \varphi_p \in \mathcal{F}'_p, \\ \forall p \in U \text{ there is an open neighborhood } U_p \ni p \\ \text{and a section } s \in \mathcal{F}'(U_p) \text{ s.t. } \varphi_q = s_q \forall q \in U_p \end{array} \right\}$$

That is, a section is a collection of germs, so that locally the germs correspond to a section of the presheaf. The notation $\varphi_q = s_q$ means that the germ of s at q is the same (under germ equivalence) as the element φ_q .

This is a presheaf since under a restriction, we simply forget the additional germs, and this is still locally a presheaf section as we would just take $V_p = U_p \cap V$, $s = s|_{V_p}$.

Example 6.13

We saw previously that looking at the collection of functions on affine varieties X that are global quotients of polynomials:

$$\mathcal{O}'_X(U) = \left\{ \varphi : U \rightarrow K, \exists f, g \in A(X) \text{ s.t. } \varphi = \frac{g}{f} \right\}$$

then this defines a presheaf, but not a sheaf. However, sheafification of this presheaf returns precisely the structure sheaf of regular functions.

Example 6.14

We saw above that the tensor presheaf $\mathcal{O}_{\mathbb{P}^1}(1) \otimes' \mathcal{O}_{\mathbb{P}^1}(-1)$ is not a sheaf, since

$$s_0 = x_0 \otimes \frac{1}{x_0}, \quad s_1 = x_1 \otimes \frac{1}{x_1}$$

agree on the overlap $\{x_0 \neq 0, x_1 \neq 0\}$ but do not glue into a global tensor section. However, in the sheafification, they do glue to the section

$$\varphi_p = \begin{cases} (s_0)_p, & p \in U_0 \\ (s_1)_p, & p \in U_1 \end{cases}$$

which is well defined because $(s_0)_p = (s_1)_p$ on the overlap. Moreover, at every point there is a neighborhood (lying in either U_0, U_1) on which φ is given by one of the s_i .

There is a natural morphism θ from a presheaf to its sheafification given by

$$\theta_U : \mathcal{F}'(U) \rightarrow \mathcal{F}(U), \quad s \mapsto (s_p)_{p \in U}$$

We can use this morphism to study how sheafification changes the properties of $\mathcal{F}'$. In particular, it should not affect local properties (i.e. those of stalks) since sheafification imposes a local condition, and if $\mathcal{F}'$ is a sheaf it should not do anything.

Proposition 6.4

Let $\mathcal{F}'$ be a presheaf on a scheme X and let $\mathcal{F} = (\mathcal{F}')^{\text{sh}}$ be the sheafification, with the natural morphism $\theta : \mathcal{F}' \rightarrow \mathcal{F}$ as above. Then for any $p \in X$, the induced morphism of stalks

$$\theta_p : \mathcal{F}'_p \rightarrow \mathcal{F}_p$$

is an isomorphism. Moreover, if $\mathcal{F}'$ is itself a sheaf then θ is an isomorphism of sheaves.

Proof. For the first claim, we construct the inverse map $\zeta_p : \mathcal{F}_p \rightarrow \mathcal{F}'_p$, by defining for any germ

$$[(U, \varphi = (\varphi_q)_{q \in U})] \mapsto \varphi_p$$

which simply extracts the germ at p . Then for $[(U, s)] \in \mathcal{F}'_p$, $\theta_p([(U, s)])$ is the germ of the section $\varphi = (\varphi_q)_q$ with germs all given by $\varphi_q = [(U, s)]$. So

$$\zeta_p \circ \theta_p([(U, s)]) = \zeta_p([(U, ((U, s))_{q \in U})]) = \varphi_p = [(U, s)]$$

so $\zeta_p \circ \theta_p$ is the identity. In the other direction, let $[(U, \varphi)] \in \mathcal{F}_p$. Up to taking a smaller open set U' , we can assume that φ is a section of $\mathcal{F}'$, so that we have $[(U, \varphi)] = [(U', \varphi = (s_q)_q)]$ for some $s \in \mathcal{F}'(U')$. Then

$$\theta_p \circ \zeta_p([(U', \varphi = (s_q)_q)]) = \theta_p(s_q) = \theta_p([(U', s)]) = [(U', ((U', s))_q)] = [(U', (s_q)_q)]$$

Thus both compositions are the identity.

Then the second claim follows because we know that a morphism of sheaves is an isomorphism if and only if all the induced stalk morphisms are isomorphisms. $\square$

The following universal property characterizes the sheafification up to unique isomorphism.

Proposition 6.5

Let $\mathcal{F}'$ be a presheaf on X and let $\theta : \mathcal{F}' \rightarrow \mathcal{F} = (\mathcal{F}')^{\text{sh}}$ be the natural morphism. Then if $f' : \mathcal{F}' \rightarrow \mathcal{G}$ is any morphism to a sheaf $\mathcal{G}$, there is a unique factorization

$$\begin{array}{ccc} \mathcal{F}' & \xrightarrow{\theta} & \mathcal{F} \\ & \searrow f' & \swarrow \exists! f \\ & \mathcal{G} & \end{array}$$

Aside from characterizing the sheafification, this universal property also allows us to detect properties of f' via the factorized f . For instance, let $f : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves of $\mathcal{O}_X$ -modules on a scheme X . Then we can say that f is **injective** if the kernel sheaf is the zero sheaf:

$$\ker f = 0$$

To characterize surjectivity, we might be tempted to say that the image of each module homomorphism f_U is all of $\mathcal{G}(U)$; that is, that the image presheaf is $\mathcal{G}$. But since the image presheaf is in general not a sheaf, this won't give us the desired properties. Instead, it is better to work with the image sheaf.

Definition 6.8

Let $f : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves of $\mathcal{O}_X$ -modules on a scheme X . Then the **image sheaf** is the sheafification of the image presheaf:

$$\text{im } f = (\text{im}' f)^{\text{sh}}$$

Then, the universal property gives us a diagram of the form

$$\begin{array}{ccccc} \mathcal{F} & \longrightarrow & \text{im}' f & \xrightarrow{\theta} & \text{im } f \\ & \searrow f & \downarrow i & \swarrow \exists! \tilde{f} & \\ & & \mathcal{G} & & \end{array}$$

where the arrow $\mathcal{F} \rightarrow \text{im}' f$ is the natural morphism induced by f , and i is the inclusion. Now, we say that f is **surjective** if $\tilde{f} : \text{im } f \rightarrow \mathcal{G}$ is an isomorphism.

Now, suppose that $f : \mathcal{F} \rightarrow \mathcal{G}$ is injective. Then for each open U , $\mathcal{F}(U)$ can be seen as a sub- $\mathcal{O}(U)$ -module of $\mathcal{G}(U)$ via f . Thus we can take the quotient presheaf

$$(\mathcal{G}/\mathcal{F})(U) = \mathcal{G}(U)/\mathcal{F}(U)$$

As with the constructions before, this results in a presheaf, but not a sheaf.

Definition 6.9

Let $f : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of $\mathcal{O}_X$ -modules on a scheme X . Then the **quotient sheaf** is the sheafification

$$\mathcal{G}/\mathcal{F} = (\mathcal{G}/'\mathcal{F})^{\text{sh}}$$

As an extension of this, consider a sequence of sheaves and morphisms of this form.

$$\cdots \longrightarrow \mathcal{F}_i \xrightarrow{f_i} \mathcal{F}_{i+1} \xrightarrow{f_{i+1}} \mathcal{F}_{i+2} \longrightarrow \cdots$$

Whenever $f_{i+1} \circ f_i$ is the zero morphism for all i , this sequence is called a **complex**. In this case, on each open set we have the inclusion

$$\text{im}(f_i)_U \subseteq \ker(f_{i+1})_U \subseteq \mathcal{F}_{i+1}(U)$$

as $\mathcal{O}_X(U)$ -modules. Hence the inclusion on each U gives a morphism $\text{im}' f_i \rightarrow \ker f_{i+1}$, and since $\ker f_{i+1}$ is already a sheaf, the universal property gives a morphism $\text{im} f_i \rightarrow \ker f_{i+1}$. We then say that this sequence is **exact** if the induced maps $\text{im} f_i \rightarrow \ker f_{i+1}$ are all isomorphisms.

Definition 6.10

Let $\mathcal{F}, \mathcal{G}$ be two sheaves on a scheme X . Then the **tensor sheaf** is the sheafification of the tensor presheaf

$$\mathcal{F} \otimes \mathcal{G} = (\mathcal{F} \otimes' \mathcal{G})^{\text{sh}}$$

Finally, we may want to consider the dual of a sheaf, which we would normally think of as maps from the sheaf into the base structure (here, the structure sheaf $\mathcal{O}_X$). The definition is a bit delicate, though.

Definition 6.11

Let $\mathcal{F}$ be a sheaf on X . Then the **dual sheaf** $\mathcal{F}^\vee$ is the sheafification of the presheaf

$$U \mapsto \text{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{O}_U)$$

Note that here we need to consider morphisms of the restricted sheaves on U . This is because such morphisms carry the data of the map on smaller sets, and hence any such morphism can be restricted to a smaller open V . If we instead considered

$$\text{Hom}_{\mathcal{O}_X(U)}(\mathcal{F}(U), \mathcal{O}_X(U))$$

as simply module homomorphisms, we would have no guarantee that the homomorphisms could be restricted, and indeed this is in general not a presheaf.

Now, we return to exact sequences and study them a bit more. First, we have the property that taking the kernel and image commutes with taking stalks:

Proposition 6.6

Let $f : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of $\mathcal{O}_X$ modules on a scheme X , and let $f_p : \mathcal{F}_p \rightarrow \mathcal{G}_p$ be the induced stalk morphism. Then

$$(\ker f)_p = \ker f_p$$

and

$$(\operatorname{im} f)_p = \operatorname{im}(f_p)$$

Lemma 6.7

Consider a sequence of morphisms of $\mathcal{O}_X$ modules of the form

$$\cdots \longrightarrow \mathcal{F}_i \xrightarrow{f_i} \mathcal{F}_{i+1} \xrightarrow{f_{i+1}} \mathcal{F}_{i+2} \longrightarrow \cdots \quad (*)$$

on a scheme X . Then the following are equivalent:

- (a) $(*)$ is an exact sequence,
- (b) The restricted sequence

$$\cdots \longrightarrow \mathcal{F}_i \Big|_U \xrightarrow{f_i|_U} \mathcal{F}_{i+1} \Big|_U \xrightarrow{f_{i+1}|_U} \mathcal{F}_{i+2} \Big|_U \longrightarrow \cdots$$

is an exact sequence for any U open,

- (c) The restricted sequence

$$\cdots \longrightarrow \mathcal{F}_i \Big|_{U_k} \xrightarrow{f_i|_{U_k}} \mathcal{F}_{i+1} \Big|_{U_k} \xrightarrow{f_{i+1}|_{U_k}} \mathcal{F}_{i+2} \Big|_{U_k} \longrightarrow \cdots$$

is an exact sequence for every U_k in some open cover of X ,

- (d) The induced sequence on stalks

$$\cdots \longrightarrow (\mathcal{F}_i)_p \xrightarrow{(f_i)_p} (\mathcal{F}_{i+1})_p \xrightarrow{(f_{i+1})_p} (\mathcal{F}_{i+2})_p \longrightarrow \cdots$$

is exact for all $p \in X$.

Note that for (b), we have to work with the entire restricted sheaves and not just the modules $\mathcal{F}_i(U)$, and (c) tells us that if this is true for some open cover then it is true for all opens.

Proof. (a) $\implies$ (d)

If $(*)$ is exact then the induced maps $\operatorname{im}(f_i) \rightarrow \ker(f_{i+1})$ are all isomorphisms. Localizing

to stalks, we get isomorphism of stalks

$$(\operatorname{im} f_i)_p \xrightarrow{\cong} (\ker f_{i+1})_p$$

Since taking images and kernels commutes with taking stalks, we get an isomorphism

$$\operatorname{im}((f_i)_p) \xrightarrow{\cong} \ker((f_{i+1})_p)$$

so the sequence on stalks is an isomorphism.

(d) $\implies$ (a)

First we show that the sequence $(*)$ is a complex. Now, note that taking stalks also commutes with composition, so that

$$(f_{i+1} \circ f_i)_p = (f_{i+1})_p \circ (f_i)_p$$

The RHS is zero by assumption, so $f_{i+1} \circ f_i$ agrees with the zero morphism on all stalks, and is therefore the zero morphism. Since $(*)$ is a complex we have the induced maps $\operatorname{im} f_i \rightarrow \ker f_{i+1}$. But then by commuting stalks with kernels and images, we see that at all stalks these maps are isomorphisms. Therefore $\operatorname{im} f_i \xrightarrow{\cong} \ker f_{i+1}$ and thus $(*)$ is exact.

Now, we know that we can always check exactness at the stalk level. Therefore running the same argument shows that (b) $\iff$ (d) and (c) $\iff$ (d) as well. $\square$

Example 6.15

Let $p = (1 : 0) \in X = \mathbb{P}^1$ and consider $i : \{p\} \rightarrow X$ the inclusion. This gives a sequence

$$0 \longrightarrow \mathcal{O}_X(-1) \xrightarrow{f} \mathcal{O}_X \xrightarrow{g} K_p \longrightarrow 0 \quad (*)$$

where f is the morphism given by multiplication by x_1 , and $g : \mathcal{O}_X \rightarrow i_*\mathcal{O}_{\{p\}} = K_p$ is the evaluation at p . We want to say that $(*)$ is exact.

First, we need to show that $\ker f$ is the zero sheaf. Let U be open and let $\varphi \in \ker(f)$ on some open set U . Then

$$\varphi x_1 = 0 \in \operatorname{Frac}(K[x_0, x_1])$$

Since there are no nonzero zero divisors in a field, we get that $\varphi = 0$.

To compute the image sheaf of f , we begin with the presheaf. $\operatorname{im}' f$ is the set of regular functions of the form φx_1 , which is equivalent to the set of functions vanishing on p . This is in fact a local condition, so $\operatorname{im}' f = \operatorname{im} f$. Thus we have

$$\operatorname{im}(f)(U) = \{\varphi \in \mathcal{O}_X(U) : \varphi(p) = 0 \text{ if } p \in U\}$$

Now, note that if $p \in U$, then $g(\varphi) = \varphi(p)$, and $g(\varphi) = 0$ otherwise. So $\operatorname{im}(f)(U) = \ker(g)(U)$. Finally, we need to say that g is surjective. So we will show that $\operatorname{im}'(g) = K_p$. Indeed, taking the constant functions gives any germ needed. So $\operatorname{im}'(g) = K_p$, and thus sheafifying does nothing:

$$\operatorname{im} g = K_p$$

so g is surjective and $(*)$ is exact.

Example 6.16

Continuing the previous example, let p, i, X be as before, and take the second point $q = (0 : 1) \in X$. Then we get a double skyscraper sequence

$$0 \longrightarrow \mathcal{O}_X(-2) \xrightarrow{f'} \mathcal{O}_X \xrightarrow{h} K_p \oplus K_q \longrightarrow 0 \quad (**)$$

where f' is now multiplication by x_1x_2 , and h is the double evaluation map at both p, q . We claim that $(**)$ is exact as well. Indeed, if we check on the open charts $U_i = \mathbb{P} \setminus V(x_i)$, then we get restrictions of the single skyscraper sequence from $(*)$. By the lemma, the restrictions must be exact, and since this forms an open cover, $(**)$ is also exact.

In particular, h is surjective as a morphism of sheaves. However, if we look at the induced evaluation map on global sections, $\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$ is only the constant functions and hence

$$h_{\mathbb{P}^1} : \mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) \rightarrow (K_p \oplus K_q)(\mathbb{P}^1)$$

is the diagonal map $k \mapsto (k, k)$ from $K \rightarrow K^2$, which is not surjective. So surjectivity of the maps on open sets is not guaranteed.

Example 6.17

Recall that for $d, e \in \mathbb{Z}$, there is a multiplication map

$$\mathcal{O}_{\mathbb{P}^n}(d)(U) \otimes_{\mathcal{O}_{\mathbb{P}^n}(U)} \mathcal{O}_{\mathbb{P}^n}(e)(U) \rightarrow \mathcal{O}_{\mathbb{P}^n}(d+e)(U), \quad \varphi \otimes \psi \mapsto \varphi\psi$$

as module homomorphisms. This is true on each open set, which are the sections of the tensor presheaf. Since these agree on overlaps, we get a morphism of sheaves m :

$$\begin{array}{ccc} \mathcal{O}_{\mathbb{P}^n}(d) \otimes' \mathcal{O}_{\mathbb{P}^n}(e) & & \\ \downarrow & \searrow m & \\ \mathcal{O}_{\mathbb{P}^n}(d) \otimes \mathcal{O}_{\mathbb{P}^n}(e) & \xrightarrow[\exists! \hat{m}]{} & \mathcal{O}_{\mathbb{P}^n}(d+e) \end{array}$$

Here $\hat{m}$ is the induced morphism from the universal property. However, on any U_i , we know that the $\mathcal{O}_{\mathbb{P}^n}(d)(U_i)$ are all isomorphic, so the restrictions of $\hat{m}$ are an isomorphisms. Then translating this into an exact sequence condition, since we can check whether $\hat{m}$ is injective and surjective on an open cover, $\hat{m}$ is in fact an isomorphism.

6.3 Quasi-Coherent Sheaves

Now, we narrow our focus slightly. It turns out that the full set of arbitrary sheaves of modules on a schemes is a bit too general to work with productively. Instead, we will restrict ourself to a slightly smaller class of sheaves which arise from modules.

To see this, let R be a ring and M an R -module. Then there is a natural way to construct an induced sheaf of modules on the scheme $\text{Spec } R$. If one has a sheaf that looks like this, at least locally, this will be a tight enough connection to algebra to be able to transport our algebraic tools to schemes.

Definition 6.12

Let R be a ring, M an R -module. Then the **sheaf associated to M** is the sheaf $\tilde{M}$ with

$$\tilde{M}(U) = \left\{ \begin{array}{l} \varphi = (\varphi_p)_{p \in U} : \varphi_p \in \tilde{M}_p, \\ \forall p \in U \text{ there is an open neighborhood } U_p \ni p \\ \text{and } g \in M, f \in R \text{ s.t. } \forall q \in U_p, f \notin q \text{ and } \varphi_q = \frac{g}{f} \in M_q \forall q \in U_p \end{array} \right\}$$

Note that this is very similar to the definition of regular functions, and for $M = R$ we actually have $\tilde{R} = \mathcal{O}_{\text{Spec } R}$. All of the arguments we made before show that $\tilde{M}$ is a sheaf and $\tilde{M}(U)$ is a $\tilde{R}(U)$ module, since the multiplication only happens in the numerator. Thus $\tilde{M}$ is indeed a module over $\mathcal{O}_{\text{Spec } R}$.

Proposition 6.8

Let $X = \text{Spec } R$ be an affine scheme and M an R -module. Then

- (a) For any $p \in X$, the stalk $\tilde{M}_p$ is isomorphic to the localization M_p ,
- (b) For any $f \in R$, the sections $\tilde{M}(D(f))$ on the distinguished open subsets are M_f . In particular for $f = 1$, $\tilde{M}(X) = M$.

Proof. We have already made these proofs for the case $M = R$ when studying the localization of structure sheaves. None of these arguments needed to multiply the numerators of the fractions, so the arguments work the same. $\square$

Example 6.18

Not every module over $\mathcal{O}_{\text{Spec } R}$ is of the form above, although most are. For instance, take $R = K[x]_{\langle x \rangle}$ (recall the localization here allows denominators that are not in the ideal $\langle x \rangle$, which is any polynomial with nonzero constant term). Then this is the stalk $\mathcal{O}_{\mathbb{A}^1, 0}$.

The prime ideals of $K[x]_{\langle x \rangle}$ are the prime ideals of $K[x]$ which are contained in $\langle x \rangle$, hence the spectrum is a two point set

$$\operatorname{Spec} R = \{\langle 0 \rangle, \langle x \rangle\}$$

$\langle x \rangle$ is a closed point, so $\{\langle 0 \rangle\}$ is a dense open subset, and the only other open sets are empty or all of $\operatorname{Spec} R$. Then we can define the following presheaf:

$$\begin{aligned}\mathcal{F}(X) &= \{0\} \\ \mathcal{F}(\{\langle 0 \rangle\}) &= K[x]_{\langle 0 \rangle} \\ \mathcal{F}(\emptyset) &= 0\end{aligned}$$

The motivation for the choice of $K[x]_{\langle 0 \rangle}$ is that on the distinguished open set $D(x)$, the sections should be the localization R_x . But $K[x]_{\langle x \rangle}$ already allows denominators of any function with nonzero constant term, and localization at x then allows any denominator except zero. Note that each $\mathcal{F}(U)$ is indeed an $\mathcal{O}_{\operatorname{Spec} R}(U)$ module, as

$$\mathcal{O}_{\operatorname{Spec} R}(X) = R, \mathcal{O}_{\operatorname{Spec} R}(\{\langle 0 \rangle\}) = K[x]_{\langle 0 \rangle}, \mathcal{O}_{\operatorname{Spec} R}(\emptyset) = \{0\}$$

This is a presheaf when the restriction maps are the zero map. Moreover, it is trivially a presheaf because there are no open covers of any open sets which do not include the set itself. Thus every section glues trivially. Yet this cannot be a sheaf of the form $\tilde{M}$, because then we would have $M = \mathcal{F}(X) = \{0\}$. But clearly this is not the sheaf generated by the zero module.

As mentioned above, it is good enough to have sheaves which only locally look like they have been induced from modules.

Definition 6.13

A sheaf $\mathcal{F}$ of $\mathcal{O}_X$ -modules on a scheme X is **quasi-coherent** if there is an affine open cover $(U_i = \operatorname{Spec} R_i)$ of X where

$$\mathcal{F}|_{U_i} \cong \tilde{M}_i$$

for some R_i -module M_i .

It turns out that if $\mathcal{F}$ is quasi-coherent then for any affine open subset $U = \operatorname{Spec} R \subseteq X$, we have

$$\mathcal{F}|_U = \widetilde{\mathcal{F}(U)}$$

Thus, as with many constructions of this type, if the property above holds for some affine open cover, then it holds for any affine open cover. Also, one can look at **coherent** sheaves instead, where the M_i are required to be finitely generated R_i -modules.

Example 6.19

The structure sheaf is quasicoherent. This is because if we take any affine open cover $\text{Spec } R_i$, we already know that $\mathcal{O}_{\text{Spec } R_i} \cong \tilde{R}_i$ (this is true intrinsically for the structure sheaf on affine schemes, so it is also true as a subset $R_i \subseteq X$).

Example 6.20

As a corollary, we know that the twisting sheaves $\mathcal{O}_{\mathbb{P}^n}(d)$ are isomorphic to the structure sheaf $\mathcal{O}_{U_i}$ on each affine open cover. Thus, the twisting sheaves are also quasi-coherent.

To avoid actually checking this condition too often, we are fortunate because most constructions using quasi-coherent sheaves will produce other quasi-coherent sheaves.

Lemma 6.9

Let $X = \text{Spec } R$ be an affine scheme.

- (a) For any two R -modules M, N there is a bijection between morphisms of sheaves $\tilde{M} \rightarrow \tilde{N}$ and R -module homomorphisms $M \rightarrow N$.
- (b) A sequence of R -modules

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

is exact if and only if the corresponding sequence of sheaves

$$0 \longrightarrow \tilde{M}_1 \longrightarrow \tilde{M}_2 \longrightarrow \tilde{M}_3 \longrightarrow 0$$

is exact (with the arrows corresponding via (a)).

- (c) $\tilde{M} \oplus \tilde{N} = \widetilde{M \oplus N}$, $\tilde{M} \otimes \tilde{N} = \widetilde{M \otimes N}$, and if M is a finitely generated R -module over a Noetherian ring R , $\tilde{M}^\sim = \widetilde{M}$.
- (d) If $f : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of quasi-coherent sheaves corresponding to the R -module homomorphism $f : M \rightarrow N$ then $\ker f, \text{im } f$ are quasicoherent. If f is injective then $\mathcal{G}/\mathcal{F}$ is quasicoherent.

In particular, images, kernels, quotients, tensor products and direct sums of quasi-coherent sheaves are quasi-coherent. Also, the bijection gives an equivalence of categories between R -modules and quasicoherent sheaves over a $\text{Spec } R$.

Proof. (a) If $\tilde{f} : \tilde{M} \rightarrow \tilde{N}$ then this comes with the data of $\tilde{f}_X : \tilde{M}(X) \rightarrow \tilde{N}(X)$, but $\tilde{M}(X) = M, \tilde{N}(X) = N$, so $f = \tilde{f}_X$ is a $\mathcal{O}_X(X) = R$ -module homomorphism from $M \rightarrow N$.

In the other direction, given $f : M \rightarrow N$, this induces a map of localizations $f_p : M_p \rightarrow N_p$ for all $p \in X$. Thus applying this pointwise we can pass any section through the localization maps and this is a morphism. One then verifies that these are inverse constructions.

- (b) Recall from commutative algebra that since localization is exact, the sequence of R -modules is exact if and only if

$$0 \longrightarrow (M_1)_p \longrightarrow (M_2)_p \longrightarrow (M_3)_p \longrightarrow 0$$

is exact. But these are precisely the stalks of $\tilde{M}_1, \tilde{M}_2, \tilde{M}_3$, and we know exactness of sequences of sheaves can be checked on stalks.

- (c) The direct sum and tensor product commute with localization, and for finitely generated modules over Noetherian rings, the dual does as well. For the direct sum, let $U \subseteq X$ be open and consider a section

$$(\varphi = (\varphi_p)_{p \in U}, \psi = (\psi_p)_{p \in U}) \in (\tilde{M} \oplus \tilde{N})(U)$$

Then we map this to

$$((\varphi_p \oplus \psi_p)_{p \in U})$$

This is an element of $\widetilde{(\tilde{M} \oplus \tilde{N})}(U)$ because $\varphi_p \oplus \psi_p \in M_p \oplus N_p = (M \oplus N)_p$. One can check that this identification produces a section. This gives a morphism $\tilde{M} \oplus \tilde{N} \rightarrow \widetilde{M \oplus N}$ of $\mathcal{O}_X$ -modules. The way we have defined this is an isomorphism on stalks, which then shows that the sheaves are isomorphic. Similar arguments hold for the rest.

- (d) Let $f : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of $\mathcal{O}_X$ -modules for $\mathcal{F}, \mathcal{G}$ quasicoherent. Let $U = \text{Spec } R \subseteq X$ be an affine open, and say that $\mathcal{F}|_U = \tilde{M}, \mathcal{G}|_U = \tilde{N}$. Then $f|_U : \mathcal{F}|_U \rightarrow \mathcal{G}|_U$ corresponds to an R -module homomorphism $F : M \rightarrow N$. Thus we get an exact sequence of R -modules

$$0 \longrightarrow \ker F \longrightarrow M \longrightarrow N$$

and by the equivalence this gives an exact sequence of sheaves

$$0 \longrightarrow \widetilde{\ker F} \longrightarrow \tilde{M} \longrightarrow \tilde{N}$$

Since this is exact, we thus have that

$$\widetilde{\ker f|_U} = \ker(F)$$

and since taking restrictions commutes with kernels, we get

$$\widetilde{\ker f} = \ker f$$

So $\ker f$ is quasicoherent.

One can prove the same for cokernels using the sequence

$$M \longrightarrow N \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Then for the image, we use the fact that $\operatorname{im}(f) = \ker(\mathcal{G} \rightarrow \operatorname{coker} f)$, and pass tildes through the kernel and cokernel.

For quotients, this is easy because given $f : \mathcal{F} \hookrightarrow \mathcal{G}$, $\mathcal{G}/\mathcal{F} = \operatorname{coker} f$ so we just use what we have proved above. $\square$

The final construction that we need to look at is the pushforward. In the general case, the pushforward of a quasi-coherent sheaf may not be quasi-coherent. However, it does work for inclusions of closed subschemes.

Lemma 6.10

Let $i : Y \rightarrow X$ be the inclusion of a closed subscheme. Then:

- (a) If $\mathcal{F}$ is a quasi-coherent sheaf on Y then $i_*\mathcal{F}$ is quasi-coherent on X .
- (b) There is a quasi-coherent sheaf $\mathcal{J}_{Y/X}$ and an exact sequence of quasi-coherent sheaves on X given by

$$0 \longrightarrow \mathcal{J}_{Y/X} \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Y \longrightarrow 0$$

where the arrow from $\mathcal{O}_X \rightarrow i_*\mathcal{O}_Y$ is induced by the pushforward; that is, it can be thought of as restricting regular functions on U to those on $U \cap Y$.

Of course, the exact sequence forces $\mathcal{J}_{Y/X}$ to be the kernel of the pullback morphism $\mathcal{O}_X \rightarrow i_*\mathcal{O}_Y$, and it is called the **ideal sheaf**. By the interpretation of the pushforward arrow, we can think of this as the sheaf of regular functions on U which restrict to zero on $U \cap Y$.

Proof. (a) Quasicoherence can be checked on an open cover so start by assuming $X = \operatorname{Spec} R$ is affine. Then the inclusion i corresponds to some natural ring homomorphism

$$R \rightarrow R/I$$

with $Y = \operatorname{Spec} R/I$ for some ideal I . Say that $\mathcal{F} = \tilde{M}$ for some (R/I) -module M . Then the map $R \rightarrow R/I$ defines an R -module structure on M , which we denote M^R .

Thus we can define $\widetilde{M^R}$ as a sheaf on $\operatorname{Spec} R$. We claim that this is the pushforward $i_*\mathcal{F}$. We check this on the stalks. The stalks of M^R are

$$M_{\mathfrak{p}}^R = \begin{cases} M_{\mathfrak{p}, R/I}, & \mathfrak{p} \in Y \subseteq \operatorname{Spec} R \iff \mathfrak{p} \supseteq I \\ 0 & \end{cases}$$

Now, let $U \subseteq X$ be open. Then the sections of $i_*\mathcal{F}$ on U are the sections of $\mathcal{F}$ on $i^{-1}(U)$. That is, they look like

$$\varphi = (\varphi_p \in M_{p, R/I})_{p \in i^{-1}(U)}$$

which are locally quotients. We can map this to

$$\varphi^R = (\varphi_p^R \in M_p^R)_{p \in U}$$

where $\varphi_p^R = \varphi_p$ if $p \in U \cap Y$ and zero otherwise. This defines an isomorphism so for the affine case, $i_*\mathcal{F} = \widetilde{M^R}$ is quasicoherent. In general, we cover X with affines and repeat.

(b) We know that for any surjective morphism $f : X \rightarrow Y$ (in any appropriate category),

$$0 \longrightarrow \ker f \longrightarrow X \xrightarrow{f} Y \longrightarrow 0$$

is an exact sequence. Since kernels are quasi-coherent, this sequence will work. So it suffices to show that the morphism $\mathcal{O}_X \rightarrow i_*\mathcal{O}_Y$ is surjective. This can be checked on an affine open cover of X , so we can assume that $X = \operatorname{Spec} R$. Then the map of sheaves

$$\mathcal{O}_X \rightarrow i_*\mathcal{O}_Y$$

is equivalent to the map of R -modules

$$R \rightarrow R/I$$

This is a surjective map so by the equivalence, the map of sheaves is also surjective. In particular, we observe that the module-level exact sequence is

$$0 \longrightarrow I \longrightarrow R \longrightarrow R/I \longrightarrow 0$$

so $\mathcal{J}_{Y/X}$ corresponds to the ideal I (seen as an R -module), which is why it is called the ideal sheaf. $\square$

Example 6.21

Consider again the inclusion of the point $p = (1 : 0) \in \mathbb{P}^1$ along $i : \{p\} \rightarrow \mathbb{P}^1$. We previously showed the exactness of the sequence using the skyscraper sheaf

$$0 \longrightarrow \mathcal{O}_X(1) \xrightarrow{\cdot x_1} \mathcal{O}_X \xrightarrow{\text{eval at } p} K_p \longrightarrow 0$$

But the ideal sequence

$$0 \longrightarrow \mathcal{J}_{\{p\}/X} \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_{\{p\}} \longrightarrow 0$$

is also exact. We know that $i_*\mathcal{O}_{\{p\}}$ is the skyscraper sheaf K_p , which shows that $\mathcal{J}_{\{p\}/X} = \mathcal{O}_X(-1)$. Moreover, on the open set U_0 , $K_p|_{U_0}$ is the sheaf associated to the ideal $K[x_1]/\langle x_1 \rangle$ which is a $K[x_1]$ module, and on U_1 it is the zero sheaf. So locally it is given by modules and hence it is quasi-coherent.

Moreover, we can reprove the exactness of the original sequence, as on U_0 the module

level equivalent is

$$0 \longrightarrow K[x_1] \xrightarrow{\cdot x_1} K[x_1] \longrightarrow K[x_1]/\langle x_1 \rangle \longrightarrow 0$$

and on U_1 the skyscraper sheaf is trivial, so we just need the isomorphism $\mathcal{O}_{\mathbb{P}^1}(1)|_{U_1} \cong \mathcal{O}_{\mathbb{P}^1}$, or equivalently the sequence

$$0 \longrightarrow K[x+1] \xrightarrow{\text{id}} K[x_0] \longrightarrow 0 \longrightarrow 0$$

Alternatively, we may use the homogeneous coordinates and the definitions of the twisting sheaves to look at

$$0 \longrightarrow \frac{1}{x_0} K \left[\frac{x_1}{x_0} \right] \xrightarrow{\cdot x_1} K \left[\frac{x_1}{x_0} \right] \longrightarrow K \left[\frac{x_1}{x_0} \right] / \left\langle \frac{x_1}{x_0} \right\rangle \longrightarrow 0$$

and

$$0 \longrightarrow \frac{1}{x_1} K \left[\frac{x_1}{x_0} \right] \xrightarrow{\cdot x_1} K \left[\frac{x_0}{x_1} \right] \longrightarrow 0 \longrightarrow 0$$

One can also formulate ideal sheaves in more general settings.

Definition 6.14

An **ideal sheaf** is an $\mathcal{O}_X$ -module $\mathcal{J}$ with an injective morphism $\mathcal{J} \rightarrow \mathcal{O}_X$.

Then by the lemma above, every inclusion $Y \rightarrow X$ induces an injection $\mathcal{J}_{Y/X} \rightarrow \mathcal{O}_X$, so these are indeed ideal sheaves in this sense. However, the converse is also true; that is, any quasi-coherent ideal sheaf $\mathcal{J}$ determines a closed subscheme Y of X so that $\mathcal{J} = \mathcal{J}_{Y/X}$. Thus one can define Y to be the vanishing locus of the ideal sheaf $\mathcal{J}$.

To see this, we pass to an affine cover $(\text{Spec } R_i)_i$, on which $\mathcal{J}|_{\text{Spec } R_i} = J_i$ for some ideals $J_i \subseteq R_i$. But then restricting the injective morphism to $\text{Spec } R_i$ determines an injection

$$J_i \rightarrow R_i$$

which is the ideal sheaf injection corresponding to the inclusion

$$\text{Spec } R_i / J_i \rightarrow \text{Spec } R_i$$

These glue together into an inclusion $Y \rightarrow X$ for which $\mathcal{J}$ is the ideal sheaf.

Using this notion of ideals, we can now blowup general sheaves. If $\mathcal{J}$ is a quasi-coherent ideal sheaf, then on an affine cover we have $J|_U = \tilde{I}$ for some ideal $I \subseteq R$. We can then blow up the cover $\text{Spec } R$ at I , and gluing these together defines the blowup $\text{Bl}_{\mathcal{J}} X$.

Example 6.22

Letting $X \subseteq \mathbb{P}^d$ be an irreducible hypersurface of degree d , the ideal sheaf $\mathcal{I}_{X/\mathbb{P}^n}$ is $\mathcal{O}_{\mathbb{P}^n}(-d)$.

Remark 6.1

From now on, every sheaf is assumed to be quasi-coherent.

As a last construction, we define a pullback of sheaves to parallel the pushforward. In theory, this should behave nicer than the pushforward. However, the construction is a bit tricky.

Beginning in the affine case $X = \operatorname{Spec} R, Y = \operatorname{Spec} S$, with some morphism $f : X \rightarrow Y$ and $\mathcal{F}$ a sheaf on Y , we know that f corresponds to some ring homomorphism $S \rightarrow R$, which gives an S -module structure on R . Additionally, $\mathcal{F}$ is the data of some S -module M . We want to use the morphism to reconstruct the data of some R -module. To do this, we define

$$i^*\mathcal{F} = \widetilde{M \otimes_S R}$$

which can be seen as an R module by defining multiplication by R to be multiplication against the second coordinate. To define this for arbitrary schemes, we have a slight problem because we want to define $i_*\mathcal{F}(U)$ to be something in terms of the sections $\mathcal{F}(f(U))$. But in general, $f(U)$ is not actually open.

Thus, we start by defining a presheaf where the sections on U are any section $\varphi \in \mathcal{F}(V)$ defined on any open set V containing $f(U)$, modulo the equivalence relation

$$[(V, \varphi)] \sim [(V', \varphi')]$$

where φ, φ' restrict to the same section in $\mathcal{F}$ on some smaller set $W \subseteq V \cap V'$. This defines a presheaf, and we denote its sheafification by $f^{-1}\mathcal{F}$.

This has the nice property that the stalks agree:

$$(f^{-1}\mathcal{F})_p = \mathcal{F}_{f(p)}$$

which is basically immediate because the definition of $f^{-1}\mathcal{F}$ uses the same definitions as the stalks. However, because all the sections here are sections on Y , it is not clear how we might multiply by sections on X .

Nevertheless, it does have the structure of an $f^{-1}\mathcal{O}_Y$ module, since we can perform the multiplication using the $\mathcal{O}_Y$ -module structure of $\mathcal{F}$. $f^{-1}\mathcal{O}_Y$ is a sheaf of rings on X (since $\mathcal{O}_Y$ is a sheaf of rings on Y). Moreover, we can pull back sections of $f^{-1}\mathcal{O}_Y$ to functions on X by setting

$$[(V, \varphi = (\varphi_p)_{p \in V})] \mapsto \psi = (\varphi_{f(q)})_{q \in U}$$

This is well defined because $f(q) \in f(U) \subseteq V$ is always part of the data of a section, and we showed the stalks match correctly so this is a function to the correct stalks. The other properties may be verified. This thus defines a homomorphism of sheaves of rings $f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$. In this way, $\mathcal{O}_X$ and $f^{-1}\mathcal{F}$ are both sheaves of modules over $f^{-1}\mathcal{O}_Y$. Thus we make the following definition, analogous to the affine case:

Definition 6.15

Let $f : X \rightarrow Y$ be a morphism and let $\mathcal{F}$ be a sheaf on Y . Then the **pullback sheaf** on X is the sheaf

$$f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$$

As with the affine case, $f^*\mathcal{F}$ can be taken to be an $\mathcal{O}_X$ module by defining

$$\varphi(\psi \otimes \phi) = \psi \otimes (\varphi\phi)$$

for $\varphi, \phi \in \mathcal{O}_X, \psi \in f^{-1}\mathcal{F}$.

This is a complicated construction but one can effectively always just pass to the affine case for computations; this construction merely verifies that there is actually a sheaf which looks like the affine case locally. To verify that this does reduce to the affine case we considered, we make the following computation:

Example 6.23

Say that $X = \operatorname{Spec} R, Y = \operatorname{Spec} S$ and $\mathcal{F} = \tilde{M}$ for an S -module M . Then at the level of stalks, because tensor products commute with localization, we verify that

$$(f^*\mathcal{F})_p = (f^{-1}\mathcal{F})_p \otimes_{(f^{-1}\mathcal{O}_Y)_p} \mathcal{O}_{X,p} = \mathcal{F}_{f(p)} \otimes_{\mathcal{O}_{Y,f(p)}} \mathcal{O}_{X,p} = M_{f(p)} \otimes_{S_{f(p)}} R_p$$

It is a commutative algebra fact that this is equal to

$$(M \otimes_S R)_p$$

so by checking at stalks, we see that $f^*\mathcal{F}, \widetilde{M \otimes_S R}$ agree. Then by passing through this isomorphism at each stalk, we can map a section of $f^*\mathcal{F}$ to a section of $\widetilde{M \otimes_S R}$:

$$\varphi \mapsto (\varphi_p)_{p \in U} \in \widetilde{M \otimes_S R}(U)$$

This is a morphism of sheaves that induces isomorphisms of stalks, so it is an isomorphism of sheaves.

Proposition 6.11

$$f^*\mathcal{O}_Y = \mathcal{O}_X.$$

Proof. Locally, the pullback looks like

$$S \otimes_S R$$

and by commutative algebra this is just R . □

In the case where regular functions were actually set theoretic functions, we saw that sections pulled back to sections. In this case, we make the following construction: let $\varphi \in \mathcal{F}(V)$ for

$V \subseteq Y$ open. Then by the definition of $f^{-1}(\mathcal{F})$, it was the sheafification of pairs $[(U, \varphi)]$ up to equivalence. Taking $U = f^{-1}(V)$, we certainly have $f(U) \subseteq V$, so

$$[(V, \varphi)] \in (f^{-1}\mathcal{F})^{\text{pre}}(f^{-1}(V))$$

and since presheaves sit inside their sheafification, we can see $[(f^{-1}(V), \varphi)] \in f^{-1}\mathcal{F}$. Finally, we define the **pullback** of φ to be

$$f^*\varphi = [(V, \varphi)] \otimes 1 \in (f^*\mathcal{F})(f^{-1}(V))$$

Therefore f can be seen to pull back sections to sections.

Proposition 6.12

If $\varphi \in \mathcal{F}(V)$ and $\psi \in \mathcal{O}_Y(V)$ then

$$f^*(\varphi\psi) = (f^*\varphi)(f^*\psi)$$

where the multiplication on the left hand side is via the $\mathcal{O}_Y$ -module structure of $\mathcal{F}$, and multiplication on the right is via the $\mathcal{O}_X$ -module structure of $f^*\mathcal{F}$.

Proof. On the left, we have

$$[(V, \varphi\psi)] \otimes 1$$

Since ψ lies in $f^{-1}(\mathcal{O}_Y)(f^{-1}(V))$, which is what the tensor product is over, it can be pulled outside, so this is

$$\psi([(V, \varphi)] \otimes 1) = [(V, \varphi)] \otimes f^*\psi$$

because $f^*\psi$ is the image of ψ under the tensoring map $f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$. Then, because multiplication by $\mathcal{O}_X$ is just in the second coordinate, this is by definition equal to

$$(f^*\psi) \cdot [(V, \varphi)] \otimes 1 = (f^*\psi)(f^*\varphi) \quad \square$$

Proposition 6.13

Given a composition of morphisms of schemes

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

and a quasi-coherent sheaf $\mathcal{F}$ on Z , the pullback sheaves satisfy

$$f^*(g^*\mathcal{F}) = (g \circ f)^*\mathcal{F}$$

as $\mathcal{O}_X$ -modules.

Basically, the pullback is functorial.

Proof. Consider the affine case $X = \operatorname{Spec} R, Y = \operatorname{Spec} S, Z = \operatorname{Spec} T, \mathcal{F} = \tilde{M}$ for M a T -module. Then

$$f^* g^* \mathcal{F} = f^* (\widetilde{M \otimes_T S}) = (\widetilde{M \otimes_T S}) \otimes_S R$$

By commutative algebra, this tensor product is just

$$\widetilde{M \otimes_T R}$$

which is the same as

$$(g \circ f)^* \mathcal{F}$$

□

Example 6.24

In the special case where we pull back along an open embedding $i_U : U \rightarrow X$, the definition of the preimage presheaf $(f^{-1}\mathcal{F})^{\text{pre}}$ is already the sheaf $\mathcal{F}|_U$, so taking the presheaf does nothing. Then

$$i_U^* \mathcal{F} = \mathcal{F}|_U \otimes_{\mathcal{O}_X|_U} \mathcal{O}_U = \mathcal{F}|_U \otimes_{\mathcal{O}_U} \mathcal{O}_U = \mathcal{F}|_U$$

so the embedding just pulls the sheaf back to the restricted sheaf, as we might expect.

Definition 6.16

Let $\mathcal{F}$ be a sheaf on X and $p \in X$ a closed point. If $f : \{p\} \rightarrow X$ is the inclusion, then the **fiber** of $\mathcal{F}$ at p is the pullback

$$\mathcal{F}|_p = (f^* \mathcal{F})(\{p\})$$

For a given closed point, we can take an affine open containing p , and make the factorization

$$\{p\} \xrightarrow{i_{p,U}} U = \operatorname{Spec} R \xrightarrow{i_U} X$$

i_p

Then we compute the fiber via the properties above:

$$i_p^* \mathcal{F} = i_{p,U}^* i_U^* \mathcal{F} = i_{p,U}^* \mathcal{F}|_U$$

Since $\mathcal{F}$ is quasi-coherent, $\mathcal{F}|_U = \tilde{M}$ for some R -module M . Moreover, since p is a closed point, we can see it as $\operatorname{Spec} R/\mathfrak{p}$ for some maximal $\mathfrak{p}$ in R . Then

$$i_{p,U}^* \tilde{M} = \widetilde{M \otimes_R R/\mathfrak{p}}$$

By commutative algebra, this is

$$\widetilde{M/\mathfrak{p}M}$$

which is a vector space over the residue field $K(p) = R/\mathfrak{p}$ (which is a field since $\mathfrak{p}$ is maximal). Moreover, for any section $\varphi \in \mathcal{F}(V)$ for V open containing p , we see that $\varphi_p \in M_{\mathfrak{p}}$, and we define the **value** of φ at p to be the interpretation in the quotient

$$\varphi(p) = \varphi_p \in M_{\mathfrak{p}/\mathfrak{p}} \cdot M_{\mathfrak{p}} \cong \mathcal{F}|_p$$

So any section has an associated collection of values in the fibers at closed points. Note that for the sheaf of regular functions, this is just the same as the definition of the value of a regular function, which was a value in the residue field. Thus the section of an arbitrary sheaf also has values, but in vector spaces over the residue field instead of the residue field itself.

Example 6.25

Let X, Y be varieties and $f : X \rightarrow Y$ a morphism. Let $\mathcal{F}$ be a quasi-coherent sheaf on Y . We consider the fiber of $f^*\mathcal{F}$ at a point $p \in X$. We have the diagram

$$\begin{array}{ccc} \{p\} & \xrightarrow{i_p} & X \xrightarrow{f} Y \\ & \searrow i_{f(p)} & \nearrow \end{array}$$

By definition,

$$f^*\mathcal{F}|_p = i_p^* f^*\mathcal{F}$$

and by functoriality this is

$$i_{f(p)}^* \mathcal{F} = \mathcal{F}|_{f(p)}$$

Thus, we can interpret the pullback as simply transporting the fibers backward onto X .

6.4 Locally Free Sheaves, Differentials, and (Co-)Tangent Bundles

The easiest class of R -modules is the free R -modules. This corresponds to a nice class of quasi-coherent sheaves.

Definition 6.17

A quasi-coherent sheaf $\mathcal{F}$ is called a **locally free sheaf** if there is an open affine cover $(U_i = \text{Spec } R_i)_i$ on which $\mathcal{F}|_{U_i} = \tilde{M}_i$ for some free R_i -module M_i of finite rank. In the case that the ranks are all the same then the **rank** of $\mathcal{F}$ is this common value.

At the level of the affine cover, if $\mathcal{F}$ is of rank r_i at U_i , we have

$$\mathcal{F}|_{U_i} = (R_i)^{\oplus r_i}$$

but it is not in general true that

$$\mathcal{F} = \mathcal{O}_X^{\oplus \text{rk } \mathcal{F}}$$

For instance, we have seen that on the open affine cover,

$$\mathcal{O}_{\mathbb{P}^1}(d)|_{U_i} \cong \mathcal{O}_{\mathbb{P}^1}^{\oplus 1}$$

is locally free of rank 1. However, it is not equal to $\mathcal{O}_{\mathbb{P}^n}$ as a global sheaf, so this provides a counterexample.

Proposition 6.14

If $\mathcal{F}$ is locally free of rank r then the fibers $\mathcal{F}|_p$ are r -dimensional vector spaces over $K(p)$.

Proof. Passing to an open neighborhood,

$$\mathcal{F}|_U = \tilde{M}_i = \widetilde{R_i^{\oplus r}} = (\tilde{R}_i)^{\oplus r} = (\mathcal{O}_U)^{\oplus r} \implies f^* \mathcal{F}|_U = (f^* \mathcal{O}_U)^{\oplus r} = (\mathcal{O}_{\{p\}})^{\oplus r} = (K(p))^r \quad \square$$

Of course, being locally free is more than the statement that these fibers are r -dimensional vector spaces, but also since they are given locally by some module, these fibers should vary nicely. Equivalently, we can do this more geometrically by constructing a new scheme called the total space $\text{Tot}(\mathcal{F})$ with a projection $\pi : \text{Tot}(\mathcal{F}) \rightarrow X$. Intuitively, this looks like the collection of vector spaces glued together (like a Möbius strip is a glued together collection of segments on S^1). Then we can see sections of $\mathcal{F}$ as $\text{Tot}(\mathcal{F})$ -valued functions, with the requirement that they take values in the “correct” spaces:

$$\mathcal{F}(U) = \{s : U \rightarrow \text{Tot}(\mathcal{F}) : \pi \circ s = \text{id}_U\}$$

As in other cases, this space is constructed locally as the direct product with the projection. That is, for U_i affine and $\mathcal{F}|_{U_i} \cong \mathcal{O}_{U_i}^{\oplus r}$, we define

$$\text{Tot}(\mathcal{O}_{U_i}^{\oplus r}) = U_i \times \mathbb{A}^r$$

with the natural projection to the first coordinate $U_i \times \mathbb{A}^r \rightarrow U_i$. To see this, note that the sections in both cases are just a collection of r regular functions. Gluing along the U_i , this glues into a scheme $\text{Tot}(\mathcal{F})$ which we call a **vector bundle**. When $\mathcal{F}$ is of rank 1 we call it a **line bundle** and call $\mathcal{F}$ an **invertible sheaf**.

Example 6.26

As in the affine case, in general if we take the trivial locally free sheaf

$$\mathcal{F} = \mathcal{O}_X^{\oplus r}$$

then

$$\text{Tot}(\mathcal{O}_x^{\oplus r}) = X \times \mathbb{A}^r$$

with the natural projection to X , and this is called the trivial vector bundle.

Example 6.27

If $\mathcal{F} = K_p$ is the skyscraper sheaf, then we see that

$$\mathcal{F}|_q = \begin{cases} K, & q = p \\ 0, & q \neq p \end{cases}$$

So unless $\{p\}$ is open, this cannot be locally free as there is no neighborhood of p where all fibers are of the same dimension. Thus the skyscraper sheaf is not locally free.

Proposition 6.15

If $\mathcal{F}, \mathcal{G}$ are locally free of rank m, n on X , and $f : Y \rightarrow X$ is a morphism of schemes, then

- (a) $\mathcal{F} \oplus \mathcal{G}$ is locally free of rank $m + n$,
- (b) $\mathcal{F} \otimes \mathcal{G}$ is locally free of rank mn ,
- (c) $\mathcal{F}^\sim$ is locally free of rank m ,
- (d) $f^*\mathcal{F}$ is locally free of rank m on Y .

Proof. Take a point $p \in X$, and take U, V to be open neighborhoods with $\mathcal{F}|_U = \mathcal{O}_U^{\oplus m}, \mathcal{G}|_V = \mathcal{O}_V^{\oplus n}$. Then on $W = U \cap V$ with $W = \operatorname{Spec} R$, $\mathcal{F}|_W = R^{\oplus m}, \mathcal{G}|_W = R^{\oplus n}$, so:

- (a) The direct sum is

$$(\mathcal{F} \oplus \mathcal{G})_W = R^{\oplus m} \oplus R^{\oplus n} = R^{\oplus(m+n)} = \mathcal{O}_W^{\oplus(m+n)}$$

- (b) For the direct sum,

$$(\mathcal{F} \otimes \mathcal{G})_W = R^{\oplus m} \otimes_R R^{\oplus n} = \mathcal{O}_W^{\oplus(mn)}$$

- (c) For the dual,

$$\mathcal{F}^\sim|_W = (\operatorname{Hom}(\mathcal{F}|_W \rightarrow \mathcal{O}_W))^\sim = (\operatorname{Hom}_R(R^{\oplus m} \rightarrow R))^\sim = \widetilde{R^m} = \mathcal{O}_W^{\oplus m}$$

- (d) Finally, for the pullback, cover X with sets of the form $\operatorname{Spec} R_i$, and cover $f^{-1}(\operatorname{Spec} R_i)$ by sets of the form $\operatorname{Spec} S_{i,j}$. Then

$$f^*\mathcal{F}|_{\operatorname{Spec} S_{i,j}} = (R_i^{\oplus m} \otimes_{R_i} S_{i,j})^\sim = ((R_i \otimes_{R_i} S_{i,j})^{\oplus m})^\sim = \widetilde{S_{i,j}^{\oplus m}} = \mathcal{O}_{\operatorname{Spec} S_{i,j}}^{\oplus m} \quad \square$$

We conclude this section by looking at how we can interpret maps to projective space in the framework of bundles. Recall that the global sections of the twisting sheaves $\mathcal{O}_{\mathbb{P}^n}(d)$ were isomorphic to $K[x_0, \dots, x_n]_d$, and in particular $\mathcal{O}_{\mathbb{P}^n}(1)$ are the homogeneous linear polynomials. Moreover, for any collection of $f_0, \dots, f_n \in S(X)_d$ homogeneous functions of the same degree on X , this defines a morphism from $X \setminus V_X(f_0, \dots, f_n) \rightarrow \mathbb{P}^n$. We can interpret each of the f_i as sections of the line bundle $\mathcal{L} = \mathcal{O}_{\mathbb{P}^n}(d)$.

Theorem 6.16

Given X a variety over K , there is a bijection between morphisms from $X \rightarrow \mathbb{P}_K^n$ and tuples $(\mathcal{L}, s_0, \dots, s_n)$, where $\mathcal{L}$ is an invertible sheaf over X and $s_0, \dots, s_n \in \mathcal{L}(X)$ are global sections which do not all vanish at the same point (for any $p \in X$, there is i with $s_i(p) \neq 0$), considered up to isomorphism.

“Considered up to isomorphism” means that two such tuples $(\mathcal{L}, s_0, \dots, s_n), (\mathcal{L}', s'_0, \dots, s'_n)$ are equivalent if there is an isomorphism $\mathcal{L} \rightarrow \mathcal{L}'$ which also takes $s_i \rightarrow s'_i$.

We can construct the forward map Ψ fairly easily, as given a morphism $f : X \rightarrow \mathbb{P}_K^n$ we pull back the twisting sheaf, setting $\mathcal{L} = \mathcal{O}_{\mathbb{P}^n}(1)$ and $s_i = f^*x_i$.

Lemma 6.17

Let Y be a scheme with $\psi : \mathcal{O}_Y \rightarrow \mathcal{O}_Y$ an automorphism of the structure sheaf. Then $s = \psi(1)$ is invertible in $\mathcal{O}_Y(Y)$ (in particular, nowhere zero), and for any global section φ , $\psi(\varphi) = s\varphi$.

Proof. The second statment is immediate from being an $\mathcal{O}_Y$ -module homomorphism. The first statement is true because if $t = \psi^{-1}(1)$, then $1 = \psi(\psi^{-1}(1)) = \psi(t) = st$. So s is invertible. $\square$

Proof of Theorem 6.16. First, we show that the map Ψ we wrote down earlier is a well defined map. That is, $\mathcal{L} = f^*\mathcal{O}_{\mathbb{P}^n}(1)$ is invertible since $\mathcal{O}_{\mathbb{P}^n}(1)$ is locally free of rank 1, and the pullback preserves this. Also, x_i are each global sections of $\mathcal{O}_{\mathbb{P}^n}(1)$, and the pullback takes sections of $\mathbb{P}^n$ to sections of $f^{-1}(\mathbb{P}^n) = X$. So the s_i are global sections, and they do not vanish all at once since

$$s_i(p) = 0 \in f^*\mathcal{O}_{\mathbb{P}^n}(1)|_p \cong \mathcal{O}_{\mathbb{P}^n}(1)|_{f(p)} \iff x_i|_{f(p)} = 0$$

Here x_i is an abstract section but we can mechanically check that $x_i|_{f(p)} = 0$ if and only if $f(p) \in V(x_i)$, where x_i is now taken as a regular function and this is the normal vanishing locus. The intersection of all the vanishing loci in $\mathbb{P}_K^n$ is empty, so Ψ is well defined.

For the inverse, let $(\mathcal{L}, s_0, \dots, s_n)$ be a tuple. We need to construct a morphism $X \rightarrow \mathbb{P}^n$. Take an open cover U_i of X on which $\mathcal{L}|_{U_i} \cong \mathcal{O}_{U_i}$ via φ_i . By the same isomorphism, s_j maps to some $f_{j,U_i} = \varphi_i s_j \in \mathcal{O}_{U_i}(U_i)$. These are regular functions, so we get a morphism

$$f_i = (f_{0,U_i} : \dots : f_{n,U_i}) : U_i \rightarrow \mathbb{P}^n$$

as the coordinates do not vanish simultaneously by assumption. These agree on overlaps since on $U_i \cap U_k$, $\mathcal{O}_{U_i \cap U_k} \cong \mathcal{L}|_{U_i \cap U_k}$ through both φ_i, φ_j . Composing the isomorphisms, we get an automorphism $\varphi_{i,k} = \varphi_k^{-1} \circ \varphi_i : \mathcal{O}_{U_i \cap U_k} \rightarrow \mathcal{O}_{U_i \cap U_k}$. By the lemma, such automorphisms are given by multiplication by some invertible function $s_{i,k} \in \mathcal{O}_{U_i \cap U_k}(U_i \cap U_k)$. Then

$$\begin{aligned} f_i|_{U_i \cap U_k} &= (f_{0,U_i} : \dots : f_{n,U_i})|_{U_i \cap U_k} \\ &= (s_{i,k} f_{0,U_k} : \dots : s_{i,k} f_{n,U_k})|_{U_i \cap U_k} = (f_{0,U_k} : \dots : f_{n,U_k})|_{U_i \cap U_k} \end{aligned}$$

where the last identity follows because of equality under homogeneous scaling. Thus we have agreement on overlaps and therefore the f_i glue into a morphism $f : X \rightarrow \mathbb{P}_K^n$.

These maps can be shown to be inverses.

Also, by the same argument, any two tuples which are isomorphic can be shown to agree via the same automorphism lemma and homogeneous scaling argument. $\square$

The theorem above holds in more generality but for varieties the argument is much simpler because equality of morphisms can be checked pointwise.

Example 6.28

Every line bundle $\mathcal{L}$ on $\mathbb{P}^n$ is of the form $\mathcal{O}_{\mathbb{P}^n}(d)$ for some d .

Example 6.29

$\text{Aut}(\mathbb{P}_K^n) \cong \text{PGL}_K(n+1)$.

Finally, we conclude by defining the module of differentials and constructing the cotangent and tangent spaces. This gives a rigorous treatment of our earlier discussion of wanting a tangent space so that sections look like vector fields taking values in the tangent space at each point, where the tangent vectors vary nicely. We will do this by constructing the tangent sheaf T_X , which will be locally free of rank $\dim X$ (reflecting the idea that at a smooth point of X , there should be $\dim X$ degrees of freedom). To actually construct the tangent sheaf, we first construct the cotangent space and then define the tangent space to be the dual.

For this treatment we have to work with varieties, so there is a fixed algebraically closed ground field K in the remainder.

Definition 6.18

Let R be a K -algebra for K an algebraically closed field. Then the R -module of **Kähler differentials** of R over K is the R -module Ω_R of formal symbols of the form df for $f \in R$, modulo the relations

$$\begin{cases} df + dg = d(f + g), & f, g \in R \\ df = f dg + g df, & f, g \in R \\ df = 0, & f \in K \end{cases}$$

Formally, we write

$$\Omega_R = \bigoplus_{f \in R} R \cdot df \Big/ \left\langle \begin{cases} df + dg = d(f + g), & f, g \in R \\ df = f dg + g df, & f, g \in R \\ df = 0, & f \in K \end{cases} \right\rangle$$

Effectively, we generalize the product rule from calculus to the formal symbols above, with the requirement that the “constant functions” $f \in K$ have zero differential. There is an obvious map $R \rightarrow \Omega_R$ which takes $f \mapsto df$. The combination of the product rule and $df = 0$ for $f \in K$ implies that this map is K -linear, but as a map of R -modules it is not R -linear due to the general form of the product rule.

We expect these differentials to look like elements of the cotangent sheaf. For instance, in differential geometry, the differential form df can be evaluated at $p \in X$ to give an element of the cotangent space $(T_p X)^*$. When this is contracted against a tangent vector $v \in T_p X$, we get $df_p(v) = \langle df_p, v \rangle$, which is just a number (namely, the derivative of f along v).

Restricting to either distinguished opens or stalks involves localization, so we need to understand the relationship between the map $R \rightarrow \Omega_R$ and localization. For instance, let $S \subseteq R$ be a multiplicatively closed subset in a K -algebra R . Then the elements of $S^{-1}R$ look like fractions r/s , so differentials look like

$$d\frac{r}{s}$$

Then using the rules above, if $f \in S$ then

$$0 = d1 = d\left(\frac{1}{f} \cdot f\right) = \frac{1}{f} df + f d\frac{1}{f} \implies d\frac{1}{f} = -\frac{1}{f^2} df$$

and hence for general $f \in S, g \in R$,

$$d\frac{g}{f} = g d\frac{1}{f} + \frac{1}{f} dg = -\frac{g}{f^2} df + \frac{1}{f} dg = \frac{f dg - g df}{f^2}$$

analogous to the normal product rule. Since we can reduce all the differentials of fractions to fractions of elements of Ω_R , we should be able to construct an isomorphism

$$\Omega_{R[S^{-1}]} \xrightarrow{\cong} S^{-1}\Omega_R$$

This can be specified via the action on generators via the product rule

$$d\frac{g}{f} \mapsto \frac{1}{f} dg - \frac{g}{f^2} df$$

One can check that this is an isomorphism. As a result, taking differentials commutes with localization.

Proposition 6.18

For R a K -algebra as above, there is a map of sheaves (not of $\mathcal{O}_{\text{Spec } R}$ modules)

$$\mathcal{O}_{\text{Spec } R} \xrightarrow{d} \widetilde{\Omega}_R$$

Note that although Ω_R (being an R -module) induces an $\mathcal{O}_{\text{Spec } R}$ module $\widetilde{\Omega}_R$, the map above is not an $\mathcal{O}_{\text{Spec } R}$ module morphism, again for the reason of the product rule.

Proof. Let $\varphi = (\varphi_p \in R_p)_{p \in U} \in \mathcal{O}_{\text{Spec } R}(U)$ be a regular function on U . Then φ_p locally looks like g/f . We can simply map this pointwise to

$$(\mathrm{d}\varphi_p \in \Omega_{R_p})_{p \in U}$$

and we know this commutes with localization as

$$(\mathrm{d}\varphi_p \in (\Omega_R)_p)_{p \in U}$$

We need to show that $\mathrm{d}\varphi_p$ is locally given as the quotient of nonvanishing elements of Ω_R , but this is true because if $\varphi = g/f$ locally, then

$$\mathrm{d}\varphi = \frac{f \mathrm{d}g - g \mathrm{d}f}{f^2}$$

locally. Thus we have defined

$$\mathrm{d}\varphi \in \widetilde{\Omega}_R(U)$$

□

Example 6.30

When $R = K[x_1, \dots, x_n]$, the product rule shows that formal differentiation of polynomials holds:

$$\mathrm{d}f = \sum_i \partial_{x_i} f \mathrm{d}x_i$$

For instance, for $f = 2x_1 + x_2^2$,

$$\mathrm{d}f = \mathrm{d}(2x_1) + \mathrm{d}(x_2^2) = 2 \mathrm{d}x_1 + x_1 \mathrm{d}2 + 2x_2 \mathrm{d}x_2 = 2 \mathrm{d}x_1 + 2x_2 \mathrm{d}x_2 = \partial_{x_1} f \mathrm{d}x_1 + \partial_{x_2} f \mathrm{d}x_2$$

So for the case of the polynomial ring, Ω_R is generated by $\mathrm{d}x_i$. Moreover, one can show that the relation $\mathrm{d}f = \sum_i \partial_{x_i} f \mathrm{d}x_i$ implies the three relations defining the Kähler differentials. So there are no relations between the $\mathrm{d}x_i$, and therefore Ω_R is actually a free R -module of rank n :

$$\Omega_R = R \mathrm{d}x_1 \oplus \cdots \oplus R \mathrm{d}x_n$$

The above is encouraging because we will eventually want our differentials to be a locally free module.

Example 6.31

Let $R = A(X)$ for $X \subseteq \mathbb{A}^n$ an affine variety, that is, $R = K[x_1, \dots, x_n]/I(X)$. In this case, we have the additional relation that $\mathrm{d}f = 0$ for any $f \in I(X)$, since such f are the zero function. However, there are no other relations, and Ω_R is still generated

by the dx_i , so if $I(X) = \langle f_1, \dots, f_m \rangle$ then

$$\begin{aligned}\Omega_R &= R dx_1 \oplus \dots \oplus R dx_n / \langle df_i = 0 : 1 \leq i \leq m \rangle \\ &= R dx_1 \oplus \dots \oplus R dx_n / \left\langle \sum_j \partial_{x_j} f_i dx_j : 1 \leq i \leq m \right\rangle\end{aligned}$$

In the case of a maximal ideal $\mathfrak{p} \subseteq R$ (corresponding to a point $p \in X$), we can take the fiber of $\widetilde{\Omega}_R$ at p , which by definition is

$$\begin{aligned}\widetilde{\Omega}_R|_p &= \Omega_R \otimes_R R/\mathfrak{p} \cong \Omega_R \otimes_R K \\ &\cong K dx_1 \oplus \dots \oplus K dx_n / \left\langle \sum_j \partial_{x_j} f_i(p) dx_j : 1 \leq i \leq m \right\rangle\end{aligned}$$

We saw in the affine Jacobi criterion that the tangent space of X at p is the zero locus of $\partial_{x_j} f_i(p)x_j$, so this becomes the cotangent space of X at p . That is, the fibers of Ω_R are given by

$$\widetilde{\Omega}_R|_p = (T_p X)^\vee$$

so the differentials have the correct fibers at points.

We now extend this construction to general varieties. It would be possible to glue these by hand but this is very messy. Instead, we will just find an alternative construction which is a global sheaf, and locally looks like the above construction. Once this exists, we can simply say that we work with the cotangent sheaf, and then whenever working locally use the explicit description above.

Lemma 6.19

Let R be a K -algebra and define the multiplication map

$$\delta : R \otimes_K R \rightarrow R, \quad f \otimes g \mapsto fg$$

Define $J = \ker \delta$. Then as R -modules,

$$J/J^2 \cong \Omega_R$$

Note that the tensor product $R \otimes_K R$ has two possible R -module structures, one as

$$a \cdot (b \otimes c) = (ab) \otimes c$$

and one as

$$a \cdot (b \otimes c) = b \otimes (ac)$$

This are two different structures, as a can only be commuted between the two sides when $a \in K$ (since the tensor product is only over K , not R for general R). Thus J has two R -module structures, so we will need to show that both structures are isomorphic.

We also need to understand what J^2 actually is. By definition, it is the module generated by products of elements of J . Elements of J themselves are sums of pure tensors, so

$$J^2 = \left\langle \sum_{i,j} f_{1i} f_{1j} \otimes g_{1i} g_{1j} : \sum_i f_{1i} \otimes g_{1i}, \sum_j f_{1j} \otimes g_{1j} \in J \right\rangle$$

This is a submodule of $R \otimes_K R$ with respect to either R -module structure. As an abelian group, we then form J/J^2 . It can be endowed with two a priori distinct R -module structures, with multiplication in either the left or right component.

We claim that these are the same. Indeed, if $h \in R$ and $\psi \in J$ is some mixed tensor

$$\psi = \sum_i f_i \otimes g_i$$

then there are two ways to multiply $h\psi$. If we show that the difference lies in J^2 , then the choice does not matter in the quotient. This is true since the difference between the definitions is

$$\sum_i (h f_i \otimes g_i - f_i \otimes h g_i) = \left(\sum_i f_i \otimes g_i \right) \cdot (1 \otimes h - h \otimes 1)$$

J is the kernel of δ , and

$$\delta(1 \otimes h) = 1h, \quad \delta(h \otimes 1) = h1 \implies \delta(1 \otimes h - h \otimes 1) = 0 \implies 1 \otimes h - h \otimes 1 \in J$$

Thus the difference lies in J^2 , meaning the two definitions of $h\psi$ are the same element of J^2 as an abelian group. So on the quotient, the two constructions coincide.

Proof of Lemma 6.19. With the description of J/J^2 as above, we now write down a map between J/J^2 and Ω_R . In the forward direction, we write down the map

$$\left[\sum_i f_i \otimes g_i \right] \mapsto \sum_i f_i dg_i$$

and in the reverse direction,

$$df \mapsto [1 \otimes f - f \otimes 1]$$

By the argument above, this indeed maps to an element of $J = \ker \delta$. One can check that these are R -module homomorphisms and inverse to each other. $\square$

Now, we can use this description of Ω_R to construct the cotangent sheaf generally. For X a variety, by separatedness we have that $\Delta_X \subseteq X \times X$ is a closed subvariety, and this gives a closed subscheme by $i = (\text{id}_X, \text{id}_X) : X \rightarrow X \times X$. On affine charts $X = \text{Spec } R$, $X \times X$ is $\text{Spec}(R \otimes_K R)$, so i is locally a morphism of affine schemes $i : \text{Spec } R \rightarrow \text{Spec}(R \otimes_K R)$. By the equivalence, this corresponds to a ring homomorphism $R \otimes_K R \rightarrow R$, and since

$$\delta(f \otimes g) = \text{id}_R(f) \text{id}_R(g)$$

One can walk through the correspondence and check that the corresponding homomorphism is precisely δ . Recall that $J = \ker \delta$, so we get an exact sequence of $R \otimes_K R$ modules

$$0 \longrightarrow J = \ker \delta \longrightarrow R \otimes_K R \xrightarrow{\delta} R \longrightarrow 0$$

and we also have an exact sequence of $\mathcal{O}_{X \times X}$ modules

$$0 \longrightarrow \mathcal{J} \longrightarrow \mathcal{O}_{X \times X} \xrightarrow{i} i_* \mathcal{O}_X \longrightarrow 0$$

where $\mathcal{J}$ is the ideal sheaf $\mathcal{J}_{X/X \times X}$. Since R corresponds to $i_* \mathcal{O}_X$ and $R \otimes_K R$ corresponds to $\mathcal{O}_{X \times X}$, we also have that $\mathcal{J} = \tilde{J}$ by the correspondence. Then, since the correspondence commutes with squares and quotients,

$$\mathcal{J}/\mathcal{J}^2 = \widetilde{J/J^2} \text{ on } X \times X$$

so via the pullback, which does not change the underlying module,

$$i^* \mathcal{J}/\mathcal{J}^2 = \widetilde{J/J^2} \text{ on } X$$

and since we interpret J/J^2 as an R -module when viewing its sheaf on X , this is the same as

$$i^* = \mathcal{J}/\mathcal{J}^2 = \widetilde{\Omega}_R \text{ on } X$$

Thus for general varieties, we make the following definition:

Definition 6.19

Let X be a variety over K and $\mathcal{J} = \mathcal{J}_{X/X \times X}$ the ideal sheaf for the inclusion via the diagonal. Then the **cotangent sheaf** of X is

$$\Omega_X = i^* \mathcal{J}/\mathcal{J}^2$$

The computation above shows that Ω_X restricts to $\widetilde{\Omega_{\text{Spec } R}}$ as we defined above. Moreover, the maps $d : \mathcal{O}_{\text{Spec } R} \rightarrow \Omega_R$ glue into a map of sheaves (but again, not $\mathcal{O}_X$ -modules) $d : \mathcal{O}_X \rightarrow \Omega_X$.

Suppose X is a smooth variety of dimension n . Then the tangent space has dimension n at each point. Moreover, we checked earlier that the fibers of the cotangent sheaf are the cotangent space at each point. Thus, if X is smooth then the cotangent sheaf should be locally free of rank n .

Proposition 6.20

Let X be a variety of pure dimension n . Then Ω_X is locally free of rank n if and only if X is smooth.

Proof. ($\implies$) If Ω_X is locally free of rank n then at each point, the fiber $\Omega_X|_p$ is of dimension n . We know that

$$\Omega_X|_p \cong (T_p X)^\vee$$

so $T_p X$ also has dimension n for all p . Thus, by a lemma before, since every tangent space has dimension $\dim X$, X is smooth.

($\impliedby$) If X is smooth, then pick $p \in X$ and take an affine neighborhood of p . Thus we can assume that $p \in X \subseteq \mathbb{A}^r$. Let

$$R = A(X) = K[x_1, \dots, x_n] / I(X) = K[x_1, \dots, x_n] / \langle f_1, \dots, f_m \rangle$$

From our computation before, we know that

$$(T_p X)^\vee = K dx_1 \oplus \dots \oplus K dx_n / \left\langle \sum_j \partial_{x_j} f_i(p) dx_j : 1 \leq i \leq m \right\rangle$$

By smoothness, $T_p X$ has dimension n . So the Jacobi matrix

$$J(p) = (\partial_{x_j} f_i(p))_{ij}$$

must have rank $r - n$, since we quotient by its image. Thus we can take an $(r - n) \times (r - n)$ submatrix of nonvanishing determinant, so we will assume that the submatrix of the last $r - n$ rows and columns has nonzero determinant. This determinant is a product of $\partial_{x_j} f_i(p) \in R$, so if h denotes the determinant at p , then $h \in R$. Via this matrix, we can then express $dx_{n+1}, \dots, dx_r$ in terms of the basis $dx_1, \dots, dx_n$ of $(T_p X)^\vee$, at least locally at p . To do this at a local level, we look at the distinguished open $D(h)$, which is an open neighborhood of p where h is invertible. On $D(h)$ we can still express $dx_{n+1}, \dots, dx_r$ as a linear combination of $dx_1, \dots, dx_n$, and the coefficients will lie in the localization R_h .

Thus $dx_1, \dots, dx_n$ generate Ω_{R_h} as an R_h module. To show that Ω_{R_h} is free, we need to show that there are no more relations. To see this, if there was some relation, then we would have some $g_i \in R_h$ with

$$g_1 dx_1 + \dots + g_n dx_n = 0$$

with (without loss of generality) $g_1 \neq 0$. But then we should be able to eliminate another generator and reduce the dimension of the tangent space, which is not possible. Precisely, by the Nullstellensatz there is a point $q \in D(h)$ where $g_1(q) \neq 0$. Evaluating the relation at the point q , we then have

$$g_1(q) dx_1 + \dots + g_n(q) dx_n \in (T_q X)^\vee$$

Since $dx_1, \dots, dx_n$ generate $T_q X$, the existence of a nontrivial relation means that $\dim T_q X \leq n - 1$, which is not possible. Thus Ω_{R_h} is a free R_h -module of rank n . The neighborhoods $\text{Spec } R_h$ cover $\text{Spec } R$, so $\Omega_{\text{Spec } R}$ is locally free of rank n . $\square$

Definition 6.20

If X is a smooth variety of pure dimension n , then the cotangent sheaf is a vector bundle and we can also call it the **cotangent bundle**. Then the **tangent sheaf** or **tangent bundle** is $T_X = \Omega_X^\vee$.

Example 6.32

In the case $X = \mathbb{P}^1$, then Ω_X is the twisting sheaf $\mathcal{O}_{\mathbb{P}^1}(-2)$. We have seen that the dual is then

$$T_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(2)$$

In particular, there are no nonzero global sections, meaning that from a differential geometry perspective, there are no nonzero algebraic differential forms on $\mathbb{P}^1$. On the other hand, any global section of $T_{\mathbb{P}^1}$ is just a homogeneous polynomial in two variables of degree 2, and thus has two zeros. Therefore any nonzero algebraic vector field on $\mathbb{P}^1$ has exactly two zeros. In the case $K = \mathbb{C}$, where $\mathbb{P}_{\mathbb{C}}^1$ is the sphere, this is related to the fact that there are not even nonvanishing continuous vector fields.

Example 6.33

Let $K = \mathbb{C}$ and take X to be a smooth projective variety which is connected and of dimension 1 over $\mathbb{C}$. Let S denote the subset of X of the closed points with the complex topology (for instance, by putting the complex topology on $\mathbb{P}_{\mathbb{C}}^n$ and taking the relative topology for $S \subseteq \mathbb{P}_{\mathbb{C}}^n$).

The algebraic properties here can be translated into complex topological properties. First, X being smooth of dimension 1 implies that S is a complex manifold of dimension 1, and therefore a real manifold of dimension 2. In particular, it is an orientable surface.

Since X is projective, and the projective space is compact under the complex topology, the closed subsets S is also compact. Finally, X being connected means S is as well.

Thus, S is a compact connected oriented real surface. But by the classification of surfaces, this is homeomorphic to the connect sum S_g of g tori for some g . Also, the value g is unique since S_g is not homeomorphic to any other $S_{g'}$.

Thus, the number $g = g(X)$ can be defined as the **genus** of X , which is an invariant that then exists for any smooth connected projective curve.

The cotangent sheaf allows us to compute the genus entirely with the methods of algebraic geometry, without needing to pass to the topological category.

Proposition 6.21

Let X be a smooth projective connected curve over K . Then $\Omega_X(X)$ is a finite dimensional $K = \mathcal{O}_X(X)$ vector space.

Definition 6.21

If X is a smooth connected projective curve over K , then we define the **arithmetic genus** of X to be

$$g_a(X) = \dim_K \Omega_X(X)$$

Theorem 6.22

If X is a smooth projective connected curve over $\mathbb{C}$, then the genus and arithmetic genus agree.

Thus, $g(X)$ can be computed entirely using the algebraic methods via $g_a(X)$. Using sheaf cohomology, one can compute this in the following special case:

Proposition 6.23

Let $X \subseteq \mathbb{P}^2$ be a smooth curve of degree d . Then

$$g_a(X) = \frac{(d-1)(d-2)}{2}$$

In particular, for $d = 1$, any linear curve just cuts out a copy of $\mathbb{P}^1$. For $d = 2$ we showed that any conic also cuts out an isomorphic copy of $\mathbb{P}^1$, so in both cases X looks like $\mathbb{P}_{\mathbb{C}}^1 = S^1$ which has genus 0. This can be seen algebraically since

$$g_a(X) = \dim_K \Omega_X(X) = \dim_{\mathbb{C}} \mathcal{O}_{\mathbb{P}^1}(-2)(\mathbb{P}^1) = \dim \{0\} = 0$$

so indeed these agree.

Finally, for cubic curves $d = 3$, X is called an elliptic curve. By the formula, the arithmetic genus of X is always 1. Thus the complex points are homeomorphic to the torus. When working with elliptic curves we defined a group law on the elliptic curves. If one chooses the correct homeomorphism, this group law becomes the group law on $S^1 \times S^1$.

Appendix A

Results in Commutative Algebra

This appendix is for results in commutative algebra which are not explicitly covered in the lecture notes. As in the main notes, every ring is assumed to be commutative with unity.

A.1 Hilbert's Basis Theorem

This result is taken from Eisenbud's *Commutative Algebra: with a View toward Algebraic Geometry*. One consequence of the basis theorem is that every ideal in a polynomial ring is finitely generated. This is an important basic result which is used in the proof that every variety is the finite union of irreducibles.

Definition A.1

A ring is said to be **Noetherian** if every strictly ascending chain of ideals terminates, or equivalently if every ideal is finitely generated.

Proposition A.1

These two definitions are equivalent.

Proof. For an ideal $I \subseteq R$, take $r_1 \in R$, $r_2 \in R \setminus \langle r_1 \rangle$, $r_3 \in R \setminus \langle r_1, r_2 \rangle$, and so on. Then $\langle r_1 \rangle \subseteq \langle r_1, r_2 \rangle \subseteq \dots$ is strictly ascending, so it terminates. It only terminates if $\langle r_1, \dots, r_k \rangle = I$, so I is finitely generated.

Take $I_1 \subseteq I_2 \subseteq \dots$. Then $\bigcup I_i$ is an ideal, and it is finitely generated. These generators are all included in some I_k , so the chain terminates. $\square$

Example A.1

Fields only have themselves and the zero ideal as ideals, so they are Noetherian. By the Chinese Remainder Theorem, $\mathbb{Z}$ is Noetherian, since any ideal is generated by the gcd.

Example A.2

Every ring is finitely generated as an ideal (since $R = \langle 1 \rangle$). However, we can also consider rings which are finitely generated as $\mathbb{Z}$ -modules. This means that every element is a finite sum over the generating set S of the form

$$n_1 s_1 + \dots + n_k s_k, \quad n_i \in \mathbb{Z}, s_i \in S$$

As a result of the basis theorem, we will see that any ring which is finitely generated as a $\mathbb{Z}$ -module, then the ring itself is also Noetherian.

Theorem A.2: Hilbert's Basis Theorem

If R is Noetherian then $R[x]$ is as well.

The basic point of the proof is that the leading terms of polynomials effectively act like the ring itself, when they share the same degree. For a polynomial

$$f = a_n x^n + \dots + a_1 x + a_0$$

a_n is called the leading coefficient and $a_n x^n$ the leading term.

Proof. Pick an ideal I of $R[x]$. Let f_1 be any nonzero polynomial of lowest degree in I . Recursively choose f_i to be a nonzero polynomial of lowest degree in $I \setminus (f_1, \dots, f_{i-1})$, terminating if this is ever empty. If it does not terminate, then we have infinitely many leading coefficients: denote the leading coefficient of f_j by a_j . Then $(a_1, a_2, \dots)$ is an ideal in R , and is in particular finitely generated by $a_1, \dots, a_m$ for some m . Then we can write

$$a_{m+1} = \sum_{j=1}^m a_j u_j$$

for some $u_j \in R$. We can then contradict our lowest degree construction by showing that the leading term of f_{m+1} is actually in $(f_1, \dots, f_m)$. By increasing the degree of our polynomials, we can cancel the leading term: define

$$g = \sum_{j=1}^m u_j a f_j x^{\deg f_{m+1} - \deg f_j}$$

which has leading term equal to $a_{m+1} x^{\deg f_{m+1}}$. Therefore $f_{m+1} - g$ has degree strictly less than f_{m+1} , but is also not in $(f_1, \dots, f_m)$, contradicting our choice of f_{m+1} . $\square$

A.2 Primary Decompositions of Noetherian Rings

This is Atiyah-MacDonald Chapter 7.

Lemma A.3

If R is Noetherian then every ideal is a finite intersection of irreducible ideals.

Proof. If there are some ideals that are not a finite intersection, then the set of such ideals is nonempty and has a maximal element $\mathfrak{a}$. This is reducible so it is the intersection of $\mathfrak{a} = \mathfrak{b} \cap \mathfrak{c}$ which both strictly contain $\mathfrak{a}$. But then $\mathfrak{b}, \mathfrak{c}$ are finite intersections of irreducible ideals and so is $\mathfrak{a}$. $\square$

Lemma A.4

In a Noetherian ring every irreducible ideal is primary.

Proof. We can quotient out by the ring, so this is equivalent to the statement that in any Noetherian ring, the zero ideal is primary if it is irreducible. Suppose the zero ideal is irreducible. Then take x, y with $y \neq 0$ and $xy = 0$. Then we need to show that $x^n = 0$ for some n .

Consider the chain of annihilators

$$\text{Ann}(x) \subseteq \text{Ann}(x^2) \subseteq \dots$$

The ring is Noetherian so there is n such that $\text{Ann}(x^n) = \text{Ann}(x^{n+1})$. We claim that $(x^n) \cap (y) = 0$. Indeed, if $a \in (x^n) \cap (y)$ then $a \in (y)$ so $ax = 0$. If also $a \in (x^n)$ then $a = bx^n$, and $ax = bx^{n+1} = 0$ so $b \in \text{Ann}(x^{n+1}) = \text{Ann}(x^n)$ so $0 = bx^n = a$. Then $(y) \neq 0$ so $(x^n) = 0$, since 0 is irreducible. So 0 is primary. $\square$

The following follows immediately:

Theorem A.5

In a Noetherian ring, every ideal has a primary decomposition.

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